M^{3}LB

A modular lattice Boltzmann solver on Kokkos

Scheme description, methodology and validation

Abstract

This document describes the numerical schemes, the software architecture and the complete validation record of `M3LB`, a portable lattice Boltzmann solver written in C++20 on top of Kokkos. The code targets NVIDIA GPUs and CPU clusters from a single source. It supports five lattices, two streaming schemes (including the in-place Esoteric Pull), two population storage conventions, four collision operators for the fluid (BGK, TRT, raw multiple relaxation time and central moments), a passive-scalar module for heat transfer a magnetohydrodynamic module built on a vector-valued distribution, and a multiphase module in which a conservative Allen–Cahn phase field drives a pressure-based fluid formulation at a density ratio, with rigid bodies that fall, float and roll under the flow's own reaction. Every scheme documented here is accompanied by the quantitative test that establishes it, and every stated limitation is one that has been measured rather than assumed.

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1 Scope and design

1.1 What the code is

`M3LB` solves the weakly compressible Navier–Stokes equations by the lattice Boltzmann method, optionally coupled to a passive scalar (temperature), to a magnetic field, or to a phase field carrying an immiscible interface between fluids of different density. It is written once and compiled for any Kokkos backend: Serial, Threads, OpenMP, CUDA, HIP or SYCL. Scalar precision is selected at configure time.

1.2 Governing equations

The fluid obeys

$$\partial_t\rho + \nabla\cdot(\rho\boldsymbol{u}) = 0, \tag{1}$$
$$\partial_t(\rho\boldsymbol{u}) + \nabla\cdot(\rho\boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla\cdot\boldsymbol{\tau} + \boldsymbol{F}, \tag{2}$$

with $p = \rho c_s^2$ and $\boldsymbol{\tau}$ the viscous stress. When the thermal module is active a scalar T is transported by

$$\partial_t T + \nabla\cdot(\boldsymbol{u}T) = D\,\nabla^2 T, \tag{3}$$

and couples back to the flow through the Boussinesq body force. When the magnetohydrodynamic module is active the magnetic field $\boldsymbol{b}$ obeys the induction equation

$$\partial_t\boldsymbol{b} = \nabla\times(\boldsymbol{u}\times\boldsymbol{b}) + \eta\,\nabla^2\boldsymbol{b}, \qquad \nabla\cdot\boldsymbol{b} = 0, \tag{4}$$

and acts on the flow through the Lorentz force $\boldsymbol{j}\times\boldsymbol{b}$ with $\boldsymbol{j} = \nabla\times\boldsymbol{b}$. When the multiphase module is active an order parameter ϕ marks the two fluids and obeys the conservative Allen–Cahn equation

$$\partial_t\phi + \nabla\cdot(\boldsymbol{u}\phi) = \nabla\cdot\Big[M\Big(\nabla\phi - \frac{4}{W}\,\phi(1-\phi)\,\boldsymbol{n}\Big)\Big], \qquad \boldsymbol{n} = \frac{\nabla\phi}{|\nabla\phi|}, \tag{5}$$

with M a mobility and W the interface width; density and viscosity are interpolated in ϕ, and the flow gains a capillary force. The fluid equation of state changes with it, and §10 is where that change is paid for.

1.3 Architectural rules

Seven rules govern the implementation. They are stated here because most of the design decisions in later sections follow from them rather than from the physics.

1. **No virtual functions in device code.** Lattice, streaming scheme, equilibrium, forcing, storage convention and collision operator are all compile-time policies that inline into a single kernel. Dynamic dispatch exists only in a host-side factory above the solver. A virtual call inside a Kokkos kernel runs, but it blocks inlining of the entire collision operator.
2. **Never pin the Kokkos memory layout.** Population fields are declared `View<Real**>` with extents `(node, q)` and the *default* layout. Kokkos then selects `LayoutLeft` on CUDA/HIP — node index fastest, hence a structure of arrays with coalesced access — and `LayoutRight` on CPU backends, giving an array of structures with good cache locality per node. One declaration is correct on both; hardcoding either costs performance on the other.
3. **Streaming and collision are one fused kernel.** The in-place streaming schemes cannot have a separate streaming pass, so nothing in the solver is written in a way that would assume one.
4. **Parity is a compile-time parameter.** Esoteric Pull addresses different storage slots on even and odd time steps. That decision must fold away at compile time, so `FluidSolver::step` dispatches to `run_step<0>` or `run_step<1>`.
5. **Periodicity is index wrapping, not halo copying.** See §3.3.
6. **Boundary conditions are alternative collision operators** applied to flagged cells, giving one branch per node rather than one per direction.
7. **The reference implementation is never deleted.** The two-lattice streaming scheme is retained permanently and cross-checked against Esoteric Pull after every step.

2 Lattices

2.1 Definition and constraints

A lattice is a set of Q velocities $\boldsymbol{c}_i$ with weights w_i and a lattice sound speed c_s. To recover the Navier–Stokes equations at second order the set must satisfy

$$\textstyle\sum_i w_i = 1, \qquad \sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}, \tag{6}$$

$$\textstyle\sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} = 0, \qquad \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} = c_s^4 (\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}). \tag{7}$$

Five lattices are provided. **D2Q5** and **D3Q7** satisfy (6) but *not* (7): they cannot support Navier–Stokes and are used only for scalar transport and for the magnetic field, where only the second moment is required.

lattice	Q	D	c_s^2	weights	Navier–Stokes
D2Q5	5	2	1/3	1/3; 1/6	no
D2Q9	9	2	1/3	4/9; 1/9; 1/36	yes
D3Q7	7	3	1/4	1/4; 1/8	no
D3Q27	27	3	1/3	8/27; 2/27; 1/54; 1/216	yes

Table 1: The four lattices. Note that D3Q7 has $c_s^2 = 1/4$, not 1/3.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0)$	$\frac{1}{3}$	0	3	$(0,1)$	$\frac{1}{6}$	4
1	$(1,0)$	$\frac{1}{6}$	2	4	$(0,-1)$	$\frac{1}{6}$	3
2	$(-1,0)$	$\frac{1}{6}$	1				

Table 2: D2Q5: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0)$	$\frac{4}{9}$	0	5	$(1,1)$	$\frac{1}{36}$	6
1	$(1,0)$	$\frac{1}{9}$	2	6	$(-1,-1)$	$\frac{1}{36}$	5
2	$(-1,0)$	$\frac{1}{9}$	1	7	$(1,-1)$	$\frac{1}{36}$	8
3	$(0,1)$	$\frac{1}{9}$	4	8	$(-1,1)$	$\frac{1}{36}$	7
4	$(0,-1)$	$\frac{1}{9}$	3				

Table 3: D2Q9: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

2.2 Velocity sets

The complete velocity sets follow, in the Esoteric Pull ordering of §2.3. These tables are generated from the same data the code compiles against, and the identities in the captions are re-verified in exact rational arithmetic at generation time.

2.3 The direction-ordering contract

Esoteric Pull walks opposite pairs as `for (i = 1; i < Q; i += 2)`. This requires that the rest velocity occupies index 0 and that opposite directions occupy *adjacent* indices, so that

$$\mathrm{opp}(0) = 0, \qquad \mathrm{opp}(i) = \begin{cases} i+1 & i \text{ odd} \\ i-1 & i \text{ even} \end{cases} \tag{8}$$

is pure integer arithmetic, with no lookup table in the kernel. This is a *contract*: every moment matrix in the code is generated against it.

Reordering the directions of a lattice is a column permutation of the moment transform matrices together with the matching permutation of f_i and f_i^{eq}. Every moment, raw or central, is therefore invariant, and only the per-direction post-collision expressions are permuted. This was verified numerically in exact rational arithmetic before the orderings were adopted.

2.4 Compile-time verification

Weights are stored as exact rationals ($w_i = $ `w_num`$[i]$/`w_den`) and c_s^2 as `cs2_num/cs2_den`, so all the identities above are checked at compile time in exact integer arithmetic rather than in floating point:

```
static_assert(check_pairing<L>(),       "opposite directions not adjacent");
static_assert(check_unique<L>(),        "duplicated velocity");
static_assert(check_w_sum<L>(),         "weights do not sum to 1");
static_assert(check_second_moment<L>(), "<cc> != cs2 I");
static_assert(check_third_moment<L>(),  "<ccc> != 0");
static_assert(check_fourth_moment<L>() == L::supports_navier_stokes, "");
```

The last line is an equality rather than a truth test: D2Q5 and D3Q7 are *required* to fail fourth-order isotropy, and a lattice whose flag disagrees with its actual moments will not compile.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0,0)$	$\frac{1}{4}$	0	4	$(0,-1,0)$	$\frac{1}{8}$	3
1	$(1,0,0)$	$\frac{1}{8}$	2	5	$(0,0,1)$	$\frac{1}{8}$	6
2	$(-1,0,0)$	$\frac{1}{8}$	1	6	$(0,0,-1)$	$\frac{1}{8}$	5
3	$(0,1,0)$	$\frac{1}{8}$	4				

Table 4: D3Q7: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{4}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0,0)$	$\frac{8}{27}$	0	9	$(1,-1,0)$	$\frac{1}{54}$	10	18	$(0,-1,1)$	$\frac{1}{54}$	17
1	$(1,0,0)$	$\frac{2}{27}$	2	10	$(-1,1,0)$	$\frac{1}{54}$	9	19	$(1,1,1)$	$\frac{1}{216}$	20
2	$(-1,0,0)$	$\frac{2}{27}$	1	11	$(1,0,1)$	$\frac{1}{54}$	12	20	$(-1,-1,-1)$	$\frac{1}{216}$	19
3	$(0,1,0)$	$\frac{2}{27}$	4	12	$(-1,0,-1)$	$\frac{1}{54}$	11	21	$(1,1,-1)$	$\frac{1}{216}$	22
4	$(0,-1,0)$	$\frac{2}{27}$	3	13	$(1,0,-1)$	$\frac{1}{54}$	14	22	$(-1,-1,1)$	$\frac{1}{216}$	21
5	$(0,0,1)$	$\frac{2}{27}$	6	14	$(-1,0,1)$	$\frac{1}{54}$	13	23	$(1,-1,1)$	$\frac{1}{216}$	24
6	$(0,0,-1)$	$\frac{2}{27}$	5	15	$(0,1,1)$	$\frac{1}{54}$	16	24	$(-1,1,-1)$	$\frac{1}{216}$	23
7	$(1,1,0)$	$\frac{1}{54}$	8	16	$(0,-1,-1)$	$\frac{1}{54}$	15	25	$(-1,1,1)$	$\frac{1}{216}$	26
8	$(-1,-1,0)$	$\frac{1}{54}$	7	17	$(0,1,-1)$	$\frac{1}{54}$	18	26	$(1,-1,-1)$	$\frac{1}{216}$	25

Table 5: D3Q27: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

3 Streaming

3.1 Two-lattice (reference)

Two arrays, $2Q$ reals per node. Semantics are *pull*: the kernel gathers the populations that have arrived at node n,

$$f_i^{\mathrm{in}}(n) = f_i^{\mathrm{src}}(n - \boldsymbol{c}_i), \tag{9}$$

collides, and writes locally to the destination array, which is then swapped. This scheme is obviously race-free and obviously correct. It is not the production scheme, but it is retained permanently: when an exotic collision operator produces incorrect results, being able to switch to a streaming scheme that cannot be wrong is what makes the fault bisectable.

3.2 Esoteric Pull

Esoteric Pull [4] stores a single copy of the populations and updates them in place. Write $A[n][s]$ for storage slot s at node n, and consider one opposite pair $(i, i+1)$ with i odd. The partner node is *always* $n + \boldsymbol{c}_i$, at both parities; only which of the two slots lives remotely alternates.

even step t	odd step t
$f_i \leftarrow A[n][i]$	$f_i \leftarrow A[n][i{+}1]$
$f_{i+1} \leftarrow A[n+\boldsymbol{c}_i][i{+}1]$	$f_{i+1} \leftarrow A[n+\boldsymbol{c}_i][i]$
$A[n+\boldsymbol{c}_i][i{+}1] \leftarrow f_i^{\mathrm{out}}$	$A[n+\boldsymbol{c}_i][i] \leftarrow f_i^{\mathrm{out}}$
$A[n][i] \leftarrow f_{i+1}^{\mathrm{out}}$	$A[n][i{+}1] \leftarrow f_{i+1}^{\mathrm{out}}$

Table 6: The Esoteric Pull access pattern, one opposite pair. Each parity reads exactly two slots and writes those same two, crossed.

Consistency. At an even step node n writes its outgoing f_i into $A[n+\boldsymbol{c}_i][i{+}1]$; at the next (odd) step node $m = n + \boldsymbol{c}_i$ reads f_i from $A[m][i{+}1]$, which is exactly the odd-step rule. The scheme closes.

Race freedom. Every slot has exactly one reader and one writer, and they are the same node. No temporary buffer and no synchronisation within a step are required.

Implicit bounce-back. Halfway bounce-back sets $f_i^{\text{out}} = f_{i+1}^{\text{in}}$ and $f_{i+1}^{\text{out}} = f_i^{\text{in}}$. Substituting into Table 6, every write lands back in the slot it was read from: the entire update is the identity. *Solid cells are therefore skipped outright* — no load, no store, no arithmetic. The population a fluid node emits toward a wall sits untouched for one step and is read back as the opposite direction on the next, placing the wall exactly midway between the two cells.

Memory. Q reals per node against $2Q$ for two-lattice. At 512^3 with D3Q27 in FP64 this is the difference between 29 GB and 58 GB, i.e. between fitting and not fitting on a 40 GB device.

3.3 Periodicity by index wrapping

Esoteric Pull writes into its neighbours' storage. A halo-copy treatment of periodicity would therefore require a *two-way* exchange whose set of exchanged slots depends on the parity. The code instead resolves periodicity by wrapping the neighbour index, which removes that entire class of bug:

$$\text{shift}(v) = \begin{cases} v+n & v<h \\ v-n & v \geq h+n \\ v & \text{otherwise} \end{cases} \quad (\text{periodic}), \qquad \text{clamp}(v,0,s-1) \quad (\text{otherwise}). \tag{10}$$

A halo layer is allocated only in non-periodic directions, where edge cells genuinely need somewhere out-of-domain to read from. A fast path skips the wrapping entirely for nodes away from every face.

4 Population storage

Two conventions are supported.

Raw the arrays hold f_i.

Shifted the arrays hold $g_i = f_i - w_i$.

The shift exists for conditioning. Populations are $O(w_i) \sim 10^{-1}$, while the collision operates on differences $O(10^{-6})$; forming $f_i - f_i^{\text{eq}}$ in FP32 therefore discards most of the mantissa. The momentum sum is worse, since $\sum_i \boldsymbol{c}_i f_i$ cancels terms of order 10^{-1} down to 10^{-2}. Storing g_i removes both cancellations, because $\sum_i \boldsymbol{c}_i w_i = 0$ exactly and hence

$$\sum_i \boldsymbol{c}_i\, g_i \;\equiv\; \sum_i \boldsymbol{c}_i\, f_i \tag{11}$$

identically, while every summand is small. The density follows from $\rho = 1 + \sum_i g_i$.

Almost nothing else changes. Streaming is a pure copy and is untouched. Halfway bounce-back is $g_i \leftarrow g_{\text{opp}(i)}$, untouched, because $w_i = w_{\text{opp}(i)}$. The Guo source term is already an $O(F)$ quantity and is untouched. Only the equilibrium and the density reconstruction differ.

Measured effect. Poiseuille flow at the magic relaxation time, $H = 32$, in FP32:

storage	FP64	FP32
raw	1.433×10^{-13}	5.969×10^{-4}
shifted	1.626×10^{-14}	$\mathbf{2.778 \times 10^{-5}}$
gain	$8.8\times$	**$21.5\times$**

The validation suite asserts that gain rather than merely tolerating its absence: a regression to raw storage fails the build.

For the passive scalar the same argument applies but there is *no universal reference*. Density is always near unity; a temperature scale is whatever the problem says it is. The scalar collision therefore exposes T_{ref} as a runtime parameter, defaulting to 0, which reproduces unshifted storage exactly.

5 Equilibria

This section states, for every lattice, the equilibrium that the code actually evaluates. The distinction matters in one place in particular: the moment-space operators of §6.3 *never form equilibrium populations at all.* Their equilibrium is defined directly in moment space, and the corresponding populations exist only implicitly, as the inverse transform of those moments.

5.1 The ϕ formulation

Each populations-space equilibrium policy supplies only $\phi_i(\boldsymbol{u})$, defined by

$$f_i^{\text{eq}} = w_i\, \rho\, (1 + \phi_i), \tag{12}$$

from which both storage conventions of §4 follow once:

$$\text{raw:} \quad f_i^{\text{eq}} = w_i\, \rho\, (1 + \phi_i), \tag{13}$$

$$\text{shifted:} \quad f_i^{\text{eq}} - w_i = w_i\big(\delta\rho + (1 + \delta\rho)\, \phi_i\big), \qquad \delta\rho = \rho - 1. \tag{14}$$

Form (14) is written that way deliberately: every term is small, so it never forms the difference of two $O(w_i)$ quantities. Writing it as $f_i^{\text{eq}} - w_i$ would be algebraically identical and numerically useless.

5.2 Second-order (truncated Hermite) — D2Q9, D3Q27

The default for BGK, TRT and the MHD fluid operator, on all three Navier–Stokes lattices:

$$\boxed{f_i^{\text{eq}} = w_i\, \rho \left[1 + \frac{\boldsymbol{c}_i \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u} \cdot \boldsymbol{u}}{2c_s^2}\right]} \tag{15}$$

With $c_s^2 = 1/3$ on all three, this is $f_i^{\text{eq}} = w_i \rho\big[1 + 3(\boldsymbol{c}_i \cdot \boldsymbol{u}) + \frac{9}{2}(\boldsymbol{c}_i \cdot \boldsymbol{u})^2 - \frac{3}{2}\boldsymbol{u}^2\big]$. It is correct to $O(u^2)$; its central moments carry $O(u^3)$ defects, which is what the Galilean-invariance test of §12.4 measures and what the moment operators remove.

5.3 Product form — D2Q9 and D3Q27

The complete (fourth-order) equilibrium, equivalent to `index_equilibrium = 4` of the symbolic generator. Writing $a_\alpha = c_{i\alpha}^2 - c_s^2$,

$$\boxed{\begin{aligned} f_i^{\text{eq}} = w_i \rho \Big[1 &+ \frac{c_{ix} u_x + c_{iy} u_y}{c_s^2} + \frac{a_x u_x^2 + a_y u_y^2 + 2 c_{ix} c_{iy} u_x u_y}{2c_s^4} \\ &+ \frac{a_x c_{iy} u_x^2 u_y + a_y c_{ix} u_x u_y^2}{2c_s^6} + \frac{a_x a_y\, u_x^2 u_y^2}{4c_s^8}\Big] \end{aligned}} \tag{16}$$

Its central moments are *exactly* Maxwellian for all nine representable k_{pq} — verified symbolically, 0 of 9 deviating — so it is fully Galilean invariant on this lattice.

D3Q27: sixth order. D3Q27 is a product lattice, so the same construction runs to sixth order — the highest its 27 velocities admit. The symbolic generator writes it as a sum of Hermite tensors of orders one to six. That sum was checked, in exact rational arithmetic, against the factorised form

$$\boxed{f_i^{\mathrm{eq}} = \rho\,\psi(c_{ix},u_x)\,\psi(c_{iy},u_y)\,\psi(c_{iz},u_z), \qquad \begin{aligned}\psi(0,u) &= 1 - c_s^2 - u^2,\\ \psi(\pm 1,u) &= \tfrac{1}{2}(c_s^2 + u^2 \pm u),\end{aligned}} \tag{17}$$

and the two are *identical, term for term.* The one-dimensional factor ψ is simply the three-point distribution carrying the exact moments $\{1,\,u,\,c_s^2+u^2\}$. Equation (17) is what the code evaluates: three multiplications instead of six Hermite tensors, bit-for-bit the same polynomial. Dividing each factor by its one-dimensional weight $\{\frac{1}{6},\frac{2}{3},\frac{1}{6}\}$ puts it in the ϕ form of §5.1, since $w_i = w^{1\mathrm{D}}(c_{ix})w^{1\mathrm{D}}(c_{iy})w^{1\mathrm{D}}(c_{iz})$ on a product lattice:

$$\phi_i = \chi(c_{ix},u_x)\,\chi(c_{iy},u_y)\,\chi(c_{iz},u_z) - 1, \qquad \chi(0,u) = 1 - \tfrac{3}{2}u^2, \quad \chi(\pm 1,u) = 1 + 3u(\pm 1 + u). \tag{18}$$

The defining property. The central moments of (16) and (17) are *exactly* Maxwellian in every representable moment:

$$k_{pqr}^{\mathrm{eq}} = \begin{cases}\rho\,(c_s^2)^n, & p,q,r \text{ all even},\; n = \#\{p,q,r \text{ equal to } 2\},\\ 0, & \text{otherwise},\end{cases} \tag{19}$$

so 8 of the 27 D3Q27 moments are nonzero and the remaining 19 vanish identically. Every Galilean-invariance defect of the truncation (15) is absent.
Where these are used. (16) and (17) are collected as `HighOrderEquilibrium<L>`, which selects the highest order each lattice admits. This is the default for the MHD operators of §6.4. The BGK and TRT defaults remain (15), and the moment-space operators of §5.4 reach the same equilibrium without evaluating any of these forms explicitly.

5.4 Equilibria in moment space — D2Q9, D3Q27

The MRT and central-moment operators specify their equilibrium as *moments*, in closed form. Writing $\boldsymbol{u}_b$ for the basis velocity ($\mathbf{0}$ for raw moments, $\boldsymbol{u}$ for central) and $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_b$, the equilibrium moment in the stored variable is a product of one-dimensional factors,

$$\boxed{E_{pqr} = \rho\,Q_p(\delta u_x)\,Q_q(\delta u_y)\,Q_r(\delta u_z) \;-\; [\text{shifted}]\;A_p(u_{bx})\,A_q(u_{by})\,A_r(u_{bz})} \tag{20}$$

with the triples depending on which basis the lattice uses:

lattice	basis	$Q_{0,1,2}$	$A_{0,1,2}$
D2Q9, D3Q27	product, $\varphi_2 = C^2 - c_s^2$	$1,\ \delta u,\ \delta u^2$	$1,\ -u_b,\ u_b^2$

Two consequences are worth stating explicitly.

Central moments on D2Q9 and D3Q27. Here $\delta\boldsymbol{u} = \mathbf{0}$, so $Q = (1,0,0)$ and (20) collapses to

$$E_{0\ldots0} = \rho, \qquad E_{pqr} = 0 \text{ otherwise} \qquad \text{(raw storage)}. \tag{21}$$

One nonzero equilibrium moment. This is the property that makes the operator cheap: no equilibrium populations, no stored K^{eq} vector, no $Q \times Q$ matrices. It was verified symbolically — 0 of 9 deviating on D2Q9, 0 of 27 on D3Q27 — and it is equivalent to relaxing toward (16) and its D3Q27 counterpart.

Raw MRT. Here $\boldsymbol{u}_b = \mathbf{0}$ and $\delta\boldsymbol{u} = \boldsymbol{u}$, so $Q = (1, u, u^2)$ on the product basis and $(1, u, c_s^2 + u^2)$ on the monomial one — the raw moments of a Maxwellian centred on $\boldsymbol{u}$. Note this gives raw MRT the *complete* equilibrium, not a truncated one, which is why it recovers most of the Galilean-invariance gain on its own (§12.4).

5.5 Scalar transport — D2Q5, D3Q7

The passive scalar carries only its zeroth moment, $T = \sum_i g_i$, and is advected by a velocity supplied from outside:

$$\boxed{g_i^{\mathrm{eq}} = w_i\, T\left(1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2}\right)} \qquad D = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{22}$$

Explicitly, with $c_s^2 = 1/3$ on D2Q5 and $c_s^2 = 1/4$ on D3Q7,

$$\begin{aligned} &\text{D2Q5:}\quad g_i^{\mathrm{eq}} = w_i T(1 + 3\,\boldsymbol{c}_i\cdot\boldsymbol{u}), && w = \big(\tfrac{1}{3};\ \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}\big), && (23)\\ &\text{D3Q7:}\quad g_i^{\mathrm{eq}} = w_i T(1 + 4\,\boldsymbol{c}_i\cdot\boldsymbol{u}), && w = \big(\tfrac{1}{4};\ \tfrac{1}{8}\ \text{(six times)}\big). && (24)\end{aligned}$$

In the shifted storage of §4 the code evaluates

$$h_i^{\mathrm{eq}} = w_i\left[\delta T + (T_{\mathrm{ref}} + \delta T)\,\frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2}\right], \qquad \delta T = T - T_{\mathrm{ref}}, \tag{25}$$

which is (22) minus $w_i T_{\mathrm{ref}}$, written so that no term is the difference of two $O(w_i T)$ quantities.

Why first order. The truncation in (22) is deliberate. D2Q5 and D3Q7 have an isotropic second moment but not an isotropic fourth-order one, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\cdot\boldsymbol{u})^2$ reduces to $c_\alpha^2 u_\alpha^2$ and the cross terms of the $\boldsymbol{uu}$ tensor cannot be represented at all. A second-order term would add an anisotropic error rather than accuracy. The cost is an $O(u^2)$ defect in the advection term, which is why these lattices suit low-Mach transport.

5.6 Magnetic induction — D2Q5, D3Q7

The magnetic field is carried by a distribution that is a *vector* at every link, with $b_\alpha = \sum_i g_i^\alpha$:

$$\boxed{g_i^{\mathrm{eq},\alpha} = w_i\left[b_\alpha + \frac{c_{i\beta}}{c_s^2}(u_\beta b_\alpha - b_\beta u_\alpha)\right]} \qquad \eta = c_s^2\Big(\frac{1}{\omega_\eta} - \frac{1}{2}\Big). \tag{26}$$

Its first two moments are $\sum_i g_i^{\mathrm{eq},\alpha} = b_\alpha$ and $\sum_i c_{i\beta} g_i^{\mathrm{eq},\alpha} = u_\beta b_\alpha - b_\beta u_\alpha$, which is exactly the antisymmetric flux of the induction equation. Only that first moment is required, which is why a five- or seven-velocity lattice suffices for the field while the flow runs on nine, nineteen or twenty-seven.

5.7 MHD fluid equilibrium — D2Q9, D3Q27

The Lorentz force enters through the equilibrium's second moment rather than as a body force. To (15) is added

$$\boxed{\delta f_i = \frac{w_i}{2c_s^4}\, M_{\alpha\beta}(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}), \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta} \tag{27}$$

so that $\sum_i c_{i\alpha}c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + (p + \tfrac{1}{2}|\boldsymbol{b}|^2)\delta_{\alpha\beta} - b_\alpha b_\beta$, while $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \mathbf{0}$ leave mass and momentum untouched.

Dimensional form. In two dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = |\boldsymbol{b}|^2 - |\boldsymbol{b}|^2 = 0$, so (27) reduces to the familiar

$$\delta f_i = \frac{w_i}{2c_s^4}\Big[\tfrac{1}{2}|\boldsymbol{c}_i|^2|\boldsymbol{b}|^2 - (\boldsymbol{c}_i\cdot\boldsymbol{b})^2\Big] \qquad \text{(2D only)}. \tag{28}$$

In three dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = \frac{1}{2}|\boldsymbol{b}|^2 \neq 0$ and the trace subtraction is retained. This is not cosmetic: it is precisely the term whose projection onto the plane governs the quasi-two-dimensional reduction test of §12.10.3.

5.8 Summary: which equilibrium is used where

operator	lattices	equilibrium	where
BGK, TRT	D2Q9, D3Q27	second order, (15)	§5.2
MRT (raw moments)	D2Q9, D3Q27	moment space, (20), $\boldsymbol{u}_b = \boldsymbol{0}$	§5.4
central moments	D2Q9, D3Q27	moment space, (20), $\boldsymbol{u}_b = \boldsymbol{u}$	§5.4
MHD fluid (BGK)	D2Q9, D3Q27	(15) + (27)	§5.7
MHD central moments	D2Q9	(16) + (27); CMs (42)	§6.4
	D3Q27	(17) + (27); CMs (45)	§6.5
	(`-eq2`)	second order (15); CMs (41)	§6.4
scalar transport	D2Q5, D3Q7	first order, (22)	§5.5
magnetic induction	D2Q5, D3Q7	Dellar, (26)	§5.6
highest order each lattice admits: (16) D2Q9, (17) D3Q27			
regularised walls rebuild f_i from the same equilibrium, §8.1			

6 Collision operators

6.1 BGK

$$f_i^\star = f_i + \omega\,(f_i^{\mathrm{eq}} - f_i) + S_i, \qquad \nu = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{29}$$

6.2 TRT

Split each population into parts symmetric and antisymmetric about the opposite direction,

$$f_i^\pm = \tfrac{1}{2}\big(f_i \pm f_{\mathrm{opp}(i)}\big), \tag{30}$$

and relax them at different rates ω^+ (which sets the viscosity) and ω^- (free). The free rate buys the *magic parameter*

$$\Lambda = \Big(\frac{1}{\omega^+} - \frac{1}{2}\Big)\Big(\frac{1}{\omega^-} - \frac{1}{2}\Big). \tag{31}$$

At $\Lambda = 3/16$ the halfway bounce-back wall sits exactly midway between the fluid and solid nodes at *any* viscosity. BGK manages this only at the single value $\tau = \frac{1}{2} + \frac{\sqrt{3}}{4}$, because BGK is TRT with $\omega^- = \omega^+$.

The direction-ordering contract (8) pays off here: the pair loop is `for (i = 1; i < Q; i += 2)` with no lookup table. The equilibrium is evaluated in a separate flat loop first, because the pair loop reads f_i and f_{i+1} together and does not vectorise; leaving the equilibrium inside it cost more than the temporary array does (§19).

6.3 Moment-space collision: raw MRT and central moments

Both are one implementation. They differ only in the velocity at which the moment basis is constructed:

$$\boldsymbol{u}_b = \mathbf{0} \quad \text{(raw moments, MRT)}, \qquad \boldsymbol{u}_b = \boldsymbol{u} \quad \text{(central moments)}, \tag{32}$$

which is precisely the `central_moments` toggle of the symbolic generator.

6.3.1 Product basis (D2Q9, D3Q27)

A *product lattice* contains every velocity in $\{-1, 0, 1\}^D$ exactly once. For these the basis is a product of the same three one-dimensional functions on each axis,

$$\varphi_0(C) = 1, \qquad \varphi_1(C) = C, \qquad \varphi_2(C) = C^2 - c_s^2, \qquad C = c - u_b, \tag{33}$$

so the transform factorises into D successive one-dimensional passes rather than one $Q \times Q$ contraction:

$$\text{D3Q27: } 1458 \to 324 \text{ multiply–adds } (4.5\times), \qquad \text{D2Q9: } 162 \to 72. \tag{34}$$

Each pass, on values at $c = -1, 0, +1$, is

$$m_0 = g_- + g_0 + g_+, \qquad m_1 = (g_+ - g_-) - u\, m_0, \tag{35}$$

$$m_2 = (g_+ + g_-) - 2u(g_+ - g_-) + (u^2 - c_s^2)\, m_0, \tag{36}$$

with an exactly invertible counterpart. The factorised transform was checked against the dense contraction in exact rational arithmetic.

Why this basis. Basis (33) diagonalises the Maxwellian. The product-form equilibrium has *exactly one* nonzero moment in it:

$$k^{\text{eq}}_{00\ldots} = \rho, \qquad k^{\text{eq}}_{pqr} = 0 \quad \text{otherwise}, \tag{37}$$

verified symbolically for D2Q9 (0 of 9 deviating) and D3Q27 (0 of 27). No equilibrium populations are ever formed and no K^{eq} vector is stored, which is what collapses the collision to a handful of lines.

6.3.2 Relaxation and forcing

Writing $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_b$, the equilibrium moments in the *stored* variable are

$$E_{pqr} = \rho\, Q_p(\delta u_x) Q_q(\delta u_y) Q_r(\delta u_z) - [\text{shifted}]\ A_p(u_{bx}) A_q(u_{by}) A_r(u_{bz}), \tag{38}$$

with $Q = \{1, \delta u, \delta u^2\}$ and $A = \{1, -u_b, u_b^2\}$ for the product basis (and the corresponding c_s^2-carrying triples for the monomial basis). The relaxation is then

$$\begin{aligned} \text{order } 0 :&\quad \text{conserved} \\ \text{order } 1 :&\quad k^\star = k + F_\alpha \\ \text{order } 2 :&\quad \text{trace at } \omega_{\text{bulk}},\ \text{deviatoric and shear at } \omega \\ \text{order} \geq 3 :&\quad k^\star = E. \end{aligned} \tag{39}$$

The Guo source has moments $\sum_i c_{i\alpha} F_i = F_\alpha$ and $\sum_i c_{i\alpha} c_{i\beta} F_i = u_\alpha F_\beta + u_\beta F_\alpha$. Transformed to the basis at $\boldsymbol{u}_b$ these become F_α and $(F_\beta\, \delta u_\alpha + F_\alpha\, \delta u_\beta)$. *For central moments $\delta\boldsymbol{u} = \mathbf{0}$, so the second-order force contribution vanishes identically* and the force enters only through the first moments — the well-known elegance of collision in the co-moving frame. For raw MRT it does not vanish and is applied with the usual $(1 - \omega/2)$ prefactor.

Bulk viscosity is an explicit `omega_bulk`, defaulting to ω.

6.4 MHD central moments

The hybrid scheme of De Rosis, Lévêque and Chahine [6] is implemented separately, because its equilibria are *not* the Maxwellian ones. Central moments are taken in that paper's non-orthogonal basis

$$\begin{aligned} &t_0 = 1, && t_1 = C_x, && t_2 = C_y, && t_3 = C_x^2 + C_y^2, \\ &t_4 = C_x^2 - C_y^2, && t_5 = C_xC_y, && t_6 = C_x^2C_y, && t_7 = C_xC_y^2, \quad t_8 = C_x^2C_y^2, \end{aligned} \tag{40}$$

with $\boldsymbol{C} = \boldsymbol{c}_i - \boldsymbol{u}$, and relaxed toward

$$\begin{aligned} &k_0^{\mathrm{eq}} = \rho, && k_1^{\mathrm{eq}} = k_2^{\mathrm{eq}} = 0, \qquad k_3^{\mathrm{eq}} = \tfrac{2}{3}\rho, \\ &k_4^{\mathrm{eq}} = b_y^2 - b_x^2, && k_5^{\mathrm{eq}} = -b_xb_y, \\ &k_6^{\mathrm{eq}} = -\rho u_x^2u_y + \tfrac{u_y}{2}(b_x^2 - b_y^2) + 2u_xb_xb_y, \\ &k_7^{\mathrm{eq}} = -\rho u_xu_y^2 + \tfrac{u_x}{2}(b_y^2 - b_x^2) + 2u_yb_xb_y, \\ &k_8^{\mathrm{eq}} = \tfrac{\rho}{9}(27u_x^2u_y^2 + 1) + \tfrac{u_x^2-u_y^2}{2}(b_x^2 - b_y^2) - 4u_xu_yb_xb_y, \end{aligned} \tag{41}$$

with $\omega_4 = \omega_5 = \omega_\nu$, ω_3 setting the bulk viscosity and $\omega_6 = \omega_7 = \omega_8 = 1$. k_0, k_1, k_2 are collision invariants.

Raising the equilibrium order in two dimensions. Substituting the product form (16) for (15) leaves k_0–k_5 of (41) untouched and removes precisely the ρ-dependent terms from the rest:

$$k_6^{\mathrm{eq}} = \tfrac{u_y}{2}(b_x^2-b_y^2)+2u_xb_xb_y, \qquad k_7^{\mathrm{eq}} = \tfrac{u_x}{2}(b_y^2-b_x^2)+2u_yb_xb_y, \qquad k_8^{\mathrm{eq}} = \tfrac{\rho}{9}+\tfrac{u_x^2-u_y^2}{2}(b_x^2-b_y^2)-4u_xu_yb_xb_y. \tag{42}$$

This is the default; (41) is selected by `-eq2` and is what the validation of §12.10 against the published Table 1 uses.

These equilibria are deliberately non-Maxwellian. They are the central moments of the *second-order truncated* equilibrium plus the Maxwell term, not of a product-form equilibrium. The terms $-\rho u_x^2u_y$ in k_6 and $3\rho u_x^2u_y^2$ in k_8 are exactly the Galilean defects of the truncation. Reproducing the published scheme means reproducing them rather than improving them away, which is why this is a separate operator rather than a configuration of §6.3.

Verification. Equation (41) was not taken on trust. The test suite constructs the paper's equilibrium (69) directly, contracts its central moments by summation, and compares: all nine agree to 1.1×10^{-16}, which also confirms the sign convention on $\boldsymbol{b}$.

6.5 Three dimensions: D3Q27, and the effect of equilibrium order

Reference [6] prescribes D3Q27 for the velocity field (with D3Q7 for the magnetic field) in three dimensions. The operator is extended accordingly. Writing $\boldsymbol{C} = \boldsymbol{c}_i - \boldsymbol{u}$ and

$$k_{pqr} = \sum_i f_i\, C_x^p\, C_y^q\, C_z^r, \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta, \tag{43}$$

the equilibrium central moments follow from contracting the three-dimensional equilibrium — hydrodynamic part plus the Maxwell term (27) — against the monomial basis. They were obtained symbolically in exact rational arithmetic, and depend on which hydrodynamic equilibrium is used.

With the second-order truncation (15). Orders two and three close in a form that holds for every index combination,

$$k^{\mathrm{eq}}_{\alpha\beta} = \rho c_s^2 \delta_{\alpha\beta} + M_{\alpha\beta}, \qquad k^{\mathrm{eq}}_{\alpha\beta\gamma} = -\rho\, u_\alpha u_\beta u_\gamma - \big(u_\alpha M_{\beta\gamma} + u_\beta M_{\alpha\gamma} + u_\gamma M_{\alpha\beta}\big), \tag{44}$$

and 24 of the 27 moments are nonzero. Setting $u_z = b_z = 0$ reduces (44) and its fourth-order companions to (41) exactly, moment for moment — the three-dimensional scheme is a genuine extension of the published one, not a different model that happens to agree in the bulk.

With the sixth-order equilibrium (17). Because (19) holds, the hydrodynamic contribution collapses to the Maxwellian and *every ρ-dependent term above second order disappears*:

$$\boxed{\begin{aligned}
k^{\mathrm{eq}}_{000} &= \rho, \qquad k^{\mathrm{eq}}_{\alpha} = 0,\\
k^{\mathrm{eq}}_{\alpha\beta} &= \rho c_s^2 \delta_{\alpha\beta} + M_{\alpha\beta},\\
k^{\mathrm{eq}}_{\alpha\beta\gamma} &= -\big(u_\alpha M_{\beta\gamma} + u_\beta M_{\alpha\gamma} + u_\gamma M_{\alpha\beta}\big),\\
k^{\mathrm{eq}}_{aacc} &= \tfrac{\rho}{9} + \tfrac{1}{3} b_d^2 + u_c^2 M_{aa} + u_a^2 M_{cc} + 4 u_a u_c M_{ac},\\
k^{\mathrm{eq}}_{aacd} &= u_c u_d M_{aa} + \big(\tfrac{1}{3} + u_a^2\big) M_{cd} + 2 u_a u_d M_{ac} + 2 u_a u_c M_{ad},\\
k^{\mathrm{eq}}_{acc\,dd} &= -\tfrac{1}{3} u_a b_a^2 - \tfrac{2}{3}\big(u_c M_{ac} + u_d M_{ad}\big) - 2 u_c u_d^2 M_{ac} - 2 u_c^2 u_d M_{ad}\\
&\quad - 4 u_a u_c u_d M_{cd} - u_a u_d^2 M_{cc} - u_a u_c^2 M_{dd},\\
k^{\mathrm{eq}}_{222} &= \tfrac{\rho}{27} + \tfrac{1}{18}|\boldsymbol{b}|^2 + \tfrac{1}{3}\sum_\alpha u_\alpha^2 b_\alpha^2 + \tfrac{4}{3}\sum_{\alpha<\beta} u_\alpha u_\beta M_{\alpha\beta}\\
&\quad + 4\big(u_x u_y u_z^2 M_{xy} + u_x u_y^2 u_z M_{xz} + u_x^2 u_y u_z M_{yz}\big)\\
&\quad + u_y^2 u_z^2 M_{xx} + u_x^2 u_z^2 M_{yy} + u_x^2 u_y^2 M_{zz},
\end{aligned}} \tag{45}$$

where (a, c, d) denotes a permutation of (x, y, z). Comparing with the second-order case, the terms lost are exactly $-\rho u_\alpha u_\beta u_\gamma$ at order three, $3\rho u_a^2 u_c^2$ and $3\rho u_a^2 u_c u_d$ at order four, and so on: the Galilean defects, and nothing else. What remains is the Maxwell stress, which is not a defect but the physics.

Implementation and verification. Rather than evaluate 27 closed forms — orders five and six do not factorise compactly — the operator builds the equilibrium populations and transforms them with the same monomial central-moment transform used for the populations themselves. The order of the hydrodynamic part is a compile-time switch, defaulting to (17); the second-order form is retained so the published results of [6] stay reproducible. The test suite checks, at *both* orders: the monomial transform against a direct contraction (2.2×10^{-16}); the closed forms (44) and (45) (5.6×10^{-17}); the reduction to the two-dimensional scheme at $u_z = b_z = 0$; and conservation of mass and all three momentum components.

Relaxation. As in two dimensions, the trace $k_{200} + k_{020} + k_{002}$ relaxes at ω_{bulk}, its five deviatoric partners at ω_ν, and everything of order three and above goes straight to equilibrium. Note that in three dimensions the trace carries a magnetic contribution that is absent in two,

$$k^{\mathrm{eq}}_{200} + k^{\mathrm{eq}}_{020} + k^{\mathrm{eq}}_{002} = \rho + \tfrac{1}{2}|\boldsymbol{b}|^2, \tag{46}$$

because $M_{\alpha\alpha} = \frac{1}{2}|\boldsymbol{b}|^2$ in 3D but vanishes in 2D — the same trace term discussed in §5.7.

7 Forcing

7.1 Guo

$$\boldsymbol{u} = \frac{1}{\rho}\Big(\sum_i \boldsymbol{c}_i f_i + \tfrac{1}{2}\boldsymbol{F}\Big), \qquad S_i = \Big(1 - \frac{\omega}{2}\Big) w_i \left[\frac{\boldsymbol{c}_i - \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i \cdot \boldsymbol{u})\,\boldsymbol{c}_i}{c_s^4}\right] \cdot \boldsymbol{F}. \tag{47}$$

The $(1 - \omega/2)$ prefactor is applied *per relaxed mode*, not once: TRT and the moment operators relax different modes at different rates, so only BGK can fold a single prefactor in. The raw source is exposed separately for that reason.

7.2 High-order Hermite forcing, and why it is already here

The Guo term (47) is the body force expanded in Hermite polynomials and truncated at second order. Nothing stops the expansion being carried further, and on these lattices it can be taken as far as the equilibrium goes: fourth order on D2Q9, sixth on D3Q27. Writing $\mathcal{H}^{(n)}$ for the Hermite tensors in $\boldsymbol{c}_i$ and building the coefficients by the product rule — replacing one $\boldsymbol{u}$ slot by $\boldsymbol{F}$ at a time,

$$a^{(1)}_{\alpha} = F_\alpha, \quad a^{(2)}_{\alpha\beta} = F_\alpha u_\beta + u_\alpha F_\beta, \quad a^{(3)}_{\alpha\beta\gamma} = F_\alpha u_\beta u_\gamma + u_\alpha F_\beta u_\gamma + u_\alpha u_\beta F_\gamma, \ \ldots \tag{48}$$

— the source term is

$$F_i = w_i \sum_{n\geq 1} \frac{1}{n!\, c_s^{2n}}\, a^{(n)}_{\alpha_1\ldots\alpha_n}\, \mathcal{H}^{(n)}_{\alpha_1\ldots\alpha_n}(\boldsymbol{c}_i). \tag{49}$$

Which components survive. Not every Hermite component may be kept. Two rules, both forced by the lattice rather than chosen:

- Components with an odd exponent ≥ 3 on one axis *vanish identically.* On D2Q9, $c_x^3 = c_x$ and $1/c_s^2 = 3$ give $\mathcal{H}^{(3)}_{xxx} = \hat{c}_x^3 - 3\hat{c}_x \equiv 0$. They may be kept or dropped freely.
- Components with an *even* exponent ≥ 4 on one axis are nonzero but *not independent*: $\mathcal{H}^{(4)}_{xxxx} = -3\,\mathcal{H}^{(2)}_{xx}$. Including them injects a spurious $O(Fu^3)$ term into the second moment and destroys $k_{20} = k_{02} = 0$. They must be omitted.

What remains is exactly the set of exponent triples in $\{0, 1, 2\}$ — the same representable set the equilibrium uses.

The result, and it is simple. Transforming (49) to *monomial* central moments $k_{pqr} = \sum_i F_i\, C_x^p C_y^q C_z^r$ collapses to one rule. If exactly one axis a carries exponent 1 and m other axes carry exponent 2, then

$$\boxed{k^{\text{force}} = c_s^{2m} F_a} \qquad \text{and every other moment is exactly zero.} \tag{50}$$

On D2Q9 that is $[0, F_x, F_y, 0, 0, 0, c_s^2 F_y, c_s^2 F_x, 0]$ in the ordering $(00, 10, 01, 20, 02, 11, 21, 12, 22)$. On D3Q27 it is three entries F_a, six at $c_s^2 F_a$, three at $c_s^4 F_a$, and fifteen exact zeros. In the co-moving frame the force touches only the moments that are odd in exactly one axis — which is the whole reason for working there.

And the code already does this. It is tempting to read (50) as saying that an operator which adds $\boldsymbol{F}$ only to the first-order moments, as §6.3 does, is missing the c_s^{2m} entries. It is not. `ProductBasis` stores $\varphi_2(C) = C^2 - c_s^2$ per axis, so expanding $C^2 = \varphi_2 + c_s^2$,

$$k^{\text{monomial}}_{122} = \tilde{k}_{122} + c_s^2 \tilde{k}_{120} + c_s^2 \tilde{k}_{102} + c_s^4 \tilde{k}_{100}, \tag{51}$$

and with only $\tilde{k}_{100} = F_x$ populated the monomial value is $c_s^4 F_x$, exactly as required — one factor of c_s^2 per squared axis, delivered by the basis function itself. The momentum source alone reproduces all of (50).

This was established the hard way. The third-order terms were added explicitly on top, and the operator then injected $1.5\, c_s^2 F_y$ where $c_s^2 F_y$ is correct — a clean 50% overshoot from double counting. No change to the collision operator was needed or kept. `validation/forcing_cm.cpp` checks both representations of the same populations on both lattices: monomial must show the full c_s^{2m} pattern, Hermite must show the force in the first-order slots and nowhere else. Worst deviation 5.6×10^{-13}.

7.3 Boussinesq buoyancy

$$\boldsymbol{F}(n) = \rho_0\, \boldsymbol{g}\, \beta \left(T(n) - T_0\right), \tag{52}$$

delivered through the same Guo machinery. Because this force varies in space, the collision interface carries a node index; that hook is also what the Lorentz force uses.

8 Boundary conditions

Boundary conditions are alternative collision operators applied to flagged cells. This gives one branch per *node* rather than one per direction, keeps the population loads unconditional and coalesced, and generalises to any wall model.

condition	rule	wall location
no-slip (halfway bounce-back)	$f_i^{\text{out}} = f_{\text{opp}(i)}^{\text{in}}$	midway
adiabatic scalar	$g_i^{\text{out}} = g_{\text{opp}(i)}^{\text{in}}$	midway
Dirichlet scalar (anti-bounce-back)	$g_i^{\text{out}} = -g_{\text{opp}(i)}^{\text{in}} + 2w_i\,(T_w - T_{\text{ref}})$	midway

There is an asymmetry worth stating. Bounce-back is the *identity* on Esoteric Pull's storage (§3.2), so those cells are skipped entirely. Anti-bounce-back flips a sign and adds a source, so those cells must be processed at every step. Both still write back into the same two slots they read, so neither disturbs the in-place scheme.

All three conditions above put the wall *midway* between nodes. The three that follow put it *on* the node. That difference is not a detail of accuracy but of geometry: a channel of N nodes is N lattice units wide under bounce-back and $N-1$ under an on-node condition, and mixing the two families in one problem means the fluid and the field disagree about where the wall is.

8.1 Regularised velocity walls

Boundary condition BC3 of Latt, Chopard, Malaspinas, Deville and Michler [1]. *Every* population on the node is replaced, using only ρ, the imposed $\boldsymbol{u}$, and the non-equilibrium stress:

$$\boxed{g_i = f_i^{\text{eq}}(\rho, \boldsymbol{u}) + \frac{w_i}{2c_s^4}\, \boldsymbol{Q}_i : \Pi^{(1)}, \qquad \boldsymbol{Q}_i = \boldsymbol{c}_i\boldsymbol{c}_i - c_s^2 \boldsymbol{I}.} \tag{53}$$

Note what (53) does *not* contain: ω. The reconstruction is a projection of a given tensor onto the lattice, so it serves BGK, TRT, MRT and the central-moment operators without change.

Density. The paper's closure is written for one particular figure and velocity numbering. It is implemented here geometrically, which is numbering independent and therefore survives the Esoteric Pull ordering: with $\hat{\boldsymbol{n}}$ the *outward* normal, the known populations are those with $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0$, and

$$\rho = \frac{1}{1 + \boldsymbol{u} \cdot \hat{\boldsymbol{n}}} \Big(2 \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0} f_i + \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} = 0} f_i \Big). \tag{54}$$

Stress. The unknown populations are given, *temporarily*, a bounce-back of their off-equilibrium part, and $\Pi^{(1)} = \sum_i \boldsymbol{c}_i\boldsymbol{c}_i(f_i - f_i^{\mathrm{eq}})$ is evaluated from that. The paper is explicit that this may not be used as the boundary condition itself — it cannot enforce $\boldsymbol{u}$ exactly — and it exists here only to give $\Pi^{(1)}$ a value.

Which populations are unknown. Not a half-space. A dot-product test $\boldsymbol{c}_i\cdot\hat{\boldsymbol{n}} < 0$ is correct on a straight wall and *wrong* on a corner: at the corner $(0,0)$ of a box, both $(1,-1)$ and $(-1,1)$ are unknown, each having its source node outside a different wall. The code therefore carries a per-node bitmask, built once from the geometry: bit i is set when the source $\boldsymbol{p} - \boldsymbol{c}_i$ lies outside the fluid. Where *both* members of a pair are unknown — possible only on corners and edges — those populations are set to equilibrium, contributing nothing to $\Pi^{(1)}$, which is the only choice that invents no stress.

Corners and edges. Equation (54) needs a single normal, which a corner does not have, so ρ is extrapolated to second order along one wall from two straight-wall neighbours — never reaching into the bulk, so no density field is required. For $\Pi^{(1)}$ there are two routes, and the default is the second:

- *local* — the same closure as a straight wall. Operator-agnostic, but a 2D corner offers only three streamed directions to build a stress from.
- *finite-difference* — the route of [1], Sec. V: $\Pi^{(1)}$ from second-order one-sided velocity gradients through
$$\Pi^{(1)} = -\frac{2c_s^2}{\omega}\,\rho\,\boldsymbol{S}, \qquad \boldsymbol{S} = \tfrac{1}{2}\big[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\big]. \tag{55}$$

Neighbour velocities are taken from the imposed value when the neighbour is itself a wall node — its own unknowns are not yet fixed, so reading its populations would be wrong — and from its populations otherwise.

This is where the condition acquires a collision dependence. Equation (55) is a constitutive law, and its coefficient is the viscosity $\nu = c_s^2(1/\omega - 1/2)$. Inferring stress from strain rate therefore *requires* naming the shear relaxation rate, whereas measuring it from the populations does not: the operator's effect is already carried by the data. The rate is read from the operator (`omega`, or `omega_p` for TRT, whose even rate sets the viscosity), never supplied by the caller; an operator exposing none cannot select this route at all, which removes by construction the possibility of silently evaluating (55) with a placeholder $\omega = 1$ and imposing the wrong viscosity at the wall.

Body forces. Guo forcing defines $\rho\boldsymbol{u} = \sum_i f_i\boldsymbol{c}_i + \boldsymbol{F}/2$, so a reconstruction must satisfy $\sum_i \boldsymbol{c}_i g_i = \rho\boldsymbol{u}_w - \boldsymbol{F}/2$. Equation (53) delivers $\rho\boldsymbol{u}_w$, because $\sum_i w_i\boldsymbol{c}_i\boldsymbol{Q}_i = \boldsymbol{0}$. Two corrections follow from the forced Chapman–Enskog expansion and both are applied by default.

First, the expansion gives (dropping $O(\mathrm{Ma}^3)$)

$$\sum_i \boldsymbol{c}_i f_i^{(1)} = -\frac{\boldsymbol{F}}{2}, \qquad \Pi^{(1)} = -\frac{2\rho c_s^2}{\omega}\boldsymbol{S} - \frac{1}{2}(\boldsymbol{u}\boldsymbol{F} + \boldsymbol{F}\boldsymbol{u}), \tag{56}$$

the second term vanishing at a no-slip wall. The second-order Hermite reconstruction of $f^{(1)}$ consistent with (56) therefore needs its own first-moment term, which $\boldsymbol{Q}_i : \Pi$ cannot supply:

$$g_i \;\to\; g_i \;-\; \frac{w_i}{2c_s^2}\,\boldsymbol{c}_i\cdot\boldsymbol{F}. \tag{57}$$

Second — and this is the larger effect — the *measurement* of $\Pi^{(1)}$ is biased. Since $f^{(1)}$ carries the odd part $-\frac{w_i}{2c_s^2}\boldsymbol{c}_i\cdot\boldsymbol{F}$, bounce-back reverses its sign, so each unknown is wrong by $(w_i/c_s^2)\,\boldsymbol{c}_i\cdot\boldsymbol{F}$. Contracting $\boldsymbol{c}_i\boldsymbol{c}_i$ against that error would cancel over the full velocity set, but the unknowns form a *half-space* and it does not. On D2Q9 at a flat wall:

force direction	$\delta\Pi_{xx}$	$\delta\Pi_{yy}$	$\delta\Pi_{xy}$
tangential	0	0	$F/6$
normal	$F/6$	$F/2$	0

The tensors are different, which is why no correction of the form $\boldsymbol{u}_w \mp \boldsymbol{F}/2\rho$ can serve both cases: a tangential force produces spurious *shear*, a normal force spurious *normal* stress. The scaffold is therefore given the correct odd part rather than the reversed one.

What remains is not a force effect. After (57) and the scaffold repair, a residual wall offset survives for $\omega \neq 1$, obeying

$$\text{offset} \times W = \frac{4}{3}\,\frac{\omega-1}{\omega^2} \qquad \Longleftrightarrow \qquad \delta u_{\text{slip}} = -\frac{2}{3}\,\frac{\omega-1}{\omega^2}\,\frac{\partial^2 u_\parallel}{\partial n^2}. \tag{58}$$

This is a *curvature*-induced slip, not a force term. It is independent of $\boldsymbol{F}$ to five digits across an eightfold range of forcing; it vanishes identically for a linear profile, so Couette flow is exact to 4×10^{-15} at every τ from 0.6 to 2.0; and it vanishes at $\omega = 1$, consistent with the factor $(1-\omega)$ by which collision erases any error in the emitted stress. It is a pre-existing consequence of truncating $f^{(1)}$ at second Hermite order, which is what "regularised" means, and no correction for it is applied. The law (58) is empirical: it was fitted at two values of τ and then *predicted* three held-out values to within 0.3%, but the coefficient 4/3 has not been derived.

Departure from the paper. Reference [1] uses BC5, Eq. (48), on corners and edges; the form implemented here is BC4, Eq. (46). The two differ only by terms nonlinear in $\boldsymbol{u}$, which — as [1] states — "cancel for symmetry reasons during the evaluation of the tensor $\Pi^{(1)}$", and the asymptotic dynamics depends on $\Pi^{(1)}$ alone. The two therefore impose the same stress and differ only in ghost-population detail.

8.2 Outflow: constant back-pressure

An open outlet needs a different closure from a wall: the velocity there is not known, it is an output of the interior flow. The condition implemented reuses the regularised reconstruction of §8.1 and differs only in where $(\rho, \boldsymbol{u})$ come from.

Why the density must be imposed. The obvious choice — copy both ρ and $\boldsymbol{u}$ from the upstream neighbour, a zero-gradient outlet — is *ill posed* in this scheme, and not marginally so. The regularised nodes overwrite every population from $(\rho, \boldsymbol{u}, \Pi^{(1)})$, so they are not mass conserving; with a velocity inlet injecting mass and no condition fixing the pressure anywhere, the total mass has no equilibrium. Run on the channel of §13.6 it settles at $\rho \approx 181$ with a density ramp along x and a centreline velocity a factor two below the analytic profile. The outlet must therefore fix the pressure:

$$\rho(\text{outlet}) = \rho_0. \tag{59}$$

Normal velocity. With ρ known, the wall closure (54) inverts. Partitioning the populations by the sign of $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}}$ with $\hat{\boldsymbol{n}}$ the *outward* normal, and writing $\Sigma_+ = \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0} f_i$ (streamed from the interior, hence known) and $\Sigma_0 = \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} = 0} f_i$,

$$\rho\,(1 + \boldsymbol{u} \cdot \hat{\boldsymbol{n}}) = 2\Sigma_+ + \Sigma_0 \qquad \Longrightarrow \qquad \boxed{\boldsymbol{u} \cdot \hat{\boldsymbol{n}} = \frac{2\Sigma_+ + \Sigma_0}{\rho} - 1} \tag{60}$$

which is the same relation as the wall's, read in the other direction. The outlet therefore passes whatever flux the interior sends it rather than prescribing one, and the condition is self-regulating.

Tangential velocity. The node's own populations carry no inflow directions, so the tangential components are taken from the upstream neighbour (zero gradient, first order). A second-order variant — linear extrapolation from two upstream nodes, `set_outflow_order(2)` — is implemented and is *not* used, because it was measured to be no better: worse on coarse grids (1.388×10^{-3} against 1.160×10^{-3} at $L_y = 41$) and indistinguishable on the finest (6.184×10^{-6} against 6.192×10^{-6} at $L_y = 641$). §13.6 shows why — the outlet is not what limits the order there.

Only the $+x$ face is implemented (`NrmOutXp`). Nothing in (60) is specific to that direction; the other five are a matter of extending `upstream_of` and `normal_of`.

The magnetic counterpart, and its defect. The same idea is available on the induction lattices (`MagOutXp`): the node takes $\boldsymbol{b}$ from its upstream neighbour instead of an imposed value, and is then closed by the moment condition (64) exactly as a Dirichlet magnetic wall is. It reuses the field pass rather than the collision pass – the upstream neighbour's populations are read where $\boldsymbol{b}$ is computed, which is race free because that kernel only reads populations, and the node's own $\boldsymbol{b}$ then serves as the target when the moment is imposed.

This condition is not validated and has a measured defect. On the inlet-driven Hartmann flow of §12.11 — whose exact solution is independent of the streamwise coordinate, so that any x variation in the answer is boundary error and nothing else – it drives $\boldsymbol{b}$ about 6% high over the last ten or so nodes, and the overshoot decays upstream only slowly:

L_x	b/b_{exact} along the channel at $L_y = 65$
21	runs $1.000 \to 1.054$ across the *whole* domain
121	flat plateau at 1.005 in the bulk, rising to 1.061 at the end
241	plateau reached well before the sampling station

The effect on a score is large: at $L_y = 129$ the error in $\boldsymbol{b}$ falls from 3.367×10^{-3} to 9.793×10^{-4} purely by moving the outlet from $L_x = 121$ to $L_x = 241$. It is specifically the magnetic outlet and not the fluid one — pinning $\boldsymbol{b}$ analytically at the outlet instead gives errors agreeing to three significant figures with the zero-gradient case at $L_y = 17, 33, 65$, so the two differ only through that contamination. The cause is not identified. The two natural suspects are the interaction with $\nabla \cdot \boldsymbol{b} = 0$, which this code does not preserve to machine precision anyway, and the fact that a pure Neumann condition does not constrain the field level at all — the same class of ill-posedness that made a zero-gradient *fluid* outlet drive ρ to 181. Until it is understood, keep the outlet far from anything being measured.

A race that does not fire, and two repairs that made it worse. The arbitrary-face closure reads its donor's populations from inside the main `stream_collide` kernel. Under Esoteric Pull the two slots a node reads are exactly the two it writes, so the donor — an ordinary fluid cell, updated in the same pass — is rewriting the slots the outflow node is reading. Whether the pre- or post-collision state is obtained is not determined by the program.

Measured on the aorta of §15, 4000 steps with 2000 outlet nodes, the output is **byte-identical on 1, 2, 4 and 8 threads**. The reason is layout, not correctness: `rebuild_lists` builds the active list by scanning node indices in ascending order, and a donor is an immediate spatial neighbour, so an outflow node and its donor fall inside one thread's contiguous chunk. Only a pair straddling a chunk boundary can race, and with eight threads there are seven boundaries against two thousand outlet nodes. **That guarantee is a property of contiguous CPU chunking and will not survive a GPU**, where every node is its own work item.

Both obvious repairs are worse than the race. The natural fix is the one the scalar's open boundary uses (§8.5): read what cannot be read safely in a separate pass, so the ordering is defined. Two versions were built and measured against the documented steady case of §15.3, 13 500 steps at Re = 50:

donor read	$\lvert \boldsymbol{u} \rvert_{\max}$	$\lvert \Delta m \rvert$	ρ_{out}	behaviour
in-kernel (current)	0.0720	$\sim 10^{-5}$	0.9934	converges by step 7000
pre-collision, prefetched	0.0400	2×10^{-3}	oscillating	**limit cycle**
post-collision, gathered	0.0335	7.3×10^{-4}	0.9929	converges

Reading the donor *pre*-collision replaces a fixed point with a limit cycle, which is unambiguously a regression. Reading it *post*-collision is the correct phase — the result is written through `store_pair`, which stores post-collision populations — and it does remove the limit cycle and bring ρ_{out} within 0.05% of the documented value. But it still leaves mass conservation seventy-three times worse and the peak speed at half. Mass conservation is the only quantity here with a known correct value, so the in-kernel read stands.

Why defining the read should cost mass conservation is not understood. One untested story: the closure rescales the donor's distribution to a fixed density, and under the racy mixed read adjacent outlet cells received slightly different treatments whose errors partly cancelled, where a uniform read makes the error systematic. That is speculation, recorded as such.

The practical conclusion is not "leave it alone" but a sharper one: the GPU port needs a closure that is *right*, not merely one that is *ordered*. Imposing an order on the existing closure is exactly what was tried, twice, and it costs one to two orders of magnitude of mass conservation.

8.3 Forcing: every component, every axis, and a fully 3D case

§12.1 drives its channel along x with $\boldsymbol{F} = (G, 0, 0)$. That exercises $k_{12} = c_s^2 F_x$ from (50) and never touches $k_{21} = c_s^2 F_y$, so on its own it validates half the forcing. Three tests close that.

Rotating the channel, D2Q9. Walls x-normal and the force along $+y$ swaps which component is live. Both orientations are exact at the central-moment magic point, and — since D2Q9 is invariant under a quarter turn — they must agree with each other, which no single orientation can check.

	$\tau = 0.6$	$\tau = 7/8$	$\tau = 1.2$
along x (tests k_{12})	3.9225×10^{-3}	3.06×10^{-14}	4.6356×10^{-3}
along y (tests k_{21})	3.9225×10^{-3}	3.14×10^{-14}	4.6356×10^{-3}
$\lvert$difference$\rvert$	5.3×10^{-14}	6.1×10^{-15}	1.1×10^{-14}

$H = 16$. Off-magic the two agree to 10^{-13}, which is rotational isotropy of the forcing, equilibrium and streaming together.

All three axes, D3Q27. Walls normal to one axis, force along the next, periodic in the third; the three cyclic combinations put every axis in both roles. At $H = 16$ they give 2.99×10^{-14}, 3.68×10^{-14}, 3.12×10^{-14} at $\tau = 7/8$, and off-magic all three are *bit-identical* — 3.9225×10^{-3} at $\tau = 0.6$ and 4.6356×10^{-3} at $\tau = 1.2$, the same values D2Q9 gives, since a plane channel is quasi-one-dimensional and D3Q27 reduces to D2Q9 exactly on it.

A fully three-dimensional forced flow. A plane channel varies along one axis only, so it never excites a moment with two squared axes and barely touches the c_s^4 group. This case does. Periodic cube, no walls, $F_x = F_0 \sin(ky)\sin(kz)$ with $k = 2\pi/N$, whose steady solution is *exact* for the full Navier–Stokes equations,

$$u_x = \frac{F_0 \sin(ky)\sin(kz)}{2\rho\nu k^2}, \qquad u_y = u_z = 0, \tag{61}$$

because u_x does not depend on x, so $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ vanishes identically and the pressure stays uniform. The nonlinear term is not small here; it is zero.

N	$\tau = 0.8$		$\tau = 1.2$	
	rel L_2	order	rel L_2	order
8	7.4948×10^{-2}	—	8.1933×10^{-2}	—
16	2.0117×10^{-2}	1.897	2.0529×10^{-2}	1.997
32	5.1128×10^{-3}	1.976	5.1381×10^{-3}	1.998
64	1.2834×10^{-3}	1.994	1.2850×10^{-3}	2.000

Second order, and the L_2 error *equals* the peak error at every entry: the profile shape is right to round-off and all the error is amplitude.

Two magic points, and they are not the same one. That amplitude error changes sign with τ, so some viscosity makes it vanish. Scanning τ from 0.6 to 1.4 at $N = 32$ locates the crossing at $\tau = 0.9994$, and the whole family collapses onto

$$\frac{\Delta A}{A} = 2c_s^2 k^2(\tau - 1) + O(k^4), \tag{62}$$

with measured-to-model ratios running $0.9946 \to 0.9999$ as the grid refines, across $\tau \in [0.6, 1.4]$ and $N \in [8, 64]$. So the forced scheme has two distinct magic points and no single τ gives both: $\tau = 7/8$ puts the *wall* exactly on the node, a boundary property; $\tau = 1$ makes the *bulk* amplitude exact, a dispersion property. Which one limits a given run depends on whether it is wall-dominated or bulk-dominated.

The tests can fail. An exactness claim is only worth as much as its sensitivity, so the third-order force moments were deliberately perturbed by +50% and the tests re-run. The D2Q9 channel degrades from 8.44×10^{-14} to 6.69×10^{-4} and its wall slip from 7.4×10^{-14} to -3.9×10^{-3}; the transverse case, where $F_x = 0$ so only k_{21} is live, degrades from 3.14×10^{-14} to 2.67×10^{-3}. Ten orders of magnitude in both. These are not tests that pass regardless.

8.4 Moment-based walls for the scalar and magnetic fields

Dellar's moment conditions [2], Eqs. (13a)–(13b). The fields carried by the advection–diffusion lattices are *zeroth moments*,

$$B_\alpha = \sum_i g_{\alpha i}, \qquad T = \sum_i h_i, \tag{63}$$

so imposing a boundary value is one linear equation in the unknown populations. On a cross lattice a straight wall leaves *exactly one* unknown — the direction pointing into the domain along the normal — and that single equation closes the system uniquely and exactly:

$$\boxed{g_{x1} = B_0 - \big(g_{x0} + g_{x2} + g_{x3} + g_{x4}\big), \qquad g_{y1} = -\big(g_{y0} + g_{y2} + g_{y3} + g_{y4}\big).} \tag{64}$$

There is no closure assumption and no free parameter. The scalar case is identical with $\sum_i h_i = T_w - T_{\text{ref}}$, the shift being the storage reference of §5.5.

Why B may simply be imposed. Maxwell's equations make both the normal and the tangential components of $\boldsymbol{b}$ continuous across the wall, so $\boldsymbol{b}$ takes its external applied value there. The velocity enjoys no such freedom, which is why $\boldsymbol{u}$ needs the machinery of §8.1 and $\boldsymbol{b}$ does not.

Corners. More than one unknown, and the single moment no longer closes the system. The deficit is then shared in proportion to the lattice weights: the moment is still reproduced exactly, only its distribution among the unknowns is a choice, and the weight-proportional one introduces no directional preference. This is a fallback, not part of the published method.

A trap worth recording. The field kernels compute B_α and T as $\sum_i$ of the streamed populations. At a moment wall those still contain the *unfixed* inward direction, so their sum is not the field. Reading it anyway feeds a wrong $\boldsymbol{b}$ into the magnetic equilibrium and the computation diverges — observed here at 10^{150} within a few hundred steps. The field at such a node is the imposed value by construction and is reported as such. The same applies to ρ and $\boldsymbol{u}$ at a regularised wall.

condition	fields	rule	wall location
regularised	$\rho, \boldsymbol{u}$	(53) with (54)	on the node
moment (magnetic)	$\boldsymbol{b}$	(64)	on the node
moment (scalar)	T	$\sum_i h_i = T_w - T_{\text{ref}}$	on the node

8.5 Open boundaries for the scalar

The two scalar wall conditions above are both closed: adiabatic reflects everything, Dirichlet pins a value. Neither lets a plume *leave*. A transported scalar on an open domain needs a third condition, zero-gradient on C, so that what the flow carries to the boundary passes through it instead of accumulating behind it.

The condition implemented is equilibrium extrapolation. Each cell marked `ScalarOutflow` is assigned a *donor*, an interior neighbour, and every outgoing population is set to

$$h_i^{\text{out}} = h_i^{\text{eq}}(C_{\text{donor}},\, \boldsymbol{u}_{\text{node}})\,, \tag{65}$$

so the node carries its neighbour's concentration at its own velocity. The node is fully prescribed: whatever streamed into it is discarded, and its inward directions carry the interior's value back into the domain. Dropping the non-equilibrium part slightly damps the *diffusive* flux at the exit, which is negligible when the exit is advection-dominated — and is measured rather than assumed in §12.6.

Why this needs a second kernel. The obvious implementation reads the donor's populations inside the main kernel, as the arbitrary-face outflow of §8.2 does for the fluid. Under Esoteric Pull that is a genuine read/write race: the two slots a node reads are exactly the two it writes, so a donor being updated concurrently is rewriting the very slots the outflow node is reading, and which state is obtained depends on thread scheduling. Outflow cells are therefore skipped by the main kernel and handled by a second one after a fence, and that pass reads the *macroscopic field* rather than populations: the main kernel has just written C at every bulk cell, donors are required to be bulk cells, and so nothing the second pass reads is written by the second pass. Every storage slot still has exactly one writer, because the main pass skips these cells entirely and the outflow pass writes only the slots they own. The result is byte-identical on 1, 2 and 4 threads.

Donor selection. The outward directions of a cell are the axes whose neighbour lies outside the field; the donor is the neighbour one step inward along all of them at once. A face cell therefore takes its axis neighbour, an edge cell the diagonal and a corner cell the body diagonal. Without that, a box edge has no purely axial interior neighbour and every edge and corner of the domain would be inert. A cell with no bulk neighbour at all is left pointing at itself, which makes the condition a no-op there — it falls back to bounce-back — and is counted and reported at setup rather than silently reading a neighbour's garbage.

9 Coupled modules

9.1 Thermal transport

A passive scalar is carried by a second distribution set on its own lattice, with

$$T = \sum_i g_i, \qquad g_i^{\mathrm{eq}} = w_i\, T\Big(1 + \frac{\boldsymbol{c}_i \cdot \boldsymbol{u}}{c_s^2}\Big), \qquad D = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{66}$$

The velocity is an *input*, not something recovered from these populations, which is why the scalar has its own collision concept.

The first-order truncation in (66) is deliberate. D2Q5 and D3Q7 have an isotropic second moment but not an isotropic fourth-order one, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\boldsymbol{u})^2$ reduces to $c_\alpha^2 u_\alpha^2$ and the cross terms of the $\boldsymbol{uu}$ tensor cannot be represented at all. A second-order term would add an anisotropic error rather than accuracy. The cost is an $O(u^2)$ defect in the advection term, which is why these lattices suit low-Mach transport — exactly the regime Boussinesq convection occupies.

Architectural significance. This module was the first test of the composition the design was built around. D3Q7 has seven populations to the fluid's nineteen, a different sound speed, a different collision concept and different boundary conditions — and nothing in `FluidSolver`, the lattices, the streaming schemes or the moment operators required modification. The only interface addition was the node index on the collision.

9.2 Magnetohydrodynamics

Induction. Following Dellar [5], the magnetic field is carried by a *vector-valued* distribution g_i^α, with

$$b_\alpha = \sum_i g_i^\alpha, \qquad g_i^{\mathrm{eq},\alpha} = W_i \left[b_\alpha + \frac{c_{i\beta}}{\theta^2}\,(u_\beta b_\alpha - b_\beta u_\alpha) \right], \qquad \eta = \theta^2\Big(\frac{1}{\omega_\eta} - \frac{1}{2}\Big). \tag{67}$$

Taking moments gives $\sum_i g_i^{\mathrm{eq}} = b_\alpha$ and $\sum_i c_{i\beta} g_i^{\mathrm{eq},\alpha} = u_\beta b_\alpha - b_\beta u_\alpha$, which is exactly the antisymmetric flux of (4). Only that first moment is needed, so the magnetic lattice can be much smaller than the fluid's: D2Q5 or D3Q7 suffices. Running the field on a different lattice from the flow is the point, not an economy.

Lorentz force. The momentum equation gains $-\nabla\cdot\big(\boldsymbol{bb} - \frac{1}{2}|\boldsymbol{b}|^2 I\big)$. This is *not* applied as a body force, which would require derivatives of $\boldsymbol{b}$ and lose an order. The fluid equilibrium is instead given the correct second moment directly,

$$\sum_i c_{i\alpha} c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + \Big(p + \tfrac{1}{2}|\boldsymbol{b}|^2\Big)\delta_{\alpha\beta} - b_\alpha b_\beta, \tag{68}$$

achieved by adding to the hydrodynamic equilibrium

$$\delta f_i = \frac{w_i}{2c_s^4}\, M_{\alpha\beta}\big(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}\big), \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta. \tag{69}$$

Since $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\, \delta f_i = \boldsymbol{0}$, this term perturbs only the stress and leaves the shifted-storage bookkeeping untouched. In two dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = 0$, so (69) reduces to the familiar $\frac{w_i}{2c_s^4}\big[\frac{1}{2}|\boldsymbol{c}_i|^2|\boldsymbol{b}|^2 - (\boldsymbol{c}_i\cdot\boldsymbol{b})^2\big]$; in three dimensions the trace subtraction matters and is retained.

9.3 Coupling order

Two-way coupling must be evaluated *simultaneously.* If the fluid is stepped before the magnetic field is refreshed, it collides against $\boldsymbol{b}(t-1)$. That is a first-order splitting error, and — crucially — it does *not* vanish under refinement: with diffusive scaling the ratio

$$\frac{\omega^2\,\Delta t}{\nu k^2} \tag{70}$$

is independent of N, so it appears as a damping offset that survives every grid refinement.

This was found in practice. The shear Alfvén damping error *grew* with resolution, $1.55 \times 10^{-2} \to 2.79 \times 10^{-2} \to 3.16 \times 10^{-2}$, while the phase speed converged cleanly at order 2; the amplitude history showed a beat. Refreshing $\boldsymbol{b}$ before stepping the fluid removed both. `MagneticSolver::step` takes a `field_is_current` flag so a coupled driver does not pay for the field pass twice.

A non-converging error sitting beside a converging one is the signature of a coupling-order problem, and any future two-way coupling should be checked the same way.

10 Multiphase flow

Every module before this one leaves the fluid solver alone. The scalar and the magnetic field ride on their own distributions, the Lorentz force enters through a second moment, and a single-phase run is bit-for-bit what it was. The multiphase module is different: it changes the equation of state, and that change propagates through the macroscopic definitions, the forcing, the boundary treatment and the initial condition.

The reason is that a two-fluid flow at a density ratio cannot recover p from ρ. Elsewhere in `M3LB` the zeroth moment of f_i is the density and $p = \rho c_s^2$ follows; here ρ is a material property of whichever fluid occupies the node, not a thermodynamic response to the pressure, and a scheme that ties the two together makes the sound speed and the acoustic time step functions of the phase. The populations therefore carry a normalised pressure instead. That single substitution is what puts a density ratio of 800 within reach — and it is also the origin of the conditioning problem in §10.6, which is this module's principal open issue and is stated there rather than buried.

The composition rule of §1.3 still holds. The multiphase operators are separate compile-time collision policies and the phase field is a separate solver on its own lattice; nothing single-phase was modified to admit them, and a single-component case takes none of this code. `FluidSolver`, `EsotericPull` and the lattices are unchanged.

The scheme follows De Rosis and Enan (2021), Sec. II.B, for the phase-field populations and Sec. II.C for the central-moment fluid operator; the rigid-body coupling of §10.7 does not, and §10.7 says why.

Most of this section is that model. §10.8 is a *second* immiscible model on a different principle, from a different paper, sharing no code with the first beyond the lattices and the moment transform — it is here because the two fail differently, and §18.12 measures both on one case. A third route, which resolves only the liquid, is §11.

10.1 Interface capture: conservative Allen–Cahn

The interface is carried by a *third* distribution set h_i, on its own lattice, with the velocity as an *input* exactly as the thermal scalar has it:

$$\phi = \sum_i h_i, \qquad h_i^* = h_i + \omega\big(h_i^{\mathrm{eq}} - h_i\big) + S_i, \tag{71}$$

with $\phi = 1$ in the heavy phase and 0 in the light one, obeying (5). The reference throughout this section is De Rosis & Enan, *Physics of Fluids* **33**, 043315 (2021), Sec. II.B; the relaxation, the equilibrium, the relation $\tau_\phi = M/c_s^2$, the source and the gradient stencil are its Eqs. (9), (11), (13), (15)–(16) and (20). Structurally (71) is the passive-scalar operator of (66) plus one source term, and the only moment carried is the zeroth.

The conservative form earns its name from an exact one-dimensional equilibrium,

$$\phi(x) = \tfrac{1}{2}\Big[1 + \tanh\frac{2x}{W}\Big], \qquad |\nabla\phi| = \frac{4}{W}\,\phi(1-\phi) \equiv \theta, \tag{72}$$

the identity that makes the diffusive flux $M\nabla\phi$ and the anti-diffusive flux $M\theta\,\boldsymbol{n}$ of (5) cancel term for term. The interface neither spreads nor sharpens, and W is a prescribed number of lattice units rather than something that emerges from a balance and has to be measured afterwards. The pointwise vanishing of the total flux at a static flat interface is the cheapest check on the whole construction, and the one the implementation is tested against first.

The equilibrium is a policy, because the right truncation is the lattice's. On a Navier–Stokes lattice the second-order form (15) is used unchanged, with ϕ in place of ρ — that is Eq. (11) of the reference exactly, and the fluid's own equilibrium serves without being retyped. On D2Q5 and D3Q7 it drops to first order,

$$h_i^{\mathrm{eq}} = w_i\,\phi\Big(1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2}\Big), \tag{73}$$

for the reason given for the scalar in §5.5: those lattices have no isotropic fourth-order moment, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\cdot\boldsymbol{u})^2$ collapses to $c_\alpha^2 u_\alpha^2$ and the cross terms of $\boldsymbol{uu}$ cannot be represented at all. A second-order term there adds anisotropy, not accuracy. The trade is cost against advection accuracy: the reduced lattice moves 5 or 7 populations against 9 or 19 and pays an $O(u^2)$ defect in the advection term. A static interface cannot see that difference; a sheared one can.

The source, and why its prefactor is not Guo's. Chapman–Enskog on the advection–diffusion LBE with a source obeying $\sum_i S_i = 0$ and $\sum_i \boldsymbol{c}_i S_i = \boldsymbol{A}_S$ gives

$$\partial_t\phi + \nabla\cdot(\phi\boldsymbol{u}) = M\nabla^2\phi - \frac{1}{\omega}\nabla\cdot\boldsymbol{A}_S, \tag{74}$$

so matching the anti-diffusion term of (5) requires $\boldsymbol{A}_S = \omega M\,\theta\,\boldsymbol{n} = c_s^2(1-\omega/2)\,\theta\,\boldsymbol{n}$, using $\omega M = c_s^2(1-\omega/2)$. With $S_i = w_i(\boldsymbol{c}_i\cdot\boldsymbol{A}_S)/c_s^2$ the c_s^2 cancels and the source is simply

$$\boxed{S_i = \left(1-\frac{\omega}{2}\right) w_i\,\theta\,(\boldsymbol{c}_i\cdot\boldsymbol{n}), \qquad \theta = \frac{4}{W}\,\phi(1-\phi)} \tag{75}$$

carrying no mobility and no speed of sound at all. Since $\sum_i S_i = 0$, ϕ is conserved exactly by the collision, and streaming conserves it too — that is the "conservative" in conservative Allen–Cahn, and it is verified rather than assumed: ϕ drifts by 8.4×10^{-13} over 10 000 steps.

The coefficient in the recovered equation is $1/\omega$, *not* the $1/(1-\omega/2)$ that Guo's force term (47) compensates, and the difference is structural rather than a convention. A body force enters the *first*-moment equation directly, where the half-step correction is absorbed by redefining the velocity. This source enters the *zeroth*-moment equation through the divergence of the first moment, so it contributes twice — once through $h_i^{(1)}$ and once through the second-order Taylor term — and $(1/\omega - \frac{1}{2}) + \frac{1}{2} = 1/\omega$ is where that lands.

It was written with Guo's prefactor and a $1/c_s^2$ first. The source then comes out a factor $M/c_s^2 = 1/\omega - \frac{1}{2}$ too small, the anti-diffusion cannot balance the diffusion, and the interface spreads as though the term were absent: a prescribed $W = 4$ measured 14.1 after 600 steps, against the $\sqrt{4Mt} = 11$ of pure diffusion. With (75) the same measurement returns 4.0011, a relative error of 2.8×10^{-4}. Nothing else misbehaves in the wrong case, which is what makes the failure worth recording: it is loud in its effect and silent in its cause.

Mobility. $M = c_s^2(1/\omega - \frac{1}{2})$, exactly as the diffusivity in (66), with $c_s^2 = 1/3$ on D2Q5 and 1/4 on D3Q7. It is a numerical parameter and not a physical one: too small and the interface cannot heal after being advected through the grid, too large and it diffuses faster than the flow deforms it. $M \in [0.01, 0.1]$ is the working range.

Gradients are taken from the field, not from neighbouring populations. $\nabla\phi$ is formed from the ϕ *field* by a lattice-weighted stencil in a pass of its own. Reading a neighbour's populations inside the fused kernel would be a genuine read/write race under Esoteric Pull (§3.2) — the two slots a node reads are exactly the two it writes — which is the same defect the scalar outflow condition avoids in the same way (§8.2). `compute_field()` ends in a fence and the derivative kernel reads only what that kernel wrote, so every slot still has exactly one writer. The stencils are Eqs. (20) and (21) of the reference,

$$\nabla\phi(\boldsymbol{x}) = \frac{1}{c_s^2}\sum_i w_i\,\boldsymbol{c}_i\,\phi(\boldsymbol{x}+\boldsymbol{c}_i), \qquad \nabla^2\phi(\boldsymbol{x}) = \frac{2}{c_s^2}\sum_i w_i\big[\phi(\boldsymbol{x}+\boldsymbol{c}_i) - \phi(\boldsymbol{x})\big], \tag{76}$$

evaluated on a *gradient lattice* that defaults to the richest velocity set of the same dimension — D2Q9 in two dimensions, D3Q27 in three — rather than the phase field's own D2Q5 or D3Q7. Spurious currents around a static droplet are dominated by the isotropy of this stencil and the axis-only set is markedly worse, so the cost of a 27-neighbour gather is accepted in a kernel that does nothing else. Both derivatives come out of that one gather: the Laplacian needs exactly the neighbours the gradient has already loaded, so it is one accumulator and one store more, not a second pass. Only the potential-form fluid operator reads it.

The anti-diffusion vector $\boldsymbol{A} = \theta\,\boldsymbol{n}$ is assembled once per node rather than once per direction — it costs a square root and a divide and contains no i — and is guarded by $|\nabla\phi| > 1\times10^{-12}$. Away from the interface $\phi(1-\phi)$ vanishes anyway, so the guard is a second line of defence rather than the only one: it stops 0/0 in the bulk, where $\nabla\phi$ is round-off noise about zero and $\boldsymbol{n}$ is meaningless.

Cost. Five extra `Real` fields per node — ϕ, three gradient components and the Laplacian — on top of the Q populations. In FP32 on D3Q7 that is 28 bytes of populations against 20 of fields, so the fields are not a rounding error in the bandwidth budget. They are the price of not differentiating populations, and it is a price worth naming.

Coupling order. ϕ and $\nabla\phi$ must be refreshed *before* the fluid steps, so that the fluid collides against the interface at time t rather than $t-1$: `pf.refresh()`, then `fluid.step()`, then `pf.step()`. This is the same first-order splitting error the magnetic module documents in §9.3, and it does not refine away for the same reason. Here it is worse in kind rather than in size: the interface *moves*, so the error is a systematically misplaced interface rather than a damping offset. One refresh serves both consumers, and that is not an economy but a consistency requirement — the fluid's capillary stress and the phase field's interface normal are the same gradient, and evaluating them from one array at one time is what stops them disagreeing about where the interface is.

What is not implemented. `PhaseWall` is zero-flux on the populations, which is the right condition for the transport but sets no contact angle; that needs a wetting condition on $\nabla\phi$ at the wall. A zero-flux wall never collides, so its slots hold whatever a neighbour emitted toward it one step ago, and that sum is not an order parameter — the field kernel therefore leaves ϕ at its initialised value there rather than writing a meaningless one, and the gradient stencil reads that value. An interface should not be put against a wall. Open boundaries for ϕ are also absent; the donor machinery of §8.5 would port, with the same second pass and the same fence.

10.2 The fluid side: a pressure-based formulation

Two fluid operators exist. `MultiphaseBGK` is the matched-density one: ρ is a single field near 1, and ϕ enters the flow only through the capillary stress of §10.3 and, optionally, through a per-node relaxation rate (`omega_n`) for a viscosity contrast. `MultiphasePotentialBGK` carries a density ratio, and to do so it changes what the zeroth moment means.

The reason is conditioning. At a ratio ρ is no longer a small perturbation about 1, and a density-carrying distribution stops being well posed for the shifted storage of §4, which assumes ρ_{ref} is exactly 1: at a ratio of 10 the stored g_i are not small and the shift buys nothing. The incompressible model removes the density from the distribution entirely. ρ becomes a function of ϕ, and the populations carry a normalised pressure:

$$\tilde{p} = \sum_i f_i, \qquad \boldsymbol{u} = \sum_i \boldsymbol{c}_i f_i + \frac{\boldsymbol{F}}{2\rho}, \qquad f_i^{\mathrm{eq}} = w_i\big[\tilde{p} + \Phi_i(\boldsymbol{u})\big], \tag{77}$$

Eqs. (24) and (10) of the reference, with Φ_i the second-order Hermite polynomial that (15) multiplies by ρ — written Φ_i here rather than ϕ_i only because ϕ is taken. Its zeroth moment vanishes and its first is $\boldsymbol{u}$, so $\sum_i f_i^{\mathrm{eq}} = \tilde{p}$ exactly and $\sum_i \boldsymbol{c}_i f_i^{\mathrm{eq}} = \boldsymbol{u}$, *not* $\rho\boldsymbol{u}$. There is consequently no division by ρ in the momentum sum; only the half-force correction carries one. Getting that wrong is a viscosity error that presents as a bad boundary condition. A unit test asserts all three moment statements, and the momentum gain $\boldsymbol{F}/\rho$ under the full force, directly against the equilibrium and to round-off.

The physical pressure follows from $\tilde{p}$ by a factor that is itself a choice, and one that turns out to matter more than its algebra suggests; §10.6 is where that is settled. The shift, meanwhile, is easier here than anywhere else in the code: f_i^{eq} is *affine* in $\tilde{p}$, so storing $f_i - w_i$ merely replaces $\tilde{p}$ by $\tilde{p}-1$ in the same expression, and one line serves both storage conventions.

Density, viscosity, and the clamp on the interpolant. With s the normalised order parameter,

$$s = \frac{\phi - \phi_L}{\phi_H - \phi_L}\bigg|_0^1, \qquad \rho = \rho_L + s\,(\rho_H - \rho_L), \qquad \mu = \mu_L + s\,(\mu_H - \mu_L), \qquad \tau = \frac{\mu}{\rho\, c_s^2}, \tag{78}$$

Eqs. (25) and (12) of the reference. Both interpolations are linear in ϕ and the prescribed viscosities are *dynamic*, so the relaxation rate $\omega = 1/(\tau + \frac{1}{2})$ varies from node to node through both μ and ρ; $\nu = \mu/\rho$ is recovered from the same single read of ϕ, for §10.4.

The clamp is on s, the interpolant, and not on ϕ itself. The conservative Allen–Cahn form keeps ϕ close to $[0, 1]$ but does not guarantee it — shear against a solid, or any under-resolved feature, overshoots locally — and interpolating off an unclamped ϕ puts the density outside the two phases it is meant to lie between. A large enough undershoot makes ρ *negative*, at which point the $1/\rho$ in (77) and in the force is unbounded. Clamping the interpolant leaves the transported field alone and bounds only the equation of state, which is the narrower repair of the two.

Four forces, one of which is not obvious. The collision is sourced by $\boldsymbol{F} = \boldsymbol{F}_s + \boldsymbol{F}_p + \boldsymbol{F}_\nu + \rho\,\boldsymbol{b}$: surface tension (§10.3), the pressure mismatch (§10.6), the viscous interface force (§10.4), and a body acceleration. The middle two are not optional extras — they are the difference between what the LBE recovers and what the variable-density, variable-viscosity momentum equation asks for — and both vanish identically at a matched density, because $\nabla\rho$ does. Under the default normalisation $\boldsymbol{F}_s$ and $\boldsymbol{F}_p$ are both multiples of $\nabla\phi$, one through the chemical potential and the other through $\nabla\rho = (\rho_H - \rho_L)/(\phi_H - \phi_L)\,\nabla\phi$ (Eq. 23), so they are assembled as a single coefficient on one vector rather than as two vectors added, and no second gradient field is needed for the density. An arbitrary external per-node force field occupies a separate slot from $\boldsymbol{F}_\nu$, so that a penalised solid and the viscous pass never have to agree about who clears the array.

The source itself is Guo's (47) in its first-order form, $F_i = (1 - \omega/2)\, w_i\, (\boldsymbol{c}_i \cdot \boldsymbol{F})/(\rho c_s^2)$, which is Eq. (14) of the reference. The central-moment version of this operator, and what it changes, is §10.5.

One cost of the fixed `Macro` shape. `Macro` carries four `Real`s and all four are spoken for, so the force — needed once to correct the velocity in `macroscopic()` and once to source the collision — is assembled twice per node instead of being carried between them, about eight extra view reads. Widening `Macro` by three defaulted fields would remove it, at the cost of three `Real`s in every other operator's hot loop. That trade is stated rather than taken, because it should be measured first.

10.3 Surface tension: two routes

Route one: a capillary stress in the equilibrium. The matched-density operator does not apply surface tension as a body force. It gives the equilibrium the right second moment directly, exactly as the MHD operator does for the Maxwell stress in (27). With $\boldsymbol{G} = \nabla\phi$,

$$\sum_i c_{i\alpha} c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + \Big(p - \tfrac{1}{2}\kappa|\boldsymbol{G}|^2\Big)\delta_{\alpha\beta} + \kappa\, G_\alpha G_\beta, \tag{79}$$

achieved by adding to the hydrodynamic equilibrium

$$\delta f_i = \frac{w_i}{2c_s^4}\, M_{\alpha\beta}\big(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}\big), \qquad M_{\alpha\beta} = \kappa\Big(G_\alpha G_\beta - \tfrac{1}{2}|\boldsymbol{G}|^2\delta_{\alpha\beta}\Big). \tag{80}$$

This is (27) with the magnetic field replaced by $\sqrt{\kappa}\,\nabla\phi$ and the sign reversed. The tensor structure is the same and so are the invariants — $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \boldsymbol{0}$, so it perturbs only the stress and leaves mass, momentum and the shifted-storage bookkeeping untouched — but the reversal is physics, not convention. The Maxwell stress is a tension *along* $\boldsymbol{b}$ and pulls field lines together; surface tension is a tension along the *interface*, which is perpendicular to $\nabla\phi$. Relative to the vector that defines the tensor, the two therefore differ by a sign.

Taking the MHD expression over unchanged is the mistake that was made, and it is not a subtle one in its consequences: with the Maxwell sign the mechanical surface-tension integral across the profile comes out negative and the droplet develops a pressure *deficit*, measured -2.5×10^{-4} where $+8.3 \times 10^{-4}$ was expected. `capillary()` is thus $-\kappa$ times `maxwell()`. The twelve duplicated lines were kept rather than factored into a shared helper, on the grounds that duplicating them is cheaper than disturbing a validated MHD path; a third stress of this shape would change that judgement.

Route two: the chemical potential. The pressure-based operator uses the potential form instead, Eqs. (18) and (19) of the reference,

$$\boldsymbol{F}_s = \mu_\phi\,\nabla\phi, \qquad \mu_\phi = 4\beta\,(\phi-\phi_L)(\phi-\phi_H)(\phi-\phi_0) - \kappa\,\nabla^2\phi, \qquad \phi_0 = \tfrac{1}{2}(\phi_L+\phi_H). \tag{81}$$

The coefficients. Integrating $\kappa(\partial_x\phi)^2$ across the tanh profile (72) of width W, for $\phi \in [0,1]$, gives

$$\sigma = \frac{2\kappa}{3W} \quad\Longleftrightarrow\quad \kappa = \frac{3\sigma W}{2}, \qquad \text{and consistently} \qquad \beta = \frac{12\sigma}{W}. \tag{82}$$

`kappa_from_sigma` returns the first and `beta_from_sigma` the second. In the stress form the double-well strength β never appears at all: the conservative Allen–Cahn equation maintains the profile geometrically, from W and the normal, so the free energy is implicit and only κ is needed. In the potential form both appear, and (82) is the pair consistent with the same σ and the same W.

Which is used where. The two are algebraically equivalent, and the stress form is the cheaper and safer of the two: it needs only $\nabla\phi$, which `PhaseFieldSolver` computes anyway for the interface normal, so it is one field cheaper and one derivative order safer — the same argument Dellar's form of the Lorentz force makes in §5.7. The pressure-based operator nonetheless uses the potential form, following the reference, and pays a Laplacian for it. What the exchange buys is a density ratio the stress form cannot reach, and the Laplacian is drawn from the neighbour gather that (76) already performs, so the cost is one accumulator rather than one pass. At matched density the two routes agree, as they must: on a droplet of radius 20 the surface tension inferred from the Laplace jump differs between them by 3.58×10^{-3}, against a prescribed $\sigma = 1 \times 10^{-2}$.

10.4 The viscous interface force

The LBE recovers $\mu\nabla^2\boldsymbol{u}$, while the momentum equation at variable viscosity wants $\nabla\cdot\left[\mu(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}})\right]$. The difference is Eq. (22) of the reference,

$$\boldsymbol{F}_\nu = \nu\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}}\right)\cdot\nabla\rho. \tag{83}$$

It is identically zero wherever $\nabla\rho$ is — that is, everywhere except the interface — so at a matched density the whole object may be left unbound and the collision reads zero for it. At a density ratio it is not a correction one may drop.

It is the only one of the four forces that cannot be assembled from node-local data: it needs the velocity *gradient*, and therefore a pass of its own. That pass uses the same isotropic stencil as (76), on the same gradient lattice.

Only the contraction is stored. The velocity gradient is nine numbers per node; contracted against $\nabla\rho$ it is three. Since $\nabla\rho$ is available in the same kernel — it is $\nabla\phi$ times the constant $(\rho_H - \rho_L)/(\phi_H - \phi_L)$ — the contraction is performed where the tensor is still in registers and only the resulting vector is written. Nine fields per node become three. The normalisation factors out of the bracket for the same reason: the accumulator holds the raw sum $\sum_i w_i c_{i\alpha} u_\beta$, and since it and its transpose need the same $1/c_s^2$, one factor multiplies the whole contraction rather than each term of it.

Coupling order, and why the pass is opt-in. The kernel reads the velocity *field*, which `FluidSolver` fills from the populations, so the order

`pf.refresh()` $\to$ `fl.compute_macroscopic()` $\to$ `vf.refresh(coll)` $\to$ `fl.step()`

builds the force from $\boldsymbol{u}(t)$ and $\phi(t)$ and consumes it in the collision at time t, with no lag anywhere — unlike the phase field itself, whose splitting error §10.1 names. The price is one extra pass over the populations per step, because the fused step writes $\boldsymbol{u}$ only as it goes, too late for a neighbour gather within the same step. That is a real cost, and it is why this pass is requested by the driver rather than performed automatically.

Not valid against a wall. The stencil reads $\boldsymbol{u}$ at every neighbour, including non-fluid ones, where the velocity field holds whatever `FluidSolver` left there, which is zero. At an interface touching a wall that is a one-sided derivative taken as though it were central. Every case that currently runs this operator is periodic. The gap has the same shape as the missing wetting condition, and the same remedy would be needed.

It earns its place. For as long as the only case running the pressure form was a static droplet — where (83) is identically zero — $\boldsymbol{F}_\nu$ was implemented and unexercised, and was documented as such rather than glossed. Layered Poiseuille flow at both a viscosity and a density ratio is the case that excites it, and switching it off there gives:

H	$L_2(u)$ with $\boldsymbol{F}_\nu$	without	ratio
32	4.2967×10^{-2}	4.2837×10^{-1}	9.97
64	4.2896×10^{-2}	5.9350×10^{-1}	13.84

A term that is exactly zero in every case previously run is precisely the kind that can stay wrong indefinitely without anything failing. This is the measurement that says it is not.

10.5 Central moments for the pressure form

BGK relaxes every mode at the same rate, so at high Reynolds number, where ω approaches 2, the non-hydrodynamic modes are over-relaxed and ring instead of decaying. That is not a margin one tunes around: at $Re = 30000$ on a 128-wide box, $\omega = 1.99795$. The symptom that prompted a moment operator here was measured rather than anticipated — a Rayleigh–Taylor run at a density ratio of 999 completed under BGK, but its velocity field was dominated by a one-cell alternating mode, 70× stronger than the same case at a ratio of 3. The point of collision in moment space is that the modes carrying no physics can be sent straight to equilibrium at rate 1 while the shear modes alone carry the viscosity.

The equilibrium moments factorise. That is the one derivation worth writing down, because it is what keeps the operator as cheap as the single-phase one. The pressure-form equilibrium splits exactly:

$$f_i^{\text{eq}} = w_i[\tilde{p} + \Phi_i(\boldsymbol{u})] = \underbrace{w_i(1 + \Phi_i(\boldsymbol{u}))}_{\text{Maxwellian at } \rho=1} + (\tilde{p} - 1)\, w_i.$$

The first bracket has central moments $\{1, 0, 0, \ldots\}$ *exactly* — that is the product-form property verified symbolically in §6.3, 0 of 9 moments deviating on D2Q9 and 0 of 27 on D3Q27. The second is $(\tilde{p} - 1)$ times the central moments of the *weights* about $\boldsymbol{u}$, and on a product lattice those factorise into the same one-dimensional triple that the shifted storage of §6.3 already carries. Hence the closed form

$$k_{pqr}^{\text{eq}} = [\, pqr = 000\,] + (\tilde{p} - 1)\, A_p(u_x)\, A_q(u_y)\, A_r(u_z), \qquad A = \{1,\ -u,\ u^2\}, \tag{84}$$

with the third factor absent in two dimensions. No dense $Q \times Q$ matrix, no stored equilibrium vector, and the same factorised transform the single-phase operator uses. Equation (84) was checked against Eq. (34) of De Rosis and Enan on four independent moments: $k_5 = (u_x^2 - u_y^2)(\tilde{p} - 1)$ and $k_7 = u_x u_y(\tilde{p} - 1)$ agree directly, and after converting this Hermite-like basis to the paper's monomial one through $k_{21}^{\text{mono}} = k_{21} + c_s^2 k_{01}$, their k_{10} and k_{16} come out identical too. The paper's odd-order moments are printed without the leading minus the derivation gives; the even ones agree outright, which is what identifies it as typography rather than a difference of scheme. The test suite does not check against the paper at all: it contracts the central moments of the equilibrium directly by summation, and all nine agree with (84) to 5.0×10^{-17}.

One deliberate departure. The reference builds its equilibrium central moments from the second-order truncated equilibrium of its Eq. (10). This operator uses the product form instead, whose central moments are exactly Maxwellian on D2Q9 and D3Q27, so the $O(u^3)$ Galilean defects of the truncation are absent rather than reproduced — the same choice, for the same reason, that §6.3 makes for single-phase flow. The gap between the two is measurable and is exactly of that size: run against BGK at $\omega = 1$, where the two operators must otherwise coincide, both recover the same $\tilde{p}$ and the same u_x to round-off, and their post-collision states differ by 1.977×10^{-4} at $|\boldsymbol{u}|^3 = 7.712 \times 10^{-4}$.

Relaxation. Following Eq. (35) of the reference, $K = \text{diag}[1, 1, 1, 1, 1, \omega, \omega, \omega, \omega, \omega, 1, \ldots, 1]$:

$$\begin{aligned} \text{order } 0 :&\quad \text{conserved — } \tilde{p} \text{ is untouched} \\ \text{order } 1 :&\quad \text{exact assignment, the force included} \\ \text{order } 2 :&\quad \text{trace at } \omega_{\text{bulk}} \text{ (the paper's 1), deviatoric and shear at } \omega \\ \text{order } \geq 3 :&\quad k^\star = k^{\text{eq}}. \end{aligned} \tag{85}$$

ω_{bulk} is exposed as a parameter because it is the bulk viscosity and nothing else in the scheme pins it.

The force is exact in the first moment, not relaxed into it. With $\boldsymbol{u} = \sum_i \boldsymbol{c}_i f_i + \boldsymbol{F}/(2\rho)$, the pre-collision first moment is

$$k_1 = \sum_i f_i\,(c_{ix} - u_x) = k_{100}^{\text{eq}} - \frac{F_x}{2\rho}, \tag{86}$$

and the post-collision one has to be $k_{100}^{\text{eq}} + F_x/(2\rho)$ for the next step to recover the right velocity. The update is therefore an assignment rather than a relaxation, and the rate-1 entry in K

is that assignment. The source of Eq. (14), $F_i = w_i(\boldsymbol{c}_i\cdot\boldsymbol{F})/(\rho c_s^2)$, is odd in $\boldsymbol{c}_i$ and so has no second moment at all: unlike Guo's there is no second-order force term to add, in any basis, and the co-moving-frame cancellation that §6.3 relies on is not even needed here. The test suite checks the consequence rather than the algebra — collision conserves $\tilde{p}$ to 2.2×10^{-16} and the momentum gains exactly F_x/ρ and F_y/ρ, to 1.4×10^{-17}, with the full multiphase force assembled.

Two constraints on where this operator may be used. It requires a product lattice: D2Q9 or D3Q27, enforced by a static assertion, since D3Q19 would need the monomial basis and its own equilibrium moments. And it takes raw storage only. The shifted convention centres the stored variable on $\tilde{p} = 1$, which is precisely the pressure gauge that must be avoided at a density ratio — the subject of §10.6.

10.6 The pressure gauge, and two normalisations

This is the most consequential finding in the module, and it is a conditioning result rather than a bug. The populations carry $\tilde{p} = p/(\rho c_s^2)$. The physical pressure p is continuous across an interface; ρ is not. So $\tilde{p}$ jumps by the whole density ratio wherever p is not near zero, and the term that permits it to, $F_p = -\tilde{p}\, c_s^2 \nabla\rho$, is proportional to the pressure *level*. It is cancelled, to whatever accuracy the two discrete operators allow, by the LBE's own $\rho c_s^2 \nabla\tilde{p}$. The physical answer is their difference; their individual sizes — and hence the size of the residual mismatch — are set by a gauge that changes no physics.

The layered channel makes it visible. Two immiscible layers driven by a uniform force G have an exact profile that does not contain ρ at all, so a density ratio alone must leave the answer unchanged. It does not. Turning the two contrasts on one at a time (D2Q9, central moments, FP64, interface width 4):

configuration	$L_2(u)$, H=32	H=64	order
single phase	2.6751×10^{-4}	6.6877×10^{-5}	2.00
viscosity ratio 5 only	3.6187×10^{-2}	2.3587×10^{-2}	0.62
density ratio 10 only	2.5078×10^{-2}	2.8995×10^{-2}	−0.21
both	4.2967×10^{-2}	4.2896×10^{-2}	0.00

The first row says the base scheme is exact and second order. The second is the diffuse interface smoothing a kink in $\mathrm{d}u/\mathrm{d}y$ over W cells, first order in W/h and converging as it should. The third should not happen at all, and extending it to $H = 128$ gives 2.4×10^{-2} — a few per cent that refinement does not remove.

It is the pressure force, and it is not spurious. Dumping the profile at a density ratio of 10 with uniform viscosity gives

$$\begin{array}{llll} \tilde{p} & \text{heavy } 2.07 \times 10^{-3} & \text{light } 2.18 \times 10^{-2} & \text{(ratio 10.5)} \\ p = \rho c_s^2 \tilde{p} & \text{heavy } 6.72 \times 10^{-3} & \text{light } 7.27 \times 10^{-3} & \text{(nearly uniform),} \end{array}$$

which is exactly what the formulation demands. F_p reaches 220 times the driving force at $H = 32$ and 408 at $H = 64$, and the reported error is the discrete mismatch between it and $\rho c_s^2 \nabla\tilde{p}$. That also explains the sign of the order directly: G scales as $1/h^2$ to hold the peak velocity fixed, so F_p/G *grows* as h^2 under refinement while the mismatch stays a fixed fraction of F_p. Refining makes the cancellation harder, not easier.

A gauge experiment confirms it. Seeding a uniform offset in $\tilde{p}$ changes no physics, since only ∇p is physical. Measured $L_2(u)$ at $H = 32$:

$\tilde{p}$ offset	single phase	viscosity only	density only
0	2.675×10^{-4}	3.619×10^{-2}	2.51×10^{-2}
0.2	2.675×10^{-4}	3.630×10^{-2}	4.34×10^{-1}

The two configurations without a density ratio are exactly gauge invariant, as they must be. The one with a density ratio degrades seventeenfold, and F_p/G goes from 220 to 4650. The conditioning, not the physics, sets the error.

Two normalisations. The reference normalises by the local density; the alternative normalises by a constant:

$$p = \rho(\phi)\, c_s^2\, \tilde{p}, \qquad F_p = -\tilde{p}\, c_s^2\, \nabla\rho \qquad \text{(default)}, \tag{87}$$

$$p = \rho_0\, c_s^2\, \tilde{p}, \qquad F_p = c_s^2\, (\rho - \rho_0)\, \nabla\tilde{p}. \tag{88}$$

Under (88) $\tilde{p}$ is uniform wherever p is, so F_p tracks the pressure *gradient* instead of the level. It needs $\nabla\tilde{p}$, which is a field the phase solver does not carry, so a separate pass differentiates the fluid's own zeroth moment on the isotropic stencil

$$\nabla f(\boldsymbol{x}) = \frac{1}{c_s^2} \sum_i w_i\, \boldsymbol{c}_i\, f(\boldsymbol{x} + \boldsymbol{c}_i), \tag{89}$$

run after the macroscopic update. Isotropic rather than axis-only for the reason the phase gradient is: this gradient is differenced against the LBE's own pressure term, and an anisotropic stencil would leave a direction-dependent residual exactly where the two are meant to cancel.

On the density-only configuration the two forms measure as

normalisation	$L_2(u)$, H=32	H=64	order	$\max\lvert F_p\rvert/G$
local ρ (default)	2.5078×10^{-2}	2.8995×10^{-2}	-0.21	$220 \to 408$
constant ρ_0	1.8572×10^{-2}	9.6381×10^{-3}	0.95	$379 \to 747$

At an offset of 0.2 the same orders are -0.37 and 0.86, and the gauge sensitivity — the factor by which the error moves under a shift that changes no physics — is 17.3× and 19.4× for the local form against 1.93× and 2.04× for the constant one. Note what does *not* improve: F_p/G is larger, 379 against 220 at $H = 32$. The gain is not that the term shrinks but that it stops tracking the pressure level. The residual factor of about two is the discrete equilibrium's own $\tilde{p}$ dependence: $f_i^{\mathrm{eq}} = w_i[\tilde{p} + \Phi_i]$ carries $(\tilde{p} - 1)$ into its higher moments through (84), so a $\tilde{p}$ offset is not a pure gauge on the lattice even when it is one in the continuum.

Neither dominates. The recovered pressure term under (88) is $-(\rho_0/\rho)c_s^2\nabla\tilde{p}$, so one phase always pays an acoustic price. With $\rho_0 = \rho_L$ the heavy phase runs at an effective sound speed $c_s\sqrt{\rho_L/\rho_H}$: stable, but the pressure term there is a cancellation of ratio size, which is tolerable where ∇p is diffuse and is not where it is concentrated. At a Laplace jump that form diverges at a ratio of 10 and above, on a case the local-ρ form handles at a ratio of 100. The opposite choice is worse and was tried first: with $\rho_0 = \rho_H$ the light phase gets $c_s\sqrt{\rho_H/\rho_L}$, which at a ratio of 10 is 1.83 lattice units per step — past the lattice speed — and the run diverged inside 500 steps. That is why ρ_0, when used at all, must be the light phase. The default remains the local-ρ form of (87), because every validated result in this tree was obtained with it; (88) is available where the pressure field is real but its gradients are diffuse, as in a driven channel, and the layered case exercises both.

Where the zero goes. Within (87) the absolute level is still free in the continuum and still not free numerically, and this is the single setting that decides whether the scheme works at all. Adding a constant P_0 to p adds $P_0/(\rho c_s^2)$ to $\tilde{p}$, which is not a constant when ρ varies, and both large terms scale with it. Seeding $p = \rho_L c_s^2$ — a uniform physical pressure, the natural-looking choice — was measured on the static droplet to return a *negative* surface tension, in error by thousands of per cent at a density ratio; the acceptance test for that case is normalised by the prescribed σ rather than by the smallest measured one precisely because a sign flip would otherwise turn a gross failure into an apparent pass, which it did. The same gauge made a Rayleigh–Taylor run at $At = 0.5$ diverge inside 424 steps, with $|\boldsymbol{u}|$ at twice the free-fall velocity at step zero, before anything had moved.

Seeding $p = 0$ where the two fluids meet keeps both terms small and leaves only the physical jump in $\tilde{p}$. With that gauge and nothing else changed, the same droplet recovers Laplace's law across the whole ratio range, and the spurious currents *fall* as the ratio rises:

ρ_H/ρ_L	σ recovered	rel. err.	max $\lvert\boldsymbol{u}\rvert$
1	9.27607×10^{-3}	7.24×10^{-2}	1.67×10^{-5}
10	9.71478×10^{-3}	2.85×10^{-2}	3.21×10^{-6}
100	9.83470×10^{-3}	1.65×10^{-2}	4.90×10^{-7}

σ varies by 5.59×10^{-2} of the prescribed value over a hundredfold ratio, and most of that spread is the matched-density row's ordinary resolution deficit rather than any dependence on the ratio. The rule the module follows is therefore to put the pressure zero at the interface — for a hydrostatic configuration, at the free surface — and it is a rule about conditioning, not about physics. Its immediate corollary is the storage restriction stated at the end of §10.5: the shifted convention centres the stored variable on $\tilde{p} = 1$, which is the one gauge that must not be chosen at a density ratio.

10.7 Rigid bodies by volume penalisation

A rigid body is coupled to the flow by volume penalisation: a smooth solid indicator, and a direct forcing that drives the fluid inside it toward the body's rigid velocity field. The two standard alternatives were considered and rejected for specific reasons. A Peskin-style immersed boundary — Lagrangian markers on the surface, interpolation and spreading kernels, a body force fed back into $\boldsymbol{F}$ — is the usual tool and it is more accurate *at the surface* than this, but it is also a great deal more machinery, and for a body whose *interior* must move rigidly, a solid square rather than a thin shell, a volumetric penalisation enforces the same condition with a fraction of it. De Rosis and Enan drive their wedge that way. Moving bounce-back geometry is the other route and is worse here on three counts: the solid mask must be rebuilt on the host every time the body advances, which is a full pass over the domain plus `rebuild_lists()`; the nodes the body uncovers need a refill scheme, their populations being meaningless; and the body's position is quantised to whole cells, so a smoothly falling square arrives in lattice-sized jumps and each one radiates a pressure pulse. Penalisation has none of these. Nothing about the geometry is ever rebuilt, the body's position *and angle* are continuous, and there are no fresh nodes. Rotation in particular is nearly free here and would be a rewrite in either of the other two.

The indicator and the force. `Rect::chi` is the product of two tanh profiles across the faces, evaluated after the point has been taken into the body frame, so the shape turns with the body and so does the slight rounding of the corners that the product form introduces. It is smoothed over about a cell and a half (`Rect::smooth`, default 1.5 cells) rather than being a sharp mask: a sharp mask is the staircase again, with the surface sitting on lattice planes and the effective shape changing as the body falls, or as it turns. The smoothing is what lets the body occupy a continuously varying pose. The forcing is

$$\boldsymbol{F}(\boldsymbol{x}) = \chi(\boldsymbol{x})\, 2\rho(\boldsymbol{x}) \Big[\boldsymbol{U} + \Omega\, \hat{\boldsymbol{z}} \times \boldsymbol{r} - \boldsymbol{u}^*(\boldsymbol{x})\Big], \qquad \boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_c, \tag{90}$$

which is direct forcing: applied through the same half-force convention as (47), one step of it takes $\boldsymbol{u}$ to the rigid field exactly where $\chi = 1$. It must run after `compute_macroscopic()` and before the fluid steps. The density is read node by node rather than taken as a constant, so the same body meets a thousandfold heavier medium when it reaches the water; a penalisation that ignored that would decelerate the body as though it were still in air.

$\boldsymbol{u}^*$ is not the stored velocity. This is the one thing that has to be right. The solver defines $\boldsymbol{u} = \big(\sum_i \boldsymbol{c}_i f_i + \frac{1}{2}\boldsymbol{F}\big)/\rho$ with $\boldsymbol{F}$ the *total* force, so the stored velocity field already contains this body's force from the previous step. Feeding it straight back into (90) applies the correction twice: the body over-shoots, the fluid over-shoots back, and the pair diverges within a few hundred steps — measured, the square left the domain at six million times its impact speed. The previous contribution is therefore removed first,

$$\boldsymbol{u}^* = \boldsymbol{u}_{\text{stored}} - \frac{\boldsymbol{F}_{\text{prev}}}{2\rho},$$

which leaves the velocity the fluid would have had from every *other* force, and that is what direct forcing is defined against. The force arrays hold $\boldsymbol{F}_{\text{prev}}$ already, so this costs one subtraction and no extra state.

Because the force is known at every node, the force and torque the fluid exerts on the body are minus its sum and minus the sum of $\boldsymbol{r} \times \boldsymbol{F}$. That closes Newton's equations, and it is the reason for doing it this way: the body falls, floats and rolls under gravity and buoyancy rather than being pushed at a prescribed speed.

The fictitious fluid, and why the obvious arrangement cannot float. The penalised region is not empty. It is full of fluid that the forcing drags along, and that fluid's inertia and weight are both already inside the reaction, so using the body's own mass alone would count them twice. Writing m_f for that fictitious fluid mass, the exact bookkeeping is

$$m_b \frac{d\boldsymbol{U}}{dt} = \boldsymbol{R} + \frac{d\boldsymbol{P}_{\text{in}}}{dt} + (m_b - m_f)\, \boldsymbol{g},$$

where $\boldsymbol{g}$ is the body force per unit mass — the same vector the collision operator is given, or the buoyancy term describes a different problem from the fluid it is meant to displace. Uhlmann's classical step approximates $d\boldsymbol{P}_{\text{in}}/dt$ as $m_f\, d\boldsymbol{U}/dt$, moves it across and divides:

$$\delta \boldsymbol{U} = \frac{\boldsymbol{R} + (m_b - m_f)\, \boldsymbol{g}}{m_b - m_f}.$$

That denominator is the whole problem. It vanishes for a neutrally buoyant body and changes sign for a light one, so a floating body — the only interesting kind, since floating is what a density ratio is *for* — is precisely the case the arrangement cannot express. This never announced itself as a failure. The old code did not crash and produced no NaN: the guard on that denominator silently zeroed the acceleration, so a body that could not be floated simply did not accelerate.

Targeting the velocity the body is about to have. The fix is not a different model. It is the observation that the reaction is *linear* in the body velocity that produces it,

$$\boldsymbol{R} = -\sum \chi\, 2\rho\, (\boldsymbol{U} + \Omega\, \hat{\boldsymbol{z}} \times \boldsymbol{r} - \boldsymbol{u}^*),$$

so if the force written into the arrays targets the velocity the body is *about* to have rather than the one it already has, $\boldsymbol{R}$ moves to the left-hand side and the same equation comes out with a different denominator,

$$(m_b + m_f)\, \delta \boldsymbol{U} = 2\, \delta \boldsymbol{P} + (m_b - m_f)\, \boldsymbol{g}, \tag{91}$$

with $\delta \boldsymbol{P} = \sum \chi\rho\, (\boldsymbol{u}^* - \boldsymbol{u}_{\text{rigid}})$ the inner fluid's momentum deficit relative to rigid motion — the hydrodynamic signal, and the only place the fluid enters. Algebraically (91) is Uhlmann's equation multiplied through: the same discrete equation, rearranged, with nothing added to it. The difference is entirely that the body and the fluid are now solved at the *same* instant instead of the fluid lagging the body by a step — the lesson of §9.3 in another guise. The denominator $m_b + m_f$ is positive for every mass, the homogeneous amplification factor is $(m_b - m_f)/(m_b + m_f)$, whose modulus is below one for every mass, and the neutrally buoyant case, previously a division by zero, is now the best-conditioned point on the line. Nothing is tuned and no limit is imposed.

Rotation, and why it does not decouple. The same substitution with rotation gives a 3×3 system rather than a scalar. It does not split into a translation part and a roll part, because the first moments $\boldsymbol{S} = \sum \chi\rho\, \boldsymbol{r}$ of the fictitious mass about the body centre are non-zero the moment the body straddles an interface — half of it is standing in water and half in air — and those moments couple sway to roll. With $A = m_b + m_f$ and $B = I_b + I_f$,

$$\begin{bmatrix} A & 0 & -S_y \\ 0 & A & S_x \\ -S_y & S_x & B \end{bmatrix} \begin{bmatrix} \delta U_x \\ \delta U_y \\ \delta \Omega \end{bmatrix} = \begin{bmatrix} 2\, \delta P_x + (m_b - m_f)\, g_x \\ 2\, \delta P_y + (m_b - m_f)\, g_y \\ 2\, \delta L_z - (S_x g_y - S_y g_x) \end{bmatrix}. \tag{92}$$

The matrix is symmetric, and it is positive definite for every body with mass and inertia of its own: Cauchy–Schwarz on the measure $\chi\rho$ gives $|\boldsymbol{S}|^2 \le m_f I_f \le AB$, so the Schur complement $B - |\boldsymbol{S}|^2/A$ is positive. There is therefore no pivoting and no case in which the solve has to give up; it is done in closed form, on the host, once a step. All seven integrals — m_f, $\boldsymbol{S}$, I_f, $\delta \boldsymbol{P}$ and δL_z — come from a single sweep, and the force is then written in a plain `parallel_for`: one reduction, not two. Locking a degree of freedom removes the coupling rather than the row, so a body held against translation solves $B\, \delta\Omega$ against the third right-hand side alone, which is not the same answer as solving (92) and discarding two rows. Finally, m_b and I_b are taken from the integrals of the indicator itself rather than from $4h_x h_y$ and the textbook $(w^2 + h^2)/12$, so that they describe the body the force actually acts on: the smoothing makes the penalised body slightly larger than the nominal rectangle, and its second moment exceeds the sharp rectangle's by 2.55% at the default smoothing width.

The last term of the third row is the righting moment. The body's own weight acts at its centre and exerts no torque about it; the displaced fluid's weight does not, because the displaced fluid is heavier on the submerged side, and $-(\boldsymbol{S} \times \boldsymbol{g})$ is exactly the hydrostatic couple that metacentric theory writes as $\rho g V\, \overline{GM} \sin\theta$. It was not put there for that purpose. It falls out of the same subtraction that produced $(m_b - m_f)\boldsymbol{g}$ in the rows above, and it is the reason a raft comes back upright. It is reported separately in `Reaction::righting` because it is the only part of the roll balance with a closed form to check against, and nothing in the way it is computed knows that form exists; §12.14 makes the comparison.

The reaction is a derived diagnostic, not a measurement. Once the solve has produced $\delta\boldsymbol{U}$ and $\delta\Omega$, the force and torque on the body follow from the same integrals in closed form,

$$\boldsymbol{R} = 2\,\delta\boldsymbol{P} - 2\big[\,m_f\,\delta\boldsymbol{U} + \delta\Omega\,(\hat{\boldsymbol{z}}\times\boldsymbol{S})\,\big], \qquad T = 2\,\delta L_z - 2\big[\,I_f\,\delta\Omega + \boldsymbol{S}\times\delta\boldsymbol{U}\,\big], \tag{93}$$

so nothing has to be summed a second time to report them. What is reported is consequently exact to round-off against $-\sum\boldsymbol{F}$ and $-\sum\boldsymbol{r}\times\boldsymbol{F}$, and is an identity rather than an independent measurement of them: the physics enters through $\delta\boldsymbol{P}$ and δL_z, and (93) is algebra on top. That identity is checked against the summed force array in §12.14.

What this does not model. There is no contact line. The phase field is advected by the fluid velocity and knows nothing about the body, so nothing sets the angle at which the free surface meets the solid; inside the body $\boldsymbol{u}$ is driven to the rigid field, so the interface is carried *with* the body rather than through it, and the conservative Allen–Cahn form keeps the profile from simply diffusing in — but the wetting physics is absent, and water entry is a contact-line problem. Read the splash, not the meniscus. The same gap means a floating body gets no surface-tension contribution to its draft, so a body small enough for that to matter, at Bond number of order one, is out of scope. There is no sub-cell surface: penalisation resolves the body to the smoothing width, so a pressure read at the face of the square is a smoothed pressure over a cell or two, and the integrated force is far more trustworthy than the local pressure, which is the usual situation with this family of methods; the draft is likewise resolved only to the interface width. There is no collision model: two bodies, or a body meeting a wall, will simply interpenetrate, because nothing here computes a contact force, and the floor stop in `demonstrator/water_entry` is a clamp in the driver rather than physics. The scheme is two-dimensional, and says so in the compiler's voice — a `static_assert` on $D = 2$, because on a 3D lattice it would silently model an infinite prism in z with a rigid-body solve carrying one angle where it needs three. Three dimensions is a 6×6 with a rotating inertia tensor and a quaternion, a different piece of work; the fluid side would not change at all. And the body is of uniform density: its centre of mass is taken to be the centre of the rectangle, which is exactly what lets its own weight drop out of the roll equation. Ballast low in the hull is the standard way to make a tall body float upright, and it is precisely what this cannot express — it would need a second centre, a $\overline{BG}$ that no longer follows from the draft, and a weight torque restored to the third row. The rectangle is also the only indicator provided, though anything carrying a signed distance function drops straight in.

10.8 A second immiscible model: the colour gradient

Everything above carries the interface in a field with its own advection equation. The colour-gradient model of Saito *et al.* (Phys. Rev. E **98**, 013305, 2018), implemented in `ColourGradient.hpp` and `ColourGradientSolver.hpp`, reaches the same physics with no interface equation at all, and it is in the tree because the two fail differently.

Two distribution sets f_i^r and f_i^b are carried. The interface is wherever both are non-zero, and it is held together by a *recolouring* step that redistributes the colour-blind post-collision state back into red and blue along the colour gradient. Nothing diffuses, nothing is reinitialised, and the interface width is an *outcome* rather than a parameter — the opposite of the conservative Allen–Cahn equation of §10.1, whose W is prescribed and whose profile is its exact one-dimensional equilibrium.

Three suboperators. Equation (10) of the paper reads right to left,

$$\Omega_i^k = (\Omega_i^k)^{(3)}\Big[(\Omega_i)^{(1)} + (\Omega_i)^{(2)}\Big]: \tag{94}$$

collide the colour-blind $f_i = f_i^r + f_i^b$, perturb it to make surface tension, then recolour. Only the third carries a colour index, which is the economy of the scheme: the expensive part, the central-moment collision, is done *once* for both fluids.

The single-phase collision is a general MRT in central moments with the relaxation grouping $s_0 = s_1 = 0$, five second-order moments at the shear rate, the trace on its own, and everything above second order straight to equilibrium. That is already what `MomentCollision.hpp` does, so `ProductBasis` is reused unchanged and only the *equilibrium* is new. The paper reaches the same operator through De Rosis's nonorthogonal basis and this code through a Hermite product basis; above second order the two cannot disagree, because when every moment of a subspace relaxes at one rate any basis of that subspace gives the same operator, and at second order the split into trace and traceless deviatoric is basis-independent for the same reason.

The density ratio lives in the rest weight. ϕ_i of Eq. (20) replaces w_i in the rest term of the equilibrium and carries a free parameter α,

$$\phi_0 = \alpha, \quad \phi_{|\boldsymbol{c}_i|^2=1} = \tfrac{2(1-\alpha)}{19}, \quad \phi_{|\boldsymbol{c}_i|^2=2} = \tfrac{1-\alpha}{38}, \quad \phi_{|\boldsymbol{c}_i|^2=3} = \tfrac{1-\alpha}{152}, \tag{95}$$

which sums to one for any α and whose second moment is $c_s^2 = 9(1-\alpha)/19$. *The sound speed is therefore a property of the phase.* Pressure continuity across the interface, $p = \rho c_s^2$, then fixes the density ratio without a single density appearing in the collision,

$$\gamma = \frac{\rho_r^0}{\rho_b^0} = \frac{1-\alpha_b}{1-\alpha_r}. \tag{96}$$

Taking $\alpha_b = 8/27$ recovers $c_s^2 = 1/3$ exactly for the blue fluid, which is the paper's choice and makes the blue phase an ordinary lattice. Note what it does *not* do: it does not make $c_s^2 = 1/3$ anywhere else, so the standard lattice sound speed is not available as a global constant here, and nothing in the operator assumes it. `ProductBasis`'s c_s^2 is a basis constant, not a physical one, and the two are deliberately different numbers.

The last term of the equilibrium, Φ_i of Eq. (21), restores Galilean invariance when the density varies, through the second moment, using $G = \frac{1}{48}[\boldsymbol{u} \otimes \nabla\rho + (\boldsymbol{u} \otimes \nabla\rho)^{\mathsf{T}}]$. It is the same physics that `ViscousInterfaceForce.hpp` adds to the phase-field module as an explicit body force $\nu(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}}) \cdot \nabla\rho$ — the same missing stress, entered by a different door, and here at no extra field beyond $\nabla\rho$ because it goes into the equilibrium rather than into F. It is also the term most easily dropped by accident, and dropping it changes nothing that any static test checks.

A factor of two that a running simulation would absorb. The perturbation is written per colour in Eq. (30) as $(A_k/2)|\nabla\phi|[w_i(\boldsymbol{c}_i \cdot \boldsymbol{n})^2 - B_i]$, and Eq. (39) sums the capillary stress over k as well as over i with $A = A_r = A_b$. Applied to the colour-blind population — which is what the paper's own text says is done — the two halves add and the coefficient is A, not $A/2$. The distinction is not cosmetic, and it was settled by deriving σ from Eq. (39) rather than copying Eq. (32): for a flat interface

$$\sigma = \int (S_{tt} - S_{nn})\,\mathrm{d}n = \frac{2A\tau}{9}\,\Delta\phi = \frac{4A\tau}{9}, \tag{97}$$

since ϕ runs from -1 to $+1$. That is Eq. (32) exactly. With $A/2$ it would have come out half as large, and a static droplet would have reported a surface tension 50% low with every other property of the model intact.

The perturbation conserves mass and momentum exactly and acts only across the interface: $\sum_i w_i(\boldsymbol{c}_i \cdot \boldsymbol{n})^4 = 3c_s^4 = 1/3$ cancels $\sum_i B_i(\boldsymbol{c}_i \cdot \boldsymbol{n})^2 = 1/3$ identically, so the *normal* component of the capillary stress vanishes for every direction of $\boldsymbol{n}$ while the tangential component does not — both direction-independent, because D3Q27's fourth moment is isotropic.

Recolouring is a partition. Equations (33)–(34) split the post-collision population back,

$$f_i^r = \frac{\rho_r}{\rho} f_i + \beta \frac{\rho_r \rho_b}{\rho^2} \cos\theta_i \, f_i^{\mathrm{eq}}(\rho, \mathbf{0}), \tag{98}$$

$$f_i^b = \frac{\rho_b}{\rho} f_i - \beta \frac{\rho_r \rho_b}{\rho^2} \cos\theta_i \, f_i^{\mathrm{eq}}(\rho, \mathbf{0}), \tag{99}$$

with $\cos\theta_i$ the cosine of the angle between $\boldsymbol{c}_i$ and $\nabla\phi$. The two sum to f_i identically, so mass and momentum survive whatever β is; all β does is push colour up the gradient. $\beta = 0.7$ is the paper's value and the largest that keeps the interface smooth. Note that $f_i^{\mathrm{eq}}(\rho, \mathbf{0})$ is simply $\rho\phi_i$: at zero velocity every other term of the equilibrium vanishes, Φ_i included.

What the operator does not own. It does not stream, it does not hold the two distributions, and it does not compute a gradient. Under Esoteric Pull the neighbour populations inside a fused kernel are a mixture of two time levels, so a gradient taken from them is meaningless; $\nabla\phi$ and $\nabla\rho$ are node fields computed by `ColourGradientSolver` in a separate pass, for exactly the reason §10.1 gives for the phase field.

Which model to use is not settled here. The phase field has a conservative advection equation and a prescribed interface width; the colour gradient has neither. What the colour gradient has instead is the density ratio: it reaches 1000 in the paper's own static tests, where the pressure form of §10.6 is already fighting its conditioning problem at 100. §18.12 measures both on one case.

11 Free-surface flow

§10 resolves both fluids, and everything difficult about it follows from that: the pressure-gauge conditioning of §10.6, the α interpolation of §10.8, the recolouring. The third route in this code changes the bargain instead of the parameters. It does not resolve the gas at all. The gas is a *void* held at a prescribed pressure — no populations, no velocity, no dynamics — and the liquid meets it across a boundary condition rather than across a diffuse interface.

What that buys is the thing the other two fight for. There is no density ratio, because there is no second density: the ratio is infinite by construction and it costs nothing, because the expensive half of the problem was deleted rather than discretised.

module	interface	fluids resolved	density ratio
phase field (§10.1)	diffuse	both	~ 100, with the gauge problem
colour gradient (§10.8)	diffuse	both	1000 in principle
free surface (this section)	sharp	liquid only	infinite, by construction

`FreeSurfaceSolver.hpp` follows the standard formulation of Körner, Thies, Hofmann, Theobald & Rüde, J. Stat. Phys. **121**, 179 (2005), with the refinements in Thürey's and Rüde's later work. Unlike §10.8 there is no paper in `SomeRefs/` to check equation by equation, so nothing below cites an equation number and every choice is justified from the mechanics.

11.1 The state, and the mass that is not a moment

Every cell is Gas, Interface or Fluid (or a wall, or halo). Fluid cells are ordinary LBM cells; Gas cells hold nothing. Interface cells are the entire method: they carry an explicit mass m alongside their populations, and a fill level

$$\varepsilon = m/\rho, \tag{100}$$

which is 1 when the cell is full and 0 when empty. The interface is a closed, one-cell-thick shell separating Fluid from Gas, and keeping it closed through arbitrary topology change is most of the work.

Mass is advected, not taken as a moment, and that is the sharpest difference from everything else in this tree. Elsewhere $\rho = \sum_i f_i$ and conservation is a property of the collision operator, checkable by algebra. Here mass moves because populations cross faces:

$$\mathrm{d}m_i(\boldsymbol{x}) = f_{\bar{\imath}}(\boldsymbol{x} + \boldsymbol{c}_i) - f_i(\boldsymbol{x}), \tag{101}$$

for both cells Fluid or Interface, scaled by the mean fill level when both are Interface, because a half-full cell can only exchange half as much. Nothing enforces conservation but the *antisymmetry* of (101): what $\boldsymbol{x}$ sends across a face is exactly what its neighbour receives. That is why the validation checks total mass to round-off first and everything else second — it is the one property that can be silently lost.

11.2 The free-surface condition

For every direction i whose source cell $\boldsymbol{x} - \boldsymbol{c}_i$ is Gas, the population that would have arrived is reconstructed as

$$f_i(\boldsymbol{x}) = f_i^{\mathrm{eq}}(\rho_G, \boldsymbol{u}) + f_{\bar{\imath}}^{\mathrm{eq}}(\rho_G, \boldsymbol{u}) - f_{\bar{\imath}}(\boldsymbol{x}), \tag{102}$$

with ρ_G fixing the gas pressure through $p = \rho_G c_s^2$. Its two moments are what a free surface *is*: the normal stress equals the gas pressure, and the tangential stress vanishes. Here $f_{\bar{\imath}}(\boldsymbol{x})$ is the post-collision population at $\boldsymbol{x}$ from the previous step, which the two-lattice scheme still holds in its source array, so no extra storage is needed.

The velocity in that reconstruction is one step old. $\boldsymbol{u}$ is needed to build the equilibria and $\boldsymbol{u}$ comes from a sum over the populations, some of which are the ones being reconstructed; the circle is broken with the previous step's velocity. That is first order in time and is what the literature does. It matters where the interface accelerates hard and is invisible in a standing wave.

11.3 Why this module alone uses two-lattice streaming

Everything else in `M3LB` streams by Esoteric Pull (§3). This solver `static_asserts` *against* it, and the reason is structural rather than a matter of effort. Three requirements, of which Esoteric Pull fails two outright:

1. Mass exchange needs $f_i(\boldsymbol{x})$ and $f_{\bar{\imath}}(\boldsymbol{x} + \boldsymbol{c}_i)$ as *post-collision* values, after the whole domain has collided. In place, those slots hold a mixture of two time levels, and which one depends on parity and on the direction's index. Under two-lattice they are simply the destination array, complete and unambiguous, once the collide pass has fenced.
2. A cell that fills up turns its gas neighbours into interface cells, which then need populations. In place, a cell's populations live in its neighbours' slots, so initialising a neighbour means writing storage that other cells are concurrently reading through their own aliases.
3. Reading a neighbour's post-collision state while writing one's own is the normal case here and impossible there.

Requirement 2 is dodged rather than met: every pass is a *gather*. A cell about to become interface initialises itself by averaging its neighbours, and a cell with excess mass publishes it in a field that its neighbours collect. Nothing ever writes another cell's memory, so there is no race to reason about and no atomics — a deliberate shape, worth keeping even if scatters were free. The cost is two population arrays instead of one, which for a method that deletes an entire phase is not the expensive part.

11.4 A standing gravity wave

The validation case is the linear dispersion relation $\omega^2 = gk\tanh(kh)$, and it is that rather than a dam break because a dam break's reference is experimental and comparing against a digitised curve tests the digitisation as much as the solver. The dispersion relation is exact, closed form, has no fitted constant, and — the part that matters — it is a statement about precisely the two conditions a free-surface scheme exists to impose: the *kinematic* condition, that the surface moves with the fluid, which here is the mass exchange and nothing else; and the *dynamic* condition, that the normal stress equals the gas pressure, which here is the population reconstruction and nothing else. Get either wrong and the period moves.

Two depths, on purpose. At $kh = 3.1$ the tanh is 0.9963 and the relation is indistinguishable from the deep-water limit $\omega^2 = gk$; at $kh = 0.79$ it is 0.657, and a scheme with the surface condition right but the depth coupling wrong would pass the first and fail the second. Testing one depth tests a number; testing two tests the relation.

The two residual errors are separate, and each was identified by making it move without moving the other. *Deep* is limited by the free surface's own discretisation: the surface is resolved to about a cell, so a wave of amplitude A carries a relative position error of order $1/A$, and refining at fixed kh took it from 6.15% at $L_x = 64$, $A = 1.5$ to 1.77% at $L_x = 128$, $A = 3$. *Shallow* is limited by the viscous boundary layer on the floor: at $h = 16$ cells the layer is $\sqrt{\nu T} \approx 4$ cells, a quarter of the depth, against a relation with no viscosity in it at all, and dropping ν fivefold took it from 8.44% to 4.23%, monotonically. Neither knob touches the other's number, which is what says these are two effects rather than one tolerance being chased.

11.5 What this does not model

- **No surface tension.** ρ_G is uniform, so the gas pressure carries no curvature term. Adding it means $p_G = p_{\text{atm}} - \sigma\kappa$ with κ from the fill-level field, and the machinery for a curvature is not here. Everything at a Bond number where surface tension competes with gravity is out of scope — which includes most droplet problems and none of the dam-break family.
- **No gas dynamics**, and this is the method rather than an omission. An enclosed bubble does not compress, because its pressure is prescribed rather than solved. Modelling that needs a volume-of-gas tracker with a bubble-merging graph.
- **The reconstruction is applied to gas-facing directions only.** The literature also reconstructs directions with $\boldsymbol{c}_i \cdot \boldsymbol{n} > 0$, which stabilises interfaces nearly tangential to the lattice. If a case shows interface roughening along a diagonal, this is the first thing to add.
- **Excess mass is redistributed uniformly** among eligible neighbours rather than weighted by the interface normal. Uniform redistribution conserves mass exactly — which matters most — but is more diffusive at a sharply curved interface.
- **No subgrid interface.** A film thinner than a cell either survives as a one-cell layer or vanishes.
- **A moving obstacle is not yet reliable**, and the reason is in the solver rather than in the caller. `mass_exchange()` conserves mass by the antisymmetry of (101), which assumes both cells agree on the mask for the whole pass. `move_obstacle()` changes the mask between passes, so every cell the body sweeps breaks the assumption. `transfer_covered_mass()` keeps the total right but does not restore the per-face antisymmetry, and the difference shows up wherever many mask cells change at once: a flat face landing on a flat surface, a corner entering, a rotated rectangle's staircase flipping cells as it turns. The fix is to make the mass exchange itself aware of the moving boundary; bounce-back, momentum exchange and refill are all in place and are not the problem.

12 Validation

Every scheme is accompanied by the test that establishes it. The suite runs in both FP64 and FP32 and consists of ten executables; the numbers below are FP64 unless stated.

12.1 Poiseuille flow: wall placement and the magic parameter

Force-driven plane Poiseuille flow, periodic in x, halfway bounce-back walls in y. With H fluid nodes the no-slip planes sit at $y = 0.5$ and $y = H + 0.5$, so

$$u(y) = \frac{G}{2\rho\nu}\,(y - 0.5)\,(H + 0.5 - y), \qquad u_{\max} = \frac{GH^2}{8\rho\nu}. \tag{103}$$

Which relaxation rates play the roles in (31) depends on the operator, so each has its own magic point. Measured wall slip (D2Q9, $H = 32$):

τ	BGK	TRT(3/16)	MRT	central moments
0.6000000	-7.40×10^{-3}	$+2.51 \times 10^{-14}$	-5.73×10^{-3}	-5.73×10^{-3}
0.8000000	-4.06×10^{-3}	$+7.53 \times 10^{-14}$	-1.56×10^{-3}	-1.56×10^{-3}
0.8750000	-1.95×10^{-3}	-1.92×10^{-14}	$\mathbf{+4.11 \times 10^{-14}}$	$\mathbf{+7.39 \times 10^{-14}}$
0.9330127	$\mathbf{-1.08 \times 10^{-14}}$	$+5.35 \times 10^{-14}$	$+1.21 \times 10^{-3}$	$+1.21 \times 10^{-3}$
1.2000000	$+1.26 \times 10^{-2}$	$+3.55 \times 10^{-15}$	$+6.77 \times 10^{-3}$	$+6.77 \times 10^{-3}$

TRT holds the wall at 0.5 to $\sim 10^{-14}$ at *every* viscosity. BGK manages it only at $\tau = \frac{1}{2} + \frac{\sqrt{3}}{4} = 0.9330127$. The moment operators relax third-order modes at unity, so their magic point is $(\frac{1}{\tau} - \frac{1}{2})\frac{1}{2} = \frac{3}{16}$, i.e. $\tau = 7/8$ exactly — *predicted before the run and confirmed to* 7×10^{-14}.

Resolution behaviour. At its magic point an operator is exact at every resolution. Away from it (FP64, D2Q9):

operator @ τ	L_2, H=16	H=32	H=64	ord(L_2)	ord(slip)
BGK @ 0.9330127	4.2e-14	1.2e-14	5.8e-13	exact	exact
TRT(3/16) @ 0.6	4.6e-14	1.1e-13	1.6e-13	exact	exact
MRT @ 0.875	3.2e-14	9.3e-14	3.1e-13	exact	exact
central moments @ 0.875	3.6e-14	8.4e-14	3.2e-13	exact	exact
BGK @ 0.6 (off-magic)	5.1e-03	1.3e-03	3.2e-04	**2.00**	**1.00**

Off-magic bounce-back is still *second* order: a lattice-aligned wall does not degrade the scheme. The pair of orders is what identifies the defect — the discrete error is a constant velocity offset $C \sim (\Lambda - 3/16)\,G/\nu$, which at fixed peak velocity scales as $1/H^2$, while the fitted wall *position* moves only as $1/H$ because the parabola's curvature shrinks with it. Seeing 2 and 1 together says "offset", not "displaced wall".

These two assertions are FP64-only by design: at $H = 64$ the discretisation error is 3×10^{-4}, which in FP32 lies below the accumulated round-off floor of about 10^{-3}. Loosening the tolerance would not fix that, it would only stop the test measuring anything.

12.2 Decaying flows with exact solutions

Three periodic flows, chosen so that each isolates something the others cannot.

Taylor–Green $\boldsymbol{u} = U(-\cos kx \sin ky,\ \sin kx \cos ky)\, e^{-2\nu k^2 t}$. Nonlinear and exact, but z-independent — so it is the *same 2D problem* on every lattice. It is a cross-lattice consistency check, not a 3D test.

ABC / Beltrami $\boldsymbol{u} = U(\sin kz + \cos ky,\ \sin kx + \cos kz,\ \sin ky + \cos kx)$. Here $\nabla \times \boldsymbol{u} = k\boldsymbol{u}$, so $\boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \boldsymbol{0}$ and the nonlinear term collapses to $\nabla(|\boldsymbol{u}|^2/2)$, absorbed by the pressure: $\boldsymbol{u}(t) = \boldsymbol{u}_0 e^{-\nu k^2 t}$ exactly. Every component varies in every direction. This is the nonlinear 3D convergence test.

Diagonal shear wave $\boldsymbol{u} = U\boldsymbol{e}\sin(\boldsymbol{K}\cdot\boldsymbol{x})e^{-\nu|K|^2 t}$ with $\boldsymbol{K} = \frac{2\pi}{L}(1,1,1)$ and $\boldsymbol{e}\cdot\boldsymbol{K} = 0$. Linear by construction, and off-axis, so it probes lattice isotropy where an axis-aligned wave cannot.

The three are complementary and none substitutes for another. The ABC flow is built from axis-aligned sinusoids — it tests nonlinear 3D convergence but says nothing about isotropy.

Interpreting the effective viscosity. ν_{eff} does *not* equal ν at finite resolution: a sinusoid decays at the rate set by the discrete Laplacian, which differs from νk^2 at $O(k^2)$. What is asserted is therefore that the error converges away at second order, not that it is small at any given N. Diffusive scaling $U \sim 1/N$ keeps the Mach-number error second order alongside the spatial error.

12.3 Grid convergence: all lattices, all operators

A dedicated sweep (`validation/grid_convergence`) covers every lattice $\times$ operator combination over a resolution ladder and reports 92 data points. Slopes are least-squares fits of log(error) against $\log N$ over the whole ladder, not pairwise ratios.

case	lattice	operator	slope ν err	slope $L_2(u)$	worst R^2
Taylor–Green	D2Q9	BGK, TRT, MRT, CM	-2.02	-1.99	
Taylor–Green	D3Q27	BGK, TRT, MRT, CM	-2.02	-1.99	
ABC/Beltrami	D3Q27	BGK, TRT, MRT, CM	-2.01	-2.01	
24 fitted slopes, range -2.023 **to** -1.994					$R^2 \geq 0.99982$

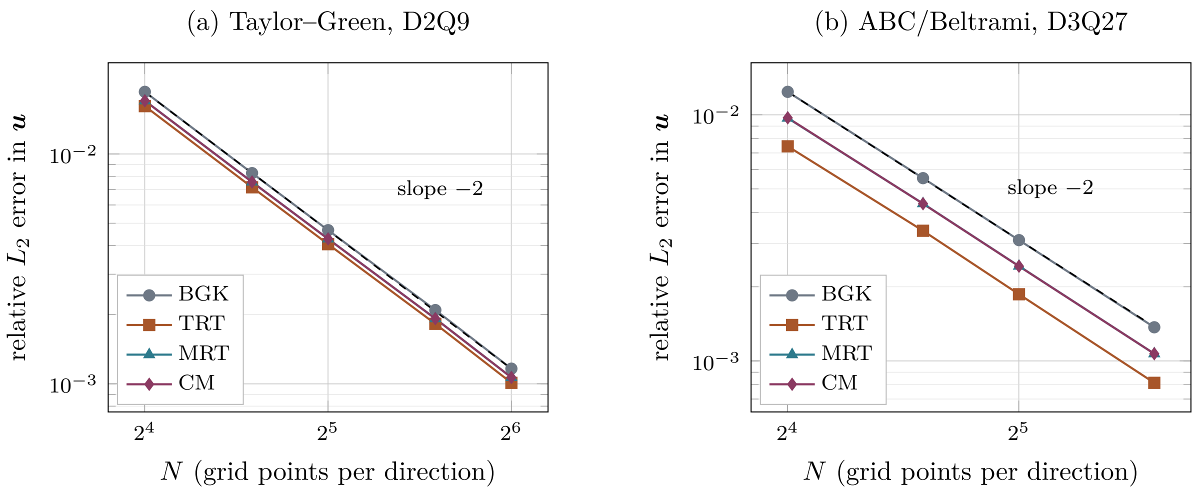


Figure 1: Grid convergence of the four collision operators, log–log, with a slope -2 reference (dashed). Diffusive scaling $U \sim 1/N$. Panel (a) is the Taylor–Green vortex on D2Q9; D3Q27 lies exactly on top of these points (spread 3.9×10^{-14}) because the field is z-independent. Panel (b) is the ABC/Beltrami flow on D3Q27. The lines are parallel: the operators differ in error *amplitude*, not in order. Fitted slopes over the whole ladder range from -1.994 to -2.023 across all 24 series.

No operator gains or loses an order. What differs is the error *amplitude*, and no operator dominates everywhere:

	ABC flow, ν error at $N = 48$	diagonal shear wave, $N = 64$
BGK	2.75×10^{-3}	1.00× (reference)
TRT(3/16)	$\mathbf{1.63 \times 10^{-3}}$	—
MRT	2.15×10^{-3}	—
central moments	2.15×10^{-3}	**1.38**× better (D3Q27)

TRT is the most accurate in a periodic bulk flow, because the magic parameter tunes the third-order modes that the moment operators send straight to equilibrium. On the off-axis wave the ordering reverses.

12.4 Galilean invariance

Superimposing a uniform velocity U_0 on a solution only translates it, so the recovered viscosity must not depend on U_0. ABC/Beltrami flow, $N = 32$, $\tau = 0.8$:

lattice	operator	U_0=0	0.05	0.10	0.15	drift vs BGK
D3Q27	BGK	0.100620	0.099864	0.097595	0.093812	—
D3Q27	TRT(3/16)	0.100368	0.099618	0.097368	0.093617	1×
D3Q27	MRT (raw)	0.100484	0.100485	0.100486	0.100490	1153×
D3Q27	central moments	0.100484	0.100484	0.100482	0.100479	**1439**×

Two findings deserve emphasis.

1. **TRT is no better than BGK here.** The magic parameter fixes where the *wall* sits, not which frame the collision prefers. These are independent defects and it takes both operators to address both.
2. **Most of the gain is the complete equilibrium, not the co-moving frame.** Raw MRT already recovers 1153×, because this implementation gives it the full product-form equilibrium; centring the basis adds a further $\sim 25\%$. The split may differ for flows whose nonlinearity is less balanced than a Beltrami field, but the effect should not be attributed wholly to centring.

Every residual scales as U_0^2 (fitted powers 2.00–2.10), so what remains is the weakly-compressible $O(\mathrm{Ma}^2)$ error of the method itself, not frame dependence of the collision.

12.5 Thermal transport

Two exact solutions of the advection–diffusion equation, with amplitude and Peclet number both scaled diffusively ($U \sim 1/N$) so that the comparison is a convergence study rather than a measurement of how far the wave travelled:

case	order (FP64)	order (FP32)	D_{eff} rel. err	phase err
D2Q5 diffusion	2.00	2.03	1.5×10^{-3}	4.5×10^{-14}
D3Q7 diffusion	2.00	2.00	1.3×10^{-3}	2.8×10^{-14}
D2Q5 advection–diffusion	2.00	2.02	1.1×10^{-3}	2.5×10^{-4}
D3Q7 advection–diffusion	1.99	2.00	7.2×10^{-4}	2.3×10^{-4}

The phase is measured as well as the amplitude: an error in the coupling term puts the wave in the wrong *place*, which an amplitude check alone would not detect.

12.6 Open boundaries: the Gaussian plume

The open boundary of §8.5 is validated against the classical plume solution. A continuous point source of strength Q at the origin in a uniform wind U along x, with isotropic diffusivity D, satisfies $U\,\partial_x C = D(\partial_{yy}C + \partial_{zz}C)$ once streamwise diffusion is neglected. That is the two-dimensional diffusion equation with x/U in place of time, so

$$C(x,y,z) = \frac{Q}{4\pi Dx}\exp\!\left(-\frac{U(y^2+z^2)}{4Dx}\right), \qquad \sigma(x) = \sqrt{\frac{2Dx}{U}}, \tag{104}$$

and integrating across the plume gives $U\iint C\,\mathrm{d}y\,\mathrm{d}z = Q$ at *every* station — which is the property the boundary condition has to preserve.

Two tests are run, and they are deliberately of different kinds.

Uniform-state invariance. A constant field under a constant wind is an exact fixed point of equilibrium, collision, streaming and (65) alike, so this needs no analytic solution and no judgement about what "close" means. On 24^3 with every face open, after 6×10^4 steps:

	$\max\lvert C - \bar{C}\rvert$	$\max\lvert\Delta C\rvert$ over 2000 steps	$\lvert\bar{C} - 1\rvert$
$u = 0$	0	0	0
$u = 0.05$	3.1×10^{-13}	1.4×10^{-15}	6.8×10^{-3}

The first two columns are the criterion and are round-off. The third is a diagnostic, not a failure: with every face zero-gradient the problem is pure Neumann, so no constant is preferred and the level is not determined. The offset is D3Q7's first-order equilibrium, the $O(u^2)$ defect of §5.5, and it scales accordingly — 5.0×10^{-4}, 1.9×10^{-3}, 6.8×10^{-3}, 2.2×10^{-2} at $u = 0.0125$, 0.025, 0.05, 0.1, i.e. as $u^{1.9}$. It is *not* the initialiser laying down equilibrium at zero velocity, which was the obvious first suspicion: attaching the wind only after the field is laid down and settled reproduces the offset to six significant figures. Any problem with a Dirichlet cell in it, the plume below included, does not see this at all.

The plume. On $128 \times 64 \times 64$ with $U = 0.05$, $D = 0.01$ ($\omega = 1.852$, $\mathrm{Pe} = 640$), a source at $x = 8$ and $Q = 1$ per step, $x = 0$ held at $C = 0$ and the five remaining faces open:

x	σ	$\sqrt{2Dx/U}$	err	$C_{\max}$	$Q/4\pi Dx$	flux/Q
20	2.370	2.191	+8.2%	5.386×10^{-1}	6.632×10^{-1}	0.99962
44	3.901	3.795	+2.8%	2.048×10^{-1}	2.211×10^{-1}	0.99962
68	4.982	4.899	+1.7%	1.266×10^{-1}	1.326×10^{-1}	0.99962
92	5.867	5.797	+1.2%	9.158×10^{-2}	9.474×10^{-2}	0.99962
122	6.813	6.753	+0.9%	6.807×10^{-2}	6.981×10^{-2}	0.99963

The flux column is the boundary-condition test, and it is flat to five decimal places from $x = 20$ to the last interior cell: worst deviation 0.038%. A reflecting exit would pile concentration up and make it climb; an over-draining one would make it fall. Neither happens, and nothing propagates back upstream. This is the same measurement as the aorta's inlet/outlet balance of §15.2, for the same reason.

The width column tests something the uniform-state test cannot see at all, namely $D = c_s^2(1/\omega - 1/2)$ with $c_s^2 = 1/4$: a wrong speed of sound appears immediately as a slope error. The agreement improves monotonically downstream, from +8.2% at $x = 20$ to +0.9% at $x = 122$, which is the signature of the finite source: (104) is a *point* source and the code releases from a

3^3 box, so the two can only agree once the plume is much wider than the box. The residual near 1% is consistent with the $O(1/\mathrm{Pe}) \approx 0.2\%$ error of neglecting streamwise diffusion in (104) together with D3Q7's $O(u^2)$ equilibrium defect.

12.7 Natural convection

Differentially heated square cavity, hot and cold vertical walls, adiabatic horizontal walls, no-slip throughout, against the benchmark of de Vahl Davis [7] at $\mathrm{Pr} = 0.71$. Fixing the characteristic velocity $u_c = \sqrt{g\beta\,\Delta T\,H}$ sets the Mach number, and then $\nu = H u_c \sqrt{\mathrm{Pr}/\mathrm{Ra}}$ follows.

Ra	N	op	Nu	benchmark	dev.	$u_{\max}$	reference
10^3	64	BGK	1.1176	1.118	−0.04%	3.65	3.649
		TRT	1.1179	1.118	−0.01%	3.65	
10^4	64	BGK	2.2431	2.243	+0.00%	16.14	16.178
		TRT	2.2479	2.243	+0.22%	16.14	
10^5	96	BGK	4.5169	4.519	−0.05%	34.47	34.730
		TRT	4.5323	4.519	+0.29%	34.52	

Within 0.3% everywhere. Note that BGK slightly edges TRT here, contrary to the expectation that the magic parameter would help: the wall placement it fixes is a *momentum* boundary, whereas what limits this case is the thermal boundary layer, resolved by the scalar lattice.

12.8 Rayleigh–Bénard convection

A layer of depth H heated from below and cooled from above, periodic horizontally, no-slip and isothermal on both plates. This exercises the scalar Dirichlet condition in the configuration it was designed for, and couples it to the buoyancy force of §7.3.

Wall pairing. The scalar uses anti-bounce-back and the fluid halfway bounce-back, so *both* wall planes sit at $y = 0.5$ and $y = H + 0.5$ and the layer is exactly H lattice units deep. Pairing an on-node condition for one field with a midway condition for the other would place the thermal and viscous boundaries half a spacing apart, which corrupts Ra itself rather than merely adding an error. The box is one critical wavelength wide, $2\pi/k_c = 2.0158\,H$.

Onset. Linear stability of a rigid–rigid layer gives $\mathrm{Ra}_c = 1707.762$ at $k_c H = 3.117$. This is the sharpest single number this configuration can be asked for: the buoyancy coupling, both boundary conditions, the diffusivity calibration and the wall placement all feed into it, and an error in any one moves it. It is measured here by bisecting on the sign of the growth rate of the critical mode.

	bounce-back + anti-bounce-back		regularised + moment	
H	Ra_c	deviation	Ra_c	deviation
16	1764.0	+3.29%	1769.5	+3.61%
24	1730.5	+1.33%	1737.4	+1.73%
32	1718.9	+0.65%	1725.7	+1.05%
48	1712.1	**+0.25%**	1716.8	+0.53%

Both families converge on 1707.762, the midway pair at $O(h^{2.3})$ and the on-node pair at $O(h^{1.6})$.

Measuring the growth rate. The obvious estimator — the total kinetic energy — does not work. Seeded at 10^{-6} it sits near round-off, and the resulting "growth rate" is not even monotonic in Ra, which is impossible for a linear instability; it put Ra_c at 1749 rather than 1719 at $H = 32$. Projecting the vertical velocity onto the critical mode $\sin(kx)\sin(\pi y)$ instead rejects acoustic transients, round-off and the spurious currents of the initial state, and recovers a growth rate that varies smoothly and crosses zero cleanly.

Nusselt number. At $H = 48$, $\mathrm{Pr} = 0.71$. Two independent estimators are reported — the wall gradient and the volume-averaged convective flux $1 + \langle vT\rangle H/(D\,\Delta T)$. They measure different things, so their agreement is a convergence check and not a tautology.

Ra	$\mathrm{Ra}/\mathrm{Ra}_c$	Nu (wall)	Nu (volume)	spread	$\lvert u\rvert_{\max}$
1 500	0.88	1.0000	1.0000	6.0×10^{-13}	0.0000
2 500	1.46	1.4692	1.4690	1.3×10^{-4}	0.0038
5 000	2.93	2.1079	2.1073	3.0×10^{-4}	0.0059
10^4	5.86	2.6490	2.6475	5.7×10^{-4}	0.0073
3×10^4	17.6	3.6144	3.6089	1.5×10^{-3}	0.0090
5×10^4	29.3	4.1523	4.1424	2.4×10^{-3}	0.0096
10^5	58.6	4.9869	4.9659	4.2×10^{-3}	0.0103

Below onset the result is exact: $Nu = 1.0000$ with the two estimators agreeing to 6×10^{-13} and $\lvert u\rvert_{\max}$ identically zero, so the conductive state is a fixed point of the coupled system to round-off. Above onset the estimators agree to better than 0.1% up to $\mathrm{Ra} = 10^4$ and to 0.42% at 10^5; the growth of that spread is the honest indicator that the top of the table is the least settled part. A power-law fit gives $Nu = 0.132\,\mathrm{Ra}^{0.319}$ overall and an exponent of 0.267 for $\mathrm{Ra} \geq 3 \times 10^4$.

What is not claimed. These Nu have *not* been compared against published tables. Values for two-dimensional rolls depend on aspect ratio and Prandtl number, and no reference set has been verified here, so no deviation column is given. What is verified is internal: the subcritical exactness, the agreement of two independent estimators, and Ra_c itself. Two further caveats apply to the highest rows: the box is one critical wavelength wide, which is the right choice for onset but restricts the roll structure at large Ra, and above $\mathrm{Ra} \sim 10^5$ two-dimensional convection may become time-dependent, in which case a single final-state Nu should be replaced by a time average.

12.9 Magnetohydrodynamics: exact solutions

case	order	recovered	rel. error
resistive decay, L_2	2.00	$\eta = 0.050110$	2.2×10^{-3}
shear Alfvén wave, phase speed	2.00	$v_A = 0.050017$	3.4×10^{-4}
shear Alfvén wave, damping	—	$(\nu + \eta)/2$	5.3×10^{-3}

The shear Alfvén wave is an exact solution of the *full nonlinear* equations, not merely the linearised ones: with $\boldsymbol{u} \perp \boldsymbol{b}_0$ and everything depending only on x, $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ vanishes identically while $(\boldsymbol{b}\cdot\nabla)\boldsymbol{b}$ does not, so the Lorentz coupling and the induction equation are both driven and both must be correct. Phase is measured as well as amplitude, because an error in the coupling produces the wrong wave *speed* rather than the wrong amplitude.

On $\nabla\cdot\boldsymbol{b}$. The two wave cases report $|\nabla\cdot\boldsymbol{b}|$ at round-off, and *this is meaningless*: in both, b_x depends only on y and b_y only on x, so the divergence is structurally zero whatever the scheme does. Orszag–Tang is the only case in the suite that exercises the property, and there $\nabla\cdot\boldsymbol{b}$ is *not* preserved to machine precision: measured against the field's own gradient scale $k|\boldsymbol{b}|$ it sits at truncation level and converges away at order 1.65. That is the honest claim, and it is stated here in place of the stronger one.

12.10 Orszag–Tang vortex

The Orszag–Tang vortex [8] has no closed-form solution, so it is not validated against a formula. It is validated against published reference data and against structural properties. Initial conditions on a periodic box of side L:

$$\boldsymbol{u}(\boldsymbol{x},0) = u_0\big[-\sin y,\ \sin x\big], \qquad \boldsymbol{b}(\boldsymbol{x},0) = b_0\big[-\sin y,\ \sin 2x\big]. \tag{105}$$

12.10.1 Comparison with published data

Reproducing the setup of [6]: $L = 2\pi\,\mathrm{m}$, $u_0 = b_0 = 2$, $\rho = 1\,\mathrm{kg\,m^{-3}}$, $N = 1024$, $\Delta t = 5\times10^{-5}\,\mathrm{s}$. The derived lattice parameters reproduce the published values exactly ($u_{0,\mathrm{lat}} = 1.6297\times10^{-2}$, $\mathrm{Ma} = 0.0282$), with $\mathrm{Re} = u_0 N/\nu$ in lattice units and $\mathrm{Pr}_m = 1$. The reference column is the high-resolution pseudo-spectral data quoted there.

	t (s)	spectral	published LB	here, BGK	here, hybrid CM
$j_{\max}$	0.5	18.24	18.24	18.23	18.23
$j_{\max}$	1.0	46.59	46.65	46.55	46.55
$\zeta_{\max}$	0.5	6.758	6.756	6.756	**6.756**
$\zeta_{\max}$	1.0	14.20	14.18	14.182	**14.180**

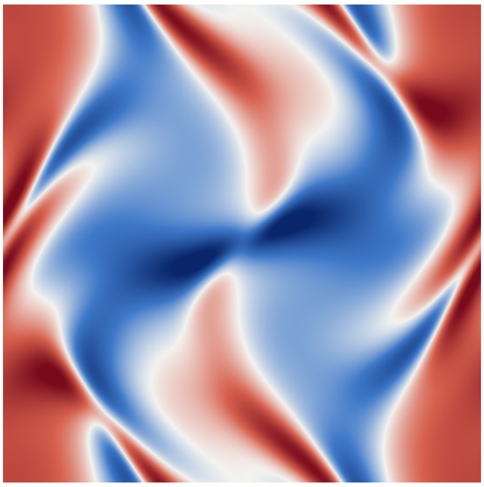
Errors against the spectral reference are 0.07/0.09/0.03/0.14 %, where the published LB errors are 0/0.13/0.03/0.14 %: the same accuracy from an independent implementation, with the hybrid scheme landing on the published vorticity values to every quoted digit. Convergence toward the reference is second order, $j_{\max}$ at $t = 1$ going -1.92% ($N = 256$), -0.46% ($N = 512$), -0.09% ($N = 1024$).

The one residual is $j_{\max}$ at $t = 1$, 0.22% below the published value. That is the quantity in which to expect it: the reference computes the current locally from the distributions whereas this code uses second-order central differences, and j is the more gradient-sensitive of the two peaks. Vorticity agreeing exactly while the current is 0.2% low points at the derivative operator rather than at the scheme.

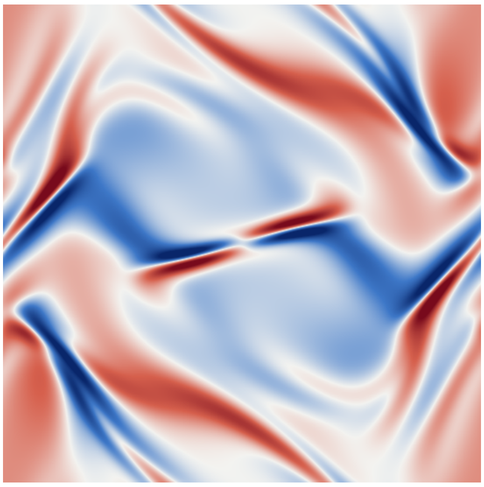
12.10.2 Stability at high Reynolds number

At $\mathrm{Re} = 5000$ the lattice viscosity gives $\tau = 0.510$ on the 1024^2 grid, close enough to 1/2 that the BGK ghost modes go unstable. Runs are terminated at the first non-finite field or at $\mathrm{Ma} > 0.5$.

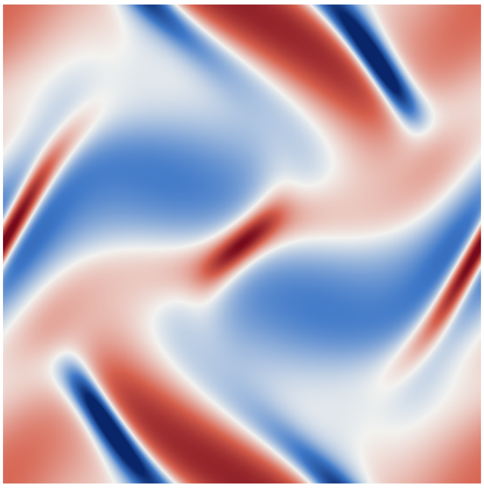
grid	τ	BGK	hybrid CM	published BGK
512^2	0.505	blow-up $t = 0.38\,\mathrm{s}$	stable to $t = 1.00\,\mathrm{s}$	—
1024^2	0.510	blow-up $\boldsymbol{t = 0.54}\,\mathrm{s}$	stable to $\boldsymbol{t = 1.20}\,\mathrm{s}$	$t \approx 0.52\,\mathrm{s}$



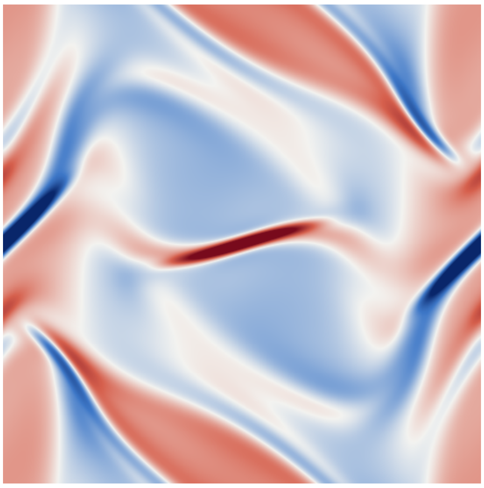
(a) vorticity ζ, $t = 0.5\,\mathrm{s}$



(b) vorticity ζ, $t = 1\,\mathrm{s}$



(c) current j, $t = 0.5\,\mathrm{s}$



(d) current j, $t = 1\,\mathrm{s}$

Figure 2: Orszag–Tang vortex at $\mathrm{Re} \approx 628$, $\mathrm{Pr}_m = 1$, on a 512^2 grid with the hybrid central-moment operator. Diverging colour scale, blue negative, red positive. The scale *saturates at the 99th percentile of* $|v|$ — 6.16 and 11.13 for ζ, 12.88 and 22.67 for j — because the current sheets are strongly localised: at $t = 1$ the peak $|j|$ is 46.4 against a 99th percentile of 22.7, so scaling by the maximum would leave the whole field pale. Between $t = 0.5$ and $t = 1$ the smooth initial vortex steepens into the thin diagonal current sheet visible in panel (d), which is the feature this benchmark exists to resolve and the reason $j_{\max}$ is the more grid-sensitive of the two reported peaks.

The BGK failure reproduces the published behaviour quantitatively — within one probe interval (0.02 s) — and shows the correct resolution dependence. Its signature is instructive: $j_{\max}$ grows smoothly to 24.0 at $t = 0.50$, triples to 73.7 at $t = 0.52$, then overflows, while the Mach number is 0.055 at the last clean sample. This is a collision instability, not a gradual compressibility breakdown.

The hybrid central-moment scheme runs 2.2× longer on the same grid and passes the $t \approx 0.99$ s mark that defeated BGK even on a finer grid in [6]. Its $j_{\max}$ peaks at 166.2 near $t = 1.00$ and then *decays* smoothly to 155.4 by $t = 1.18$, with the Mach number bounded at 0.069 throughout. That turnover is the clearest evidence the run is genuinely resolved rather than merely not-yet-diverged: an unstable run grows monotonically, it does not turn over.

12.10.3 Quasi-two-dimensional reduction: 3D lattices with $n_z = 1$

The Orszag–Tang vortex is a two-dimensional problem, so running it on a 3D lattice with a *single* cell in z is a reduction test: with $n_z = 1$ and periodic z the neighbour wrap sends the out-of-plane neighbour back to the node itself, the corresponding populations stream in place, and the answer must agree with the native 2D lattice. No code change was required — the wrap of §3.3 and the pair swap of Table 6 handle it.

The magnetic field follows the fluid's dimensionality by default (D2Q5 beside D2Q9, D3Q7 beside D3Q27), but since $b_z \equiv 0$ for this flow a 2D magnetic lattice is a legitimate pairing and isolates its contribution. Error against the spectral reference, per cent:

fluid	magnetic	$j_{\max}(0.5)$	$j_{\max}(1)$	$\zeta_{\max}(0.5)$	$\zeta_{\max}(1)$
$N = 256$					
D2Q9	D2Q5	−0.89	−1.92	−0.25	−1.57
D3Q27	**D2Q5**	**−0.89**	**−1.92**	**−0.25**	**−1.57**
D3Q27	D3Q7	−0.77	−1.66	−0.24	−1.51
$N = 512$					
D2Q9	D2Q5	−0.25	−0.46	−0.07	−0.48
D3Q27	D3Q7	−0.22	−0.39	−0.07	−0.46

Two results, one exact and one asymptotic.

D3Q27 reduces to D2Q9 exactly. Not approximately: the agreement in the highlighted row is structural, and both conditions for it can be checked in exact arithmetic. First, summing the D3Q27 weights over z reproduces the D2Q9 weights term for term,

$$\tfrac{8}{27} + 2 \cdot \tfrac{2}{27} = \tfrac{4}{9}, \qquad \tfrac{2}{27} + 2 \cdot \tfrac{1}{54} = \tfrac{1}{9}, \qquad \tfrac{1}{54} + 2 \cdot \tfrac{1}{216} = \tfrac{1}{36}, \tag{106}$$

so for a z-invariant field the projected equilibrium and the projected BGK update are exactly D2Q9's. Second — and this is the part that is not automatic — the out-of-plane Maxwell-stress contribution cancels within every in-plane group:

$$\sum_{c_z} w_i \left(c_z^2 - c_s^2\right) = 0 \qquad \text{for each } (c_x, c_y) \text{ on D3Q27.} \tag{107}$$

The spread converges away. Between $N = 256$ and $N = 512$ the lattice-to-lattice spread falls from 0.12, 0.26 and 0.06 percentage points (on $j_{\max}(0.5)$, $j_{\max}(1)$, $\zeta_{\max}(1)$) to 0.03, 0.07 and 0.02 — factors of 4.0, 3.7 and 3.0, i.e. second order. The lattice pairings are therefore converging to the same answer, and what separates them at finite resolution is truncation error rather than different physics. Of the two choices, the magnetic lattice matters more than the fluid one: swapping D2Q5 for D3Q7 moves $j_{\max}$ by about 0.25 percentage points at $N = 256$, which is unsurprising given $c_s^2 = 1/3$ against $1/4$ and correspondingly different truncation in the induction equation.

12.11 Boundary conditions

Four tests, one per claim. All are second order, which is the accuracy of the scheme itself, so what distinguishes the conditions is *where the wall is*, and that is what each table measures rather than assumes.

Regularised walls: order and wall placement. Poiseuille flow with regularised walls on the boundary rows, $\tau = 0.8$. The wall position is obtained by fitting a parabola to the three interior nodes nearest the wall and taking its root; reading $\boldsymbol{u}$ back at the wall node would test nothing, since the solver reports the imposed value there by construction.

W	L_2 error	order	fitted wall position
8	1.708×10^{-2}	—	2.68×10^{-2}
16	4.419×10^{-3}	1.950	1.33×10^{-2}
32	1.123×10^{-3}	1.976	6.67×10^{-3}

The offset halves with each refinement: first order in lattice units, hence second order relative to the channel width. It is entirely the curvature slip (58): at $\tau = 0.8$, $\frac{4}{3}(\omega - 1)/\omega^2 = 0.2133$ against 0.2128 measured. At $\tau = 1$ the same run is exact to 1×10^{-15}.

These numbers are worse than they were before the force corrections of §8.1 were applied, and that is worth stating rather than hiding: without them the offset at $\tau = 0.8$ was 1.42×10^{-2}, because the force defect and the curvature slip carry opposite signs there and partially cancelled. Two errors cancelling is not accuracy, and removing the one that is understood is the right move even though a headline number rises. The order of convergence is unchanged. D2Q9 and D3Q27 agree to seven significant figures on this flow, the plane channel being quasi-one dimensional so that D3Q27 with periodic span reduces exactly.

Corners: the two closures compared. Lid-driven cavity, 129^2, against Ghia, Ghia and Shin [3].

Re	corner closure	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$
100	local	0.0044	0.0071
100	finite diff.	0.0046	0.0065
1000	local	0.0187	0.0149
1000	finite diff.	**0.0123**	**0.0117**

At $Re = 100$ the two are indistinguishable — four corner nodes out of 16 641 do not move the centrelines. At $Re = 1000$ the finite-difference closure is better at *every* station of the u profile, by roughly a third in both the maximum and the rms. The improvement sits in the near-wall regions and vanishes in the core, which is the expected signature: the corner closure perturbs the boundary layers, and at $Re = 1000$ those are thin enough for the perturbation to reach the centreline. This is why the finite-difference route is the default.

Magnetic walls: Hartmann flow. The MHD analogue of Poiseuille flow, $\mathrm{Ha} = 10$, $\nu = \eta$, against the exact solution [2]

$$b(\xi) = \frac{FL}{B_0}\left[\frac{\sinh(H\xi)}{\sinh H} - \xi\right], \qquad u(\xi) = \frac{FL}{B_0}\sqrt{\frac{\eta}{\nu}}\coth H\left[1 - \frac{\cosh(H\xi)}{\cosh H}\right], \tag{108}$$

with $\xi = x/L$ and $H = B_0 L/\sqrt{\nu\eta}$. Velocity uses regularised walls, $\boldsymbol{b}$ uses the moment condition (64); both put the wall on the node, so they agree about where it is.

n	$\ell_2(u)$	order	$\ell_2(b)$	order	$\lvert u\rvert$ wall	$\lvert b\rvert$ wall
17	3.873×10^{-4}	—	4.117×10^{-4}	—	0	0
33	8.831×10^{-5}	2.133	1.009×10^{-4}	2.029	0	0
65	2.136×10^{-5}	2.047	2.544×10^{-5}	1.988	0	0
129	5.476×10^{-6}	1.964	6.572×10^{-6}	1.953	0	0

Second order in both fields, and both vanish at the walls exactly rather than to $O(h^2)$ — the property the moment condition exists to provide. The computed profile reproduces the flat Hartmann core, the $O(L/H)$ boundary layers, and $b(0) = 1.1 \times 10^{-17}$, antisymmetry at round-off.

Inlet-driven Hartmann flow: the same problem without a body force. The test above drives the channel with a uniform force through a streamwise-periodic domain. Removing the force and driving it from an inlet instead changes what is being tested in three ways, and each is the reason for doing it:

- It is the only way to put `MhdCentralMoments` against an analytic profile at all. That operator hardcodes `NoForcing` and cannot be body-force driven, so the force-driven test covers `MhdBGK` and could never have covered it.
- It is free of the Guo half-force interaction and the curvature slip (58), exactly as the inlet Poiseuille of §13.6 is.
- It is the only test in the suite exercising the fluid and magnetic outflow conditions at once.

Dellar's axes are rotated so the streamwise direction is x: walls on the two y-normal planes, half-width $L = (L_y - 1)/2$, $\xi = (y - L)/L$, and the applied field B_0 wall-normal along y, stretched by the flow into a streamwise b_x. The exact solution is (108) with x and y interchanged. The velocity inlet imposes the analytic u, the outlet is the constant back-pressure condition of §8.2, and $\boldsymbol{b}$ is imposed at the inlet but left free at the outlet — pinning it at both ends would tell the induction equation the answer at both boundaries and make the interior test partly circular.

L_y	rel $L_2(u)$	order	rel $L_2(b)$	order	$\lvert u\rvert$ wall	$\lvert b\rvert$ wall
17	2.281×10^{-2}	—	6.499×10^{-2}	—	0	0
33	5.391×10^{-3}	2.081	1.590×10^{-2}	2.031	0	0
65	1.286×10^{-3}	2.068	3.916×10^{-3}	2.022	0	0
129	3.071×10^{-4}	2.066	9.793×10^{-4}	2.000	0	0

$\mathrm{Ha} = 10$, $\nu = \eta = 0.1$, $u_{\max} = 0.005$, $L_x = 241$, $L_z = 1$. Second order in both fields, and both vanish at the wall node exactly. The Hartmann profile has a flat core with layers of thickness L/Ha, a tenth of the channel here, so this is a considerably harder test of the wall region than the parabolic Poiseuille profile.

Why L_x is so large. Not for physics: the magnetic outflow condition contaminates a long stretch upstream, as §8.4 records. With the $L_x = 21$ that the Poiseuille case uses, the contamination covers the entire domain and the measured order is meaningless. The Mach floor of §13.6 appears here too but is the smaller effect: at $u_{\max} = 0.02$ the last rate in u falls to 1.461, and dropping to $u_{\max} = 0.005$ recovers it.

Scalar walls: placement, measured. Steady one-dimensional conduction between walls at $T = 0$ and $T = 1$. The profile is exactly linear, so the residual is round-off ($\sim 10^{-14}$ throughout) and the only meaningful measurement is where a least-squares line through the *interior* nodes reaches the wall values.

	anti-bounce-back		moment condition	
N	$T = 0$ at	$T = 1$ at	$T = 0$ at	$T = 1$ at
9	0.500000	7.500000	−0.000000	8.000000
17	0.500000	15.500000	−0.000000	16.000000
33	0.500000	31.500000	−0.000000	32.000000
65	0.500000	63.500000	−0.000000	64.000000

Anti-bounce-back places its wall half a spacing outside the node, the moment condition exactly on it, at every resolution and to machine precision.

12.12 Laplace's law

A static droplet is the first two-phase validation and deliberately so: it has an exact answer, it needs no reference table, and the two quantities it measures fail independently. The run below is `validation/laplace`, registered under `ctest`, on a triply periodic 96^2 box with $\sigma = 0.01$, $W = 4$, $\tau = 0.8$, mobility $M = 0.05$ and 10 000 steps, D2Q9 fluid against a D2Q9 phase field.

The flat interface, which must do nothing. A slab has zero curvature, so it is the control: the width must be maintained and no flow may be driven. The conservative Allen–Cahn equation (5) has the tanh profile of prescribed width as its exact one-dimensional equilibrium, and W is recovered from an integral rather than from a fit,

$$\int \phi\,(1-\phi)\,\mathrm{d}y = \frac{W}{4} \quad \text{per interface,}$$

because for $\phi = \frac{1}{2}[1 + \tanh(2y/W)]$ the integrand is $\frac{1}{4}\,\mathrm{sech}^2(2y/W)$, whose integral is $W/4$ exactly. An integral measure uses every node in the profile instead of the two or three near the steepest point, so it does not degrade as W shrinks towards the grid. Measured against a prescribed $W = 4.0000$: width 4.0011, relative error 2.79×10^{-4}; $\max|\boldsymbol{u}| = 2.224 \times 10^{-9}$; drift in $\sum \phi$ of 8.440×10^{-13}. The width is second-order accurate rather than exact — the *continuum* equilibrium is the tanh profile and the lattice converges to it — so round-off is not what is expected here, and is not what the tolerance asks for.

The pressure jump. Across a curved interface $\Delta p = \sigma(D-1)/R$, so in two dimensions $\Delta p = \sigma/R$ and measuring Δp at several radii recovers σ with no free parameter. What makes this a test of the capillary stress rather than of the initial condition is that σ is never handled as such anywhere in the code: it enters only through $\kappa = 3\sigma W/2$, a coefficient in a tensor added to the equilibrium's second moment, so the number coming back out is a statement about the whole chain. Far from the interface $\nabla\phi$ vanishes and the capillary stress with it, leaving the total normal stress at ρc_s^2; the jump is therefore read straight off the zeroth moment, averaged over $r < R - 2W$ against $r > R + 2W$, with the interface region excluded from both.

R	Δp (half)	Δp (final)	$\sigma = \Delta p\, R$	rel. err	$\max|\boldsymbol{u}|$
12.0	7.81376×10^{-4}	7.82300×10^{-4}	9.38760×10^{-3}	6.12×10^{-2}	5.05×10^{-5}
16.0	5.81377×10^{-4}	5.83552×10^{-4}	9.33683×10^{-3}	6.63×10^{-2}	5.33×10^{-5}
20.0	4.61264×10^{-4}	4.65468×10^{-4}	9.30935×10^{-3}	6.91×10^{-2}	5.34×10^{-5}
24.0	3.79953×10^{-4}	3.87025×10^{-4}	9.28860×10^{-3}	7.11×10^{-2}	5.65×10^{-5}

Convergence is shown rather than assumed: Δp is reported at half the run and at the end, and if those two disagreed the droplet would still be equilibrating and the fitted σ would mean nothing whatever it happened to equal. The droplet also has to still be the droplet that was asked for. The area-equivalent radius $\sqrt{\sum \phi/\pi}$ comes back at 12.136, 16.102, 20.082 and 24.068 against nominal 12, 16, 20 and 24, and the plateaus hold at ϕ_{in} between 0.999816 and 0.999931 against $\phi_{\text{out}} \sim 5 \times 10^{-6}$, so a jump measured against the nominal radius is being asked the right question. Fitting the slope of Δp against $1/R$ through the origin over all four radii at once gives $\sigma = 9.350098 \times 10^{-3}$ against a prescribed 1.000000×10^{-2}, a relative error of 6.50×10^{-2}, while the spread of $\Delta p\, R$ across the sweep is 1.06×10^{-2}.

The law is asserted tightly, the constant is not. The recovered σ sits 6–7% low at every radius, and that is a discretisation error rather than a wrong coefficient. The two failures are distinguishable: a broken stress destroys the $1/R$ law, whereas a coarse lattice only shifts the constant in front of it, which is exactly what a 1% spread accompanying a 7% deficit says. Holding the droplet's shape fixed — $R/W = 5$ and R/N both constant, so that only the lattice spacing changes — and refining settles it:

N	R	W	σ	rel. err
96	20	4	9.316×10^{-3}	6.84×10^{-2}
144	30	6	9.689×10^{-3}	3.11×10^{-2}
192	40	8	9.837×10^{-3}	1.63×10^{-2}

with fitted orders 1.94 and 2.24, which is the order the scheme is. This ladder is a separate mode of the executable (`-refine`) and carries its own assertion, that the worst order exceed 1.7.

Refining W at fixed R would not measure an order, and tuning it would be worse than useless. At fixed R two errors of opposite sign compete: under-resolution of the profile, $O((h/W)^2)$ and negative, and the finite thickness of the interface against the curvature, $O(W/R)$ and positive. Measured at $R = 28$, the deficit runs -6.51×10^{-2} at $W = 4$, -2.68×10^{-2} at $W = 6$, -9.7×10^{-3} at $W = 8$ and $+1.00 \times 10^{-2}$ at $W = 12$. It passes through zero near $W = 10$, and a width chosen there would look excellent while measuring nothing but the cancellation.

Three operator forms, and the density ratio. Surface tension reaches the flow by two independent routes, and the case runs both. The stress form (`MultiphaseBGK`) puts the capillary tensor into the equilibrium's second moment and carries density in the zeroth, as every single-phase operator in this code base does; it is matched-density only. The potential form (`MultiphasePotentialBGK`) applies $F_s = \mu_\phi \nabla \phi$ as a body force on a distribution whose zeroth moment is a normalised *pressure*, and it is the one that reaches a ratio. The third form is that same potential physics collided in central moments (`MultiphaseCentralMoments`, selected by `-op cm`), which is what §12.13 runs by default. At a matched density the first two share no code beyond the phase field that feeds them, so their disagreement bounds both.

form	ρ_H/ρ_L	Δp	σ	rel. err	$\max \lvert \boldsymbol{u} \rvert$
stress	1	4.65468×10^{-4}	9.30935×10^{-3}	6.91×10^{-2}	5.34×10^{-5}
potential	1	4.63803×10^{-4}	9.27607×10^{-3}	7.24×10^{-2}	1.67×10^{-5}
potential	10	4.85739×10^{-4}	9.71478×10^{-3}	2.85×10^{-2}	3.21×10^{-6}
potential	100	4.91735×10^{-4}	9.83470×10^{-3}	1.65×10^{-2}	4.90×10^{-7}

The two forms differ by 3.58×10^{-3} at $R = 20$ with matched densities. Surface tension is a property of the interface, so the recovered value must not move with the density ratio; the total spread over a hundredfold ratio is 5.59×10^{-2}, and the table shows where it comes from.

The measurement gets *better* as the ratio rises, from 7.24×10^{-2} low at a matched density to 1.65×10^{-2} at a ratio of 100, so most of the spread is the matched-density row's ordinary resolution deficit rather than any dependence on the ratio itself. The spurious currents fall by more than an order of magnitude across the same sweep.

The pressure column is worth reading against §10.6. Under the stress form the jump rides on a background of ρc_s^2: $p_{\text{in}} = 0.334116$ against $p_{\text{out}} = 0.333333$ at $R = 12$. Under the potential form, seeded at $p = 0$, the far field sits at -6.8×10^{-5} and the interior at $+3.96 \times 10^{-4}$, so the whole of what is stored is the Laplace jump. Reconstructing that pressure is where a mistake was made and then measured: the potential form stores $\tilde{p}$, and $p = \rho(\phi)c_s^2\tilde{p}$ requires the *local* density under the default normalisation. Using a constant instead turned a 1.7% error at a ratio of 100 into 19.5% with the physics untouched — a diagnostic reading the wrong quantity, not a scheme going wrong. The spread above is likewise normalised by the *prescribed* σ rather than by the smallest measured one, because a measured value returning negative would otherwise flip the sign of the quotient and turn a gross failure into an apparent pass; it did, once.

The phase lattice. `-plat` selects a D2Q5 phase field with a first-order equilibrium or the D2Q9 default with the full second-order one. A static droplet cannot tell them apart in σ — 9.738×10^{-3} against 9.722×10^{-3} at $R = 20$, $W = 6$, since $\boldsymbol{u}$ is zero and the equilibrium's $\boldsymbol{u}$-truncation has nothing to act on — but the spurious currents halve between them, 3.63×10^{-5} to 1.64×10^{-5}. The advection accuracy the second-order form buys needs a moving interface to show, which this case does not have.

Acceptance. Eight criteria, all passing on the run tabulated above:

criterion	tolerance	measured
interface width maintained	1%	2.79×10^{-4}
flat interface drives no flow	10^{-6}	2.22×10^{-9}
$\Delta p\, R$ constant across R	2%	1.06×10^{-2}
σ at this resolution	10%	6.50×10^{-2}
spurious currents	10^{-3}	5.65×10^{-5}
ϕ conserved to round-off	10^{-11}	8.72×10^{-13}
stress and potential forms agree	2%	3.58×10^{-3}
σ independent of ρ_H/ρ_L	8%	5.59×10^{-2}

The tolerance on σ is deliberately loose because `-refine` asserts the order that the absolute value at any one resolution cannot.

12.13 Layered Poiseuille flow

`ViscousInterfaceForce` sat in the tree from the moment the pressure form landed, implemented and unexercised: it is identically zero in a static droplet, and the cases that do excite it — Rayleigh–Taylor, water entry — have no exact answer to check against. This is the case that does, and it is registered under `ctest` as `layered_poiseuille`.

Two immiscible layers lie between parallel plates, driven along the channel by a uniform force G per unit volume, fluid 1 (μ_1, ρ_1) below and fluid 2 above, no-slip at $y = \pm h$. The flow is fully developed, $\boldsymbol{u} = u(y)\hat{x}$ with $\partial_x u = 0$, so the nonlinear term vanishes *identically* rather than approximately and the steady profile solves the full Navier–Stokes equations for any pair of densities:

$$u_k(y) = -\frac{Gy^2}{2\mu_k} + a_k y + c, \qquad a_1 = \frac{Gh}{2}\,\frac{\mu_1 - \mu_2}{\mu_1(\mu_1 + \mu_2)}, \qquad a_2 = a_1\frac{\mu_1}{\mu_2}, \qquad c = \frac{Gh^2}{\mu_1 + \mu_2}, \quad (109)$$

from $u(\pm h) = 0$, continuity of u, and continuity of the *shear stress* $\mu\,\mathrm{d}u/\mathrm{d}y$ at the interface. The interface velocity c is the cleanest number in the problem: one closed form, no fitting, and dependent on both viscosities. That the density does not appear anywhere in (109) is the whole point of using it here.

What each ingredient is on the hook for. The interpolation $\mu(\phi)$ owns the kink in $\mathrm{d}u/\mathrm{d}y$: get it wrong and the two parabolas join at the wrong slope ratio. $F_\nu = \nu(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\top)\cdot\nabla\rho$ is proportional to $\nabla\rho$, so it is active here and only here among the cases with exact answers — a viscosity ratio alone leaves it zero. And $\rho(\phi)$ enters through F_ν and through the forcing prefactor but not through the answer, so a density ratio changes what the code must do while leaving what it must produce fixed. That is the kind of contrast that catches a term rather than a constant.

The driving force is applied through the operator's external force slot as a uniform G, not through the body-force acceleration $\boldsymbol{b}$: $F_b = \rho\boldsymbol{b}$ would drive the two layers unequally and give a different, much less clean exact solution. G is chosen so that c hits a fixed target of 0.02, which holds the Mach number as the grid refines — otherwise a finer grid is also a faster flow and the fitted order picks up the Mach error along with the discretisation. The reference fluid is $\mu_2 = 0.1$, $\rho_2 = 1$; the contrasts are a viscosity ratio of 5 and a density ratio of 10, interface width 4, mobility 0.02, on a channel 16 cells long with solid rows top and bottom.

Two things measured before anything is scored. The field is seeded with (109) rather than with rest. Relaxation to steady state is diffusive and takes $O(H^2/\nu)$ steps, and the first version of this case duly reported an unconverged answer as a converged one; starting from the analytic profile leaves only the discretisation difference to relax. This does not beg the question, because the scheme relaxes to *its own* steady state and the error measured is the distance between that and the exact one — and the steadiness detector, a relative change below 10^{-10} over 500 steps, still has to fire. The single-phase control reaches it at 18 500 and 63 000 steps; every configuration with a contrast runs to the 120 000-step cap.

The interface also does not stay exactly where it is placed. Its $\phi = 1/2$ crossing settles a fraction of a cell off centre while the conserved integral of ϕ is unchanged, and that offset does not shrink with resolution. The exact solution is therefore generalised to an arbitrary interface height s, with $h_1 = h + s$ and $h_2 = h - s$,

$$a_1 = \frac{G\,(h_2^2\mu_1 - h_1^2\mu_2)}{2\mu_1(\mu_2 h_1 + \mu_1 h_2)}, \qquad c = \frac{G h_1^2}{2\mu_1} + a_1 h_1, \tag{110}$$

which reduces to (109) at $s = 0$, and every profile is scored against the interface position it actually has. Comparing against a centreline solution would measure where the interface ended up rather than whether the profile around it is right, and those are separate questions deserving separate answers.

The contrasts, one at a time.

configuration	H	$L_2(u)$	order	u_{iface} err	y_{iface}	$\max\lvert F_p\rvert/G$
single phase	32	2.6751×10^{-4}	—	1.13×10^{-3}	0.1025	0
	64	6.6877×10^{-5}	2.00	2.85×10^{-4}	0.0905	0
viscosity only	32	3.6187×10^{-2}	—	5.02×10^{-2}	0.1093	0
	64	2.3587×10^{-2}	0.62	2.60×10^{-2}	0.0893	0
density only	32	2.5078×10^{-2}	—	3.41×10^{-2}	0.1967	2.20×10^{2}
	64	2.8995×10^{-2}	-0.21	3.80×10^{-2}	0.2992	4.08×10^{2}
both	32	4.2967×10^{-2}	—	1.99×10^{-2}	0.1840	7.33×10^{1}
	64	4.2896×10^{-2}	0.00	1.56×10^{-2}	0.2683	1.36×10^{2}

The first pair is the control and says the base scheme is exact and second order on a problem it should solve exactly. The second is the diffuse interface smoothing a kink in $\mathrm{d}u/\mathrm{d}y$ over W cells, which is first order in W/h and converges accordingly; with `-fine` adding $H = 128$ the column reads 3.6×10^{-2}, 2.4×10^{-2}, 1.3×10^{-2} at orders 0.62 and 0.82, against 2.7×10^{-4}, 6.7×10^{-5}, 1.6×10^{-5} at 2.00 and 2.08 for the control. The third pair should not happen at all. The exact profile does not contain ρ, so a density ratio alone must leave the answer unchanged, and instead it leaves a few per cent that refinement does not remove — 2.5×10^{-2}, 2.9×10^{-2}, 2.4×10^{-2} at $H = 32$, 64, 128, with no order to fit. With both contrasts on, that residual dominates and the total stops converging as well.

Splitting the error by distance from the interface separates a term that lives *at* the interface from one that biases the whole profile, and the two need different explanations. For the control at $H = 64$ the split is 3.343×10^{-5} near against 5.792×10^{-5} far. For the density-only case it starts concentrated, 2.386×10^{-2} near against 7.706×10^{-3} far at $H = 32$, and spreads across the channel under refinement, 2.254×10^{-2} against 1.824×10^{-2} at $H = 64$. That is the fingerprint of a force that must cancel globally rather than of a local stencil error.

It is conditioning, and it is reported rather than asserted. The diagnosis — that F_p scales with the pressure *level*, must cancel against $\rho c_s^2 \nabla \tilde{p}$ to a few hundred times the driving force, and so becomes harder to cancel as the grid refines — is given in §10.6, along with the gauge experiment that confirms it. What this case contributes is the measurement. Under the default local-ρ normalisation the density-only residual runs at order -0.21 with $|F_p|/G$ reaching 2.20×10^2 at $H = 32$ and 4.08×10^2 at $H = 64$; under the constant-ρ_0 normalisation the same configuration converges at order 0.95:

normalisation	H	$L_2(u)$	order	$\max \lvert F_p\rvert/G$
local ρ (default)	32	2.5078×10^{-2}	—	2.20×10^2
	64	2.8995×10^{-2}	-0.21	4.08×10^2
constant ρ_0	32	1.8572×10^{-2}	—	3.79×10^2
	64	9.6381×10^{-3}	0.95	7.47×10^2

Note what does not improve: $|F_p|/G$ is *larger* under the constant reference, 3.79×10^2 against 2.20×10^2 at $H = 32$. The gain is not that the term shrinks but that it stops tracking the pressure level. Neither normalisation dominates and neither is asserted here; the default remains local-ρ, and §10.6 says where each belongs.

Does F_ν earn its place? Switching it off with both contrasts on is the measurement that answers this, and it is the reason the case was written.

H	L_2 with F_ν	L_2 without	ratio
32	4.2967×10^{-2}	4.2837×10^{-1}	9.97
64	4.2896×10^{-2}	5.9350×10^{-1}	13.84

An order of magnitude, growing with resolution, on top of a residual that the pressure force already keeps from converging. The form was checked as well as the presence: the LBE with this equilibrium recovers $\nabla\cdot(\nu S)$, not $(1/\rho)\nabla\cdot(\mu S)$, and the difference is exactly $\nu S\cdot\nabla\rho$, so the implemented expression is the right one. Substituting the plausible alternative $(\nabla\mu)\cdot S$ makes the error larger, 1.8×10^{-1} at $H = 32$; that reading was tried and reverted.

One limitation of the term is not tested here. Its stencil reads $\boldsymbol{u}$ at every neighbour including non-fluid ones, where the velocity field holds whatever the solver left there, so an interface touching a wall takes a one-sided derivative as though it were central. In this case the interface sits at mid-channel and never meets the plates, so the defect cannot appear — which means the case establishes the term's form and its necessity, not its behaviour at a wall.

Acceptance. Three assertions, all passing: the single-phase control is second order, 2.00 against a threshold of 1.8; F_ν is necessary, the best ratio being 13.8 against a threshold of 3; and ϕ is conserved to 1.19×10^{-11} against a tolerance of 10^{-9}. The non-convergence at a density ratio is printed with the results and deliberately not asserted, because it is a known property of the formulation rather than a regression to be guarded.

12.14 A floating body: Archimedes and the metacentre

Water entry has no closed form to check against, which is why it is a demonstrator. Hydrostatics of a floating body has two, both sharp, and neither carries a tunable constant; `validation/floating_body` puts the volume penalisation of §10.7 against both.

The first is the draft. A box of height H and density ρ_b floating between fluids of density ρ_w and ρ_a sits at

$$d = H\,\frac{\rho_b - \rho_a}{\rho_w - \rho_a}, \tag{111}$$

which is Archimedes with the air's buoyancy *kept* rather than dropped. At the density ratio of 50 used here the air carries 2 per cent of the answer, so discarding it would be a larger error than anything measured below.

The second is the stability. A box floats upright only if its metacentric height is positive; in two dimensions, with waterplane second moment $B^3/12$ and displaced area Bd,

$$GM = BM - BG = \frac{B^2}{12\,d} - \frac{H-d}{2}, \tag{112}$$

and the *sign* of that number decides whether a heeled box returns to upright or lies down.

Why the second one is the test worth having. Two boxes are run: a raft with $B = 2H$ and a pillar with $B = H/2$. They have the same area, the same density, the same gravity, the same resolution and the same code path, and differ only in which side is longer — $GM = +13.61$ against -6.80 from (112). A computed quantity has to change sign between them, and no tolerance has to be chosen to notice. Almost any error in the coupling — a sign in the righting moment, the first moments $\boldsymbol{S}$ dropped, the inertia taken from the nominal rectangle rather than the smoothed one — would make both boxes do the same thing, and that is precisely what this pair cannot tolerate.

Both cases use $\rho_w/\rho_a = 50$, $\rho_b = 0.5\,\rho_w$, $g = 2 \times 10^{-4}$, $\nu = 0.02$, an interface width of 4 cells and $H = 32$, on 192×128 (raft, upright) and 128×128 (pillar), with the pressure-form central-moment operator and a D2Q9 phase field. Surface tension is set to exactly zero on purpose: both answers are hydrostatic, the Bond number at these proportions is in the thousands, and capillarity would contribute far less than the discretisation while adding its own spurious currents to the measurement. The interface stays sharp regardless — that is the conservative Allen–Cahn operator's work, and it does not go through κ. Run lengths are set from the linearised roll stiffness $(\rho_w - \rho_a)\,g\,Bd\,|GM|$ against the body's own inertia: 2538 steps to the period for the raft, 1796 for the pillar, where the same number is instead the e-folding time of the capsize. Averages are taken over the second half of each run only, because the tank is closed and periodic in x, so the body's own wake wraps round and comes back; a single instantaneous draft carries that wave and the mean does not.

The wall-sided correction, and a discrepancy that was nearly charged to the solver. The righting couple is read at step 0, where the fluid is still at rest and hydrostatic, so there is nothing in it but the integral of $\chi\rho\,\boldsymbol{r}$ and it can be put straight against metacentric theory. The raft is released at 15°, which is not a small angle: as it heels the emerging and immerging wedges stop being mirror images and the arm grows to

$$GZ = \Big(GM + \tfrac{1}{2}BM\tan^2\theta\Big)\sin\theta, \qquad T = -(\rho_w - \rho_a)\,g\,B\,d\,GZ. \tag{113}$$

For the raft the quadratic term is +5.7 per cent of the linear arm, and it holds while the deck stays dry and the bilge stays wet — out to 26° at these proportions. Against the textbook small-angle formula the measured couple was about 7 per cent out, of which (113) accounts for most; the correction was applied only after the discrepancy had been found, and without it the remainder would have been charged to the solver. With it the measured −37.070 stands against −36.628, 1.2 per cent. For the pillar at 5° the quadratic term is negligible and the comparison is 1.4205 against 1.4559, 2.4 per cent; the upright box at zero heel returns -6.65×10^{-16}, which is the couple being zero rather than small.

	measured	expected
draft $/H$ (Archimedes)	0.4991	0.4898
waterline slip on the hull $/d$	0.0175	0
righting couple, raft (wall-sided)	−37.070	−36.628
righting couple, pillar (wall-sided)	+1.4205	+1.4559
raft tilt, rms over second half (deg)	1.96	<7.5
pillar tilt, final (deg)	64.51	>45

The draft and slip are held to 6 per cent, the two couples to 5. The raft, released at 15°, settles to an rms tilt of 1.96°. The pillar, released at 5°, is at 64.5° after three e-folding times and still turning; its resting state is a raft with B and H exchanged, $GM = +6.8$, but reaching that far equilibrium is a much longer run than a regression test should be, so what is asserted is only that it has left upright for good — past 45°, where the rotated raft's righting arm has taken over — and the final angle is printed for whoever wants to look.

The draft is measured twice, and the pair is the interesting part. The geometric route takes the body's centre height against the free surface, the latter read as the water column height $\int\phi\,dy$ far from the body, which absorbs the level rise the body's own displacement causes in a closed tank. The mass route inverts (111) through $m_f = \int\chi\rho$, the fictitious fluid mass the rigid-body solve actually uses for buoyancy. The second is nearly circular — $m_f = m_b$ is the fixed point of the solve, so on its own its $0.4905\,H$, 0.14 per cent off Archimedes, says only that the solve converged. The *difference* is not circular: at $0.4991\,H$ against $0.4905\,H$ it is 1.8 per cent, and it *bounds* the amount by which the phase field inside the penalised region has drifted away from the free surface outside it. That is the one failure mode this coupling has that nothing else in the suite would catch, because there is no contact-line model: the waterline on the hull is advected with the body rather than pinned to the water, and if it slides then buoyancy is being computed for a body that is not the one on screen. It is a bound and not a measurement, because the same gap also contains the smoothed hull edge, the diffuse interface and the column integral of ϕ, and those do not separate here. What can be said is that the gap is small, and that it is the right quantity to watch if a case ever puts a body somewhere the interface has to move along it.

The unit test. `tests/test_body` runs no flow. It pins down the three things a simulation would hide rather than expose, in 29 checks. First, that the indicator turns: χ is evaluated in the body frame, so a rotation applied to the wrong side of the transform gives a shape that turns the wrong way, which in a symmetric case looks like nothing at all. The shape used is deliberately not square — a square would rotate onto itself and every check would pass without the transform doing anything — and its area is rotation invariant to 6.1×10^{-6} at 30° and 2.2×10^{-16} at 90°, matching $2h_x{\cdot}2h_y$ to 4.8×10^{-5}. Second, that the 3×3 sway–heave–roll system, which is solved in closed form through its Schur complement, actually satisfies its own equations: the solution is substituted back and the largest row residual, scaled by the size of the terms in the row, is zero to round-off at m_b/m_f of 5, 1 and 0.1. The middle one is the case the classical Uhlmann arrangement divides by zero on and the last is the one it gets the sign wrong on, and the check that this is nevertheless the *same* equation is that with the first moments zeroed and rotation locked the closed-form reaction still satisfies $(m_b - m_f)\, d\boldsymbol{U} = \boldsymbol{R} + (m_b - m_f)\boldsymbol{g}$ to 6.0×10^{-17} wherever that relation is defined. Third, that the reported reaction is the force that was written: $\boldsymbol{R}$ and the torque are returned in closed form rather than reduced from the force array, which is worth a full sweep a step and is exact only if the closed form is right, so the test sums the array that was actually populated — in a sheared velocity field over a stratified density, so that all seven integrals are non-zero and nothing cancels by symmetry — and finds R_x and R_y agreeing to 7.1×10^{-15} and 1.4×10^{-14} on a reaction of magnitude 17.469, and the torque to 1.4×10^{-14} on 58.544. Alongside these: a body alone with gravity falls at exactly g and does not start spinning, locking rotation leaves no roll, locking translation leaves the roll row standing alone rather than solving the 3×3 and discarding two rows, the out-of-plane force is identically zero, and no force is left behind on ground the body has vacated.

The second moment is larger than the rectangle's, and must be. The indicator's area matches the sharp rectangle to 5×10^{-5}, but its second moment exceeds it by 2.55 per cent at a smoothing width of 1.5 cells, and the test asserts that excess is positive and below 6 per cent rather than asserting it away. The tanh smoothing moves equal amounts of indicator across each face, and the r^2 weight values what went out more highly than what came in, so a larger second moment is the correct answer. This is the whole reason mass and inertia are taken as integrals of χ rather than from textbook formulae: a body given the nominal rectangle's inertia would roll a few per cent too fast.

FP32. Every acceptance row reproduces to the digits printed above except the waterline slip, 0.0176 against 0.0175, and the mass draft's departure from Archimedes, 0.15 per cent against 0.14. The unit test passes all 29 checks; the 3×3 residuals rise from zero to 7.0×10^{-9}, 1.5×10^{-8} and 5.8×10^{-8} at the three mass ratios, against a tolerance of 2×10^{-5}. Everything algebraic in the test is checked relative to the size of the terms in it — the 3×3 carries masses of order 10^5 against accelerations of order 10^{-4}, so an absolute tolerance would be meaningless in FP64 and unsatisfiable in FP32 — and the Schur complement is nowhere near cancelling for these numbers, $|\boldsymbol{S}|^2/A$ being a few per cent of B, so single precision loses only its own epsilon.

What is not tested. The roll *period*, which would need the added inertia of the entrained water and has no closed form; only the sign of the stiffness and the magnitude of the couple at release are asserted. The draft of a body small enough for surface tension to matter, at Bond number of order one, which this formulation cannot produce because it has no contact line and therefore no surface-tension contribution to the draft at all. Ballast, since the body's centre of mass is taken to be the centre of the rectangle — which is what lets its own weight drop out of the roll equation, and is exactly the standard way of making a tall body float upright that this cannot express. And any of it in three dimensions, where the rigid-body solve would be a 6×6 with a rotating inertia tensor and a quaternion; the fluid side would not change.

13 Validation campaign

The tests of §12 were written alongside the features they exercise. This section is different: it works through a published test suite without modification, so that the numbers can be put beside someone else's.

The suite is Sec. III of De Rosis and Coreixas [12], tests III.A to III.E, plus the inlet-driven Poiseuille flow of Sec. III.A of De Rosis [13]. Test III.F is out of scope: the campaign was specified to exclude applied body forces, and III.F has one.

Settings, and where they depart from the papers. Everything runs on the central-moments operator with the highest-order equilibrium each lattice admits (§5.8) — sixth order on D2Q9 and D3Q27. BGK and MRT were deliberately not run. The two-dimensional cases are run on D2Q9 and on D3Q27 with a single periodic point in z; III.E is genuinely three-dimensional and uses D3Q27. Fixed $u_0 = 0.02$ throughout rather than the papers' sweeps, and reduced resolutions where cost demanded: $D = 64$ rather than 128 in III.E, $D_{\mathrm{lb}} = 512$ only in III.D, $N = 129$ rather than 201 in III.C. These are stated per test below because in two cases they change what the number means.

13.1 Taylor–Green vortex

Periodic square of side 2π, N^2 points, $\mathrm{Re} = 1000$, error measured at $t = T$ against the exact decaying solution.

N	8	16	32	64	128	256
D2Q9	1.106×10^{-1}	2.566×10^{-2}	6.259×10^{-3}	1.444×10^{-3}	2.427×10^{-4}	6.010×10^{-5}
D3Q27	1.106×10^{-1}	2.566×10^{-2}	6.259×10^{-3}	1.444×10^{-3}	2.427×10^{-4}	6.010×10^{-5}

Fitted rate 2.186 on both lattices. The paper reports an optimal rate of 2 at its lowest u_0.

A correction to the published initial condition. Reference [12] prints

$$\boldsymbol{u}_0 = u_0[\cos\gamma x \, \sin\gamma y, \; \sin\gamma x \, \cos\gamma y, \; 0],$$

whose divergence is $-2u_0\gamma \sin\gamma x \sin\gamma y \neq 0$. Almost certainly a lost minus sign in typesetting; the standard solenoidal field, with $-\cos\gamma x \sin\gamma y$ in the first component, is used here. The printed density likewise carries a leading factor 3 that would give $\rho \approx 3$, and the standard form is used instead.

13.2 Double shear layer

$L = 256$, $\mathrm{Re} = 30\,000$, $\tau = 0.500512$ — a stability test rather than an accuracy one, run to $t/t_0 = 1$. All three lattices complete without divergence at that τ, and the normalised invariants agree closely:

	E/E_0	Ψ/Ψ_0
D2Q9	0.986436	0.622701
D3Q27	0.986436	0.622701

13.3 Lid-driven cavity

129^2, $u_{\text{lid}} = 0.02$, against Ghia, Ghia and Shin [3]. Reported as the largest absolute deviation over the 17 tabulated stations, in lid-velocity units.

	Re = 100		Re = 400		Re = 1000	
	$\max|\Delta u|$	$\max|\Delta v|$	$\max|\Delta u|$	$\max|\Delta v|$	$\max|\Delta u|$	$\max|\Delta v|$
D2Q9	0.0049	0.0067	0.0069	0.0071*	0.0094	0.0099
D3Q27	0.0049	0.0067	0.0069	0.0071*	0.0094	0.0099

* *One published reference value is anomalous, and it is Ghia's, not a transcription error here.* All three tables in `validation/cavity_cm.cpp` have been checked entry-for-entry against an independent digitisation of [3] and they match. But the v entry at Re = 400, $x = 0.9063$ reads -0.23827, and three things say it cannot be right:

- Ghia's own extremum table gives $\min v = -0.44993$ at $x = 0.8594$ for this Reynolds number. The three tabulated neighbours are then -0.44993 (0.8594), -0.23827 (0.9063), -0.22847 (0.9453): essentially the whole recovery from the minimum happens in one interval, after which the profile is flat.
- The same three stations are smooth at Re = 100 (-0.22445, -0.16914, -0.10313) and at Re = 1000 (-0.42665, -0.51550, -0.39188). Only Re = 400 kinks, and at exactly one point.
- Every other station agrees here to 0.0071 or better; this one disagrees by 0.1415, a factor of twenty. The value computed here, -0.37974, is what a smooth profile through Ghia's own minimum would give.

The entry is left in the table — it is what the paper prints, and quietly substituting a number nobody published would be worse — but it is excluded from the headline score above and reported separately by the run. It could not be checked against the original paper, which is not among the references to hand. An earlier draft of this document blamed the discrepancy on a faulty transcription on this side; that was wrong, and the correction is recorded here rather than silently applied.

13.4 Dipole–wall collision

$D_{\text{lb}} = 512$ (a 1025^2 grid), walls regularised, corners on the finite-difference closure — the only test in the campaign where the corner treatment matters, since III.A, III.B and III.E are periodic. Against Table II of [12], which reproduces the lattice Boltzmann results of Mohammed, Graham and Reis [14] and the spectral reference of Clercx and Bruneau [15]:

		energy E			enstrophy Ψ		
Re	t	here	[14]	[15]	here	[14]	[15]
625	0.25	1.4976	1.494	1.501	466.3	467.2	472.1
	0.5	1.0167	1.010	1.013	387.2	374.0	382.6
	0.75	0.7702	0.765	0.767	251.3	244.8	256.0
2500	0.25	1.8355	1.838	1.848	689.6	705.3	725.6
	0.5	1.5379	1.534	1.540	889.1	898.1	917.6
	0.75	1.3274	1.320	1.325	790.3	790.2	809.9

Energy agrees to between 0.1% and 0.7% everywhere. Enstrophy is the more demanding invariant and the agreement is between 0.01% and 3.5%; in most rows the value computed here falls *between* the published lattice Boltzmann result and the spectral one, and at Re = 2500, $t = 0.75$ it matches the published lattice Boltzmann enstrophy to four significant figures (790.3 against 790.2). D2Q9 and D3Q27 are identical to every digit shown.

An error of ours, and how it was caught. The published initial condition is written in physical units with $|\omega_e| = 299.56$, which makes $E(0) = 2$ and $\Psi(0) = 800$ exactly. It was first scaled so that the *peak* speed was u_0, which is wrong: the Reynolds number is built on the rms velocity, and for this field the peak-to-rms ratio is 11.019. The nominal Reynolds number and the elapsed physical time were both a factor eleven out, the flow dissipated far too slowly, and the discrepancy grew with t — a signature that pointed at the time scale rather than at the wall. Verifying Eq. (32) numerically ($E(0) = 2.0004$, $\Psi(0) = 799.96$, $U_{\mathrm{rms}} = 1.0001$) settled it. The results above are from the corrected normalisation.

13.5 Three-dimensional Taylor–Green vortex

$D = 64$, run to $t^* = tu_0/D = 10$. There is no exact solution; the comparison is between lattices. At the campaign's fixed $u_0 = 0.02$ the Mach number is 0.035 rather than the paper's 0.2–0.6, so a unit of physical time costs up to seventeen times more steps than in the paper.

	Re = 1600	Re = 30 000
D3Q27	completes, $E/E_0 = 0.0145$ at $t^* = 10$	completes, $E/E_0 = 0.0399$ at $t^* = 10$

At Re = 30 000 and this Mach number $\tau = 0.50026$. That case was initially judged infeasible on the available hardware; the judgement was wrong, and D3Q27 runs it to $t^* = 10$ without difficulty.

13.6 Inlet-driven Poiseuille flow

Sec. III.A of [13]. Laminar flow along x; no-slip on the two y-normal planes; the parabolic profile $u_x(0, y) = 4U_0\,(y/L_y)(1 - y/L_y)$ with $U_0 = 0.005$ imposed at the west face; outflow opposite; $L_x = 21$; $\mathrm{Re} = U_0 L_y/\nu = 10$; started from rest and run to steady state. The error is the relative L_2 norm, Eq. (15) there, between the analytic profile — which equals the inlet profile, the flow being fully developed — and a y-aligned cross section at mid-domain. $L_z = 1$ rather than the paper's 21; the flow has no z dependence and z is periodic either way. The outflow is the constant back-pressure condition of §8.2.

This test is worth having precisely because it is driven by the inlet and not by a body force, so it is free of both the Guo half-force interaction and the curvature slip (58) that dominate the forced channel of §12.1.

L_y	41	81	161	321	641	fitted rate
D2Q9	1.160×10^{-3}	3.251×10^{-4}	7.261×10^{-5}	1.558×10^{-5}	6.192×10^{-6}	1.964
D3Q27	1.160×10^{-3}	3.251×10^{-4}	7.261×10^{-5}	1.558×10^{-5}	6.192×10^{-6}	1.964

Against the published 2.063 that is 4.8% low. The shortfall is entirely in the last refinement, where the pairwise rate drops from 2.231 to 1.334, and it is understood — see below. It is worth noting first that the paper states its error at the finest grid, $L_y = 1281$, is of order 10^{-5}; the error here at $L_y = 641$ is 6.2×10^{-6}, already below that.

What limits the order — three falsified hypotheses. The outlet was the obvious suspect, and it is not the cause.

1. *Per-station error.* Breaking the error down by x at $L_y = 641$ makes the outlet look guilty: against $L_y = 161$ the inlet station converges at 1.99 and the bulk minimum at 1.92, both clean second order, while the outlet station converges at 0.70.
2. *Moving the outlet away does not help.* Repeating at $L_x = 41$ and $L_x = 81$ makes the last pairwise rate *worse*, not better: $1.334 \to 0.885 \to 0.007$. At $L_x = 81$ the error stops responding to refinement altogether — 5.974×10^{-6} at $L_y = 321$ against 5.945×10^{-6} at $L_y = 641$.
3. *A second-order outlet does not help.* Replacing the zero-gradient tangential extrapolation by a linear one (§8.2) is worse on the coarse grids and indistinguishable on the finest.

What it actually is: a non-refining, velocity-dependent error. Sweeping U_0 at fixed $\mathrm{Re} = 10$ moves the onset of the degradation, and always toward *coarser* grids as U_0 rises.

U_0	Ma	fitted rate	pairwise rates			
0.0025	0.0043	**2.067**	1.720	2.026	2.205	2.263
0.005	0.0087	1.964	1.868	2.182	2.231	1.334
0.01	0.0173	1.523	2.091	2.179	0.974	0.812
0.02	0.0346	0.844	2.097	0.150	0.644	0.947

At $U_0 = 0.0025$ the velocity-dependent term stays below the discretisation error across the whole range and the fitted rate is 2.067, against the published 2.063 — agreement to 0.2%. The scheme is second order; the shortfall at $U_0 = 0.005$ is an artefact of the test's Mach number, not a defect of the wall or the outlet.

This is not a novel observation. Reference [12] reports the same behaviour on its own Taylor–Green convergence study — "for the lowest value of u_0, an optimal convergence value equal to 2 is found. As u_0 increases, the accuracy and convergence properties of the method deteriorate as well" — and attributes it to the cubic aliasing defect $c_{i\zeta}^3 = c_{i\zeta}$, which prevents the diagonal terms u_ζ^3 from entering the third-order equilibrium moments and can only be removed by explicit correction terms. The paper on this Poiseuille case [13] instead asserts that $U_0 = 0.005$ gives a Mach number "sufficiently low to annihilate detrimental compressibility effects"; at the accuracy reached here that assertion does not hold, which is unsurprising given that the error here at $L_y = 641$ is already below the error there at $L_y = 1281$.

Fitting $\varepsilon = A/L_y^2 + B$ to the two finest grids gives $B < 0$ at $U_0 = 0.0025$ — no floor reached — and $B = 3.1 \times 10^{-6}, 9.2 \times 10^{-6}, 2.1 \times 10^{-5}$ at $U_0 = 0.005, 0.01, 0.02$. That is growth somewhere between linear and quadratic in U_0, but a two-point extraction is far too noisy to pin the exponent and none is claimed.

13.7 Orszag–Tang vortex in three dimensions

Section III.D and Fig. 6 of [13], with the initial condition from Mininni, Pouquet and Montgomery [11], that paper's Ref. [38]. Cubic periodic box of side 2π on M^3 points, $\delta = 2\pi/M$:

$$\boldsymbol{u}(\boldsymbol{x}, 0) = v_0 \left[-2\sin(\delta y),\ 2\sin(\delta x),\ 0\right], \tag{114}$$

$$\boldsymbol{b}(\boldsymbol{x}, 0) = 0.8\, b_0 \left[-2\sin(2\delta y) + \sin(\delta z),\ 2\sin(\delta x) + \sin(\delta z),\ \sin(\delta x) + \sin(\delta y)\right]. \tag{115}$$

Both are solenoidal: each component depends only on coordinates it is not differentiated by, so every term of the divergence vanishes identically. $\mathrm{Re} = 100$ on the lattice definition $\mathrm{Re} = v_0 M/\nu$, $\mathrm{Pr}_m = 1$, D3Q27 for the fluid and D3Q7 for the field, central moments throughout.

Three things the paper leaves open. Stated here because they change the numbers, not buried in the code.

- *A sign.* Equation (28) as printed has $+2\sin(\delta y)$ in the first velocity component; Ref. [11] has $-2\sin y$, which is what is used here. Unlike the two-dimensional Taylor–Green case in the same paper, the divergence does *not* discriminate — both forms are solenoidal — so this rests on the cited source rather than on a consistency check. It is a genuine ambiguity, not a demonstrated typo.
- v_0. The paper says only that "the values of v_0 and b_0 lead to a Mach number, Ma ~ 0.034", and that depends on which speed Ma is built on. Taken here on the peak initial speed $2\sqrt{2}\,v_0$, giving $v_0 = 6.94 \times 10^{-3}$.
- b_0. Not given; set to v_0, the usual equipartition choice.

What can be checked. Figure 6 is a plot against a high-resolution pseudospectral run and those values are not available numerically, so no per-point comparison is possible. What the figure *does* supply is its axes — panel (a) plots the volume-averaged kinetic energy Γ normalised from 1, panel (b) runs from 5 to 25 in $J_{\max}$ — and the text states that "the current shows a maximum at about $t = 1$".

M	$J_{\max}$ peak	at t	$\Gamma(4)$
32	15.71	1.194	0.1240
64	19.19	1.198	0.1356
128	20.43	1.200	0.1374

Every check the figure permits comes out right. The peak time converges to $t = 1.200$ against the stated $t \approx 1$; $J_{\max}$ reaches 20.4 on the finest grid, inside the 5–25 axis; Γ falls from 1 to 0.137, inside the 0–1 axis; and the curves converge monotonically upward with M, the finer grids resolving the current sheet better, which is the trend the figure reports.

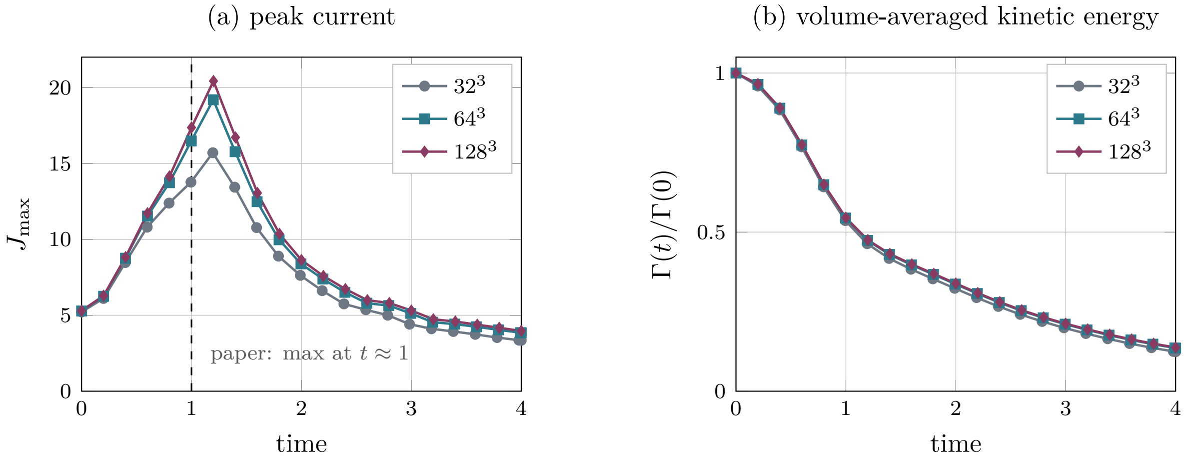


Figure 3: Three-dimensional Orszag–Tang vortex, D3Q27 with central moments, to be read against Fig. 6 of [13]. (a) The peak current magnitude rises with resolution as the current sheet is resolved, and its maximum converges to $t = 1.200$ against the paper's stated $t \approx 1$; the value on the finest grid, 20.4, sits inside that figure's 5–25 axis. (b) The energy, normalised as the paper normalises it, falls from 1 to 0.137. Both panels use a linear time axis; the published figure uses a logarithmic one.

Convergence, and what it does and does not establish. Equation (30) of [13] defines χ as the L_∞ norm of the relative discrepancy against the spectral reference, quoted there as of order 10^{-2} for the finest grid. With no reference values to hand, the same measure is computed against $M = 128$ as a stand-in:

against $M = 128$	Γ	$J_{\max}$
$M = 32$	1.56×10^{-2}	2.31×10^{-1}
$M = 64$	1.9×10^{-3}	6.05×10^{-2}

The successive differences in the $J_{\max}$ peak are 0.171 and 0.061, a ratio of 2.83 — close to second order on a quantity that is a derivative of the field and therefore the hardest thing here to converge. The $M = 64$ figures, 6.1×10^{-2} on $J_{\max}$ and 1.9×10^{-3} on Γ, bracket the 10^{-2} the paper quotes.

This is self-convergence, not agreement. Substituting our own finest grid for the spectral reference flatters systematically, because it cannot see any error common to all three grids. It establishes that the scheme converges and roughly how fast; it does not independently establish that it converges to the right answer. Read the table above as a consistency check, not as a score.

One incidental result worth recording: at $M = 32$ this runs at $\tau = 0.507$, close enough to 1/2 that BGK ghost modes are normally a problem, and it completes the full $t \in [0, 4]$ without difficulty — consistent with the stability advantage [13] claims for central moments.

13.8 A finding that cuts across the campaign

D2Q9 and D3Q27 with one point in z are bit-identical. Not merely close: identical to every digit printed, in the Taylor–Green convergence, the shear layer, the dipole, the cavity and the inlet Poiseuille — five independent tests. The reduction of D3Q27 to D2Q9 under a single periodic z point is exact, which is a useful correctness check in both directions: it exercises the 3D code path against the 2D one on every one of these flows.

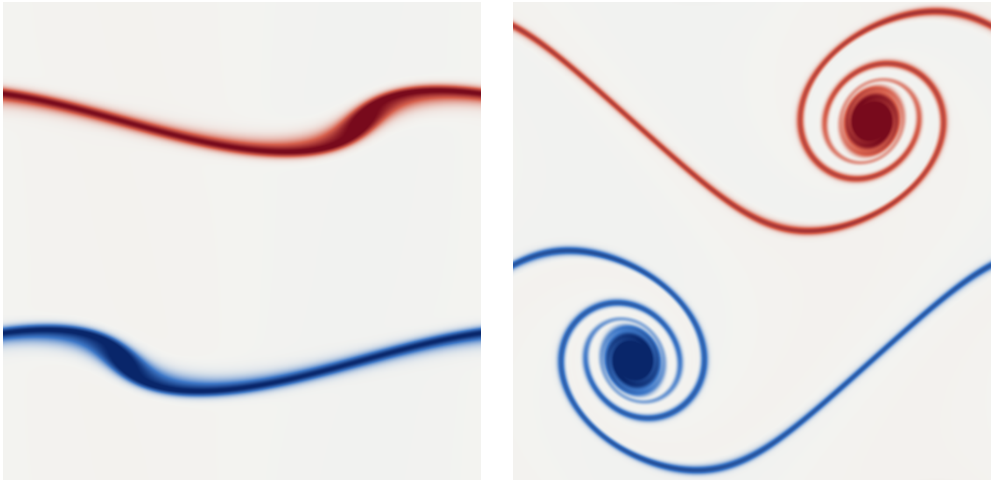
13.9 Field snapshots

The tables elsewhere in this section say whether a number is right. These say whether the *flow* is right, which is a different question and one a norm cannot answer: a scheme can hit an integral quantity while getting the structure wrong. Every panel below is the field the corresponding test actually converged to, dumped from the run that produced the numbers quoted for it.

Signed quantities — vorticity, current — use a diverging map centred on zero; quantities with no meaningful zero — temperature, speed — use a sequential one. Where the colour saturates below the peak, the caption says so: these fields are strongly localised, and scaling by the maximum leaves the structure that matters invisible.

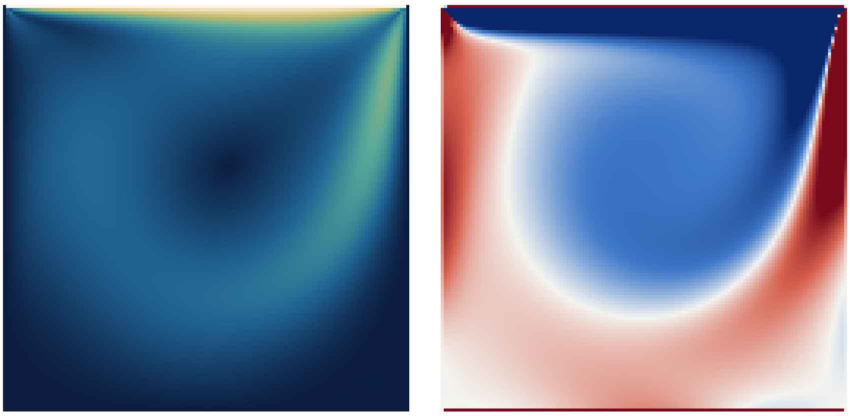
14 Thermal benchmarks of Zhou, De Rosis and Revell (2026)

The suite of §13 is a fluid-only campaign. This section works through a second published suite, this time a thermal one: Sec. 3 of Zhou, De Rosis and Revell [16], five of its seven cases. Sections 3.3 (a side-open cavity packed with conductive blocks) and 3.7 (Rayleigh–Bénard with a melting boundary) are out of scope — the first needs a porous-media drag model and the second an enthalpy formulation for phase change, and neither exists here.



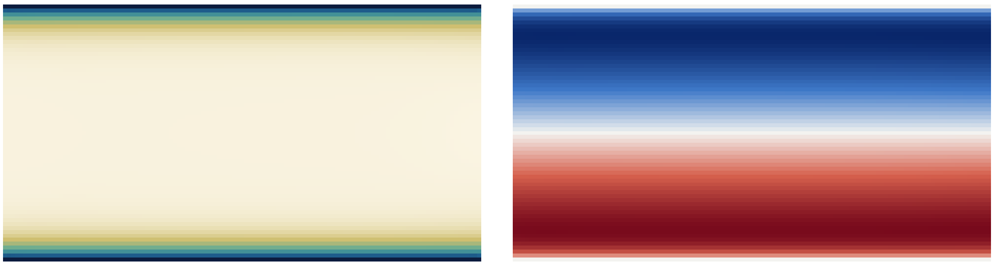
(a) $t/t_0 = 0.5$ (b) $t/t_0 = 1$

Figure 4: Double shear layer, vorticity, $L = 512$, $\mathrm{Re} = 30\,000$, D2Q9 central moments (§13). By (a) the perturbation has begun to distort the two layers; by (b) each has rolled into a vortex, and the two are joined by a single unbroken braid. That the braid stays thin and connected is the whole test: at $\tau = 0.50102$ the flow is barely resolved, and an under-dissipative or unstable scheme shows it here first, by fragmenting the braid into secondary vortices that are numerical rather than physical. Doubling the resolution from the $L = 256$ of the campaign table sharpens the structure and, more usefully, quantifies the numerical dissipation: the enstrophy retained at $t/t_0 = 1$ rises from 0.6227 to 0.6432, and the energy from 0.98644 to 0.98696. The physical dissipation at this Reynolds number is negligible over one turnover, so nearly all of that difference is the scheme's own, and it shrinks with resolution as it should. Colour saturates at the 99th percentile of $|\zeta|$.



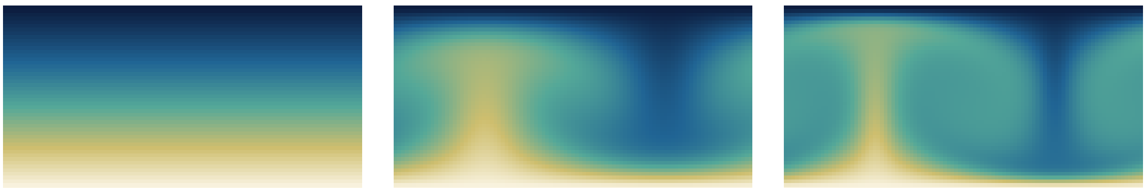
(a) speed $|\boldsymbol{u}|/u_{\mathrm{lid}}$ (b) vorticity

Figure 5: Lid-driven cavity at $\mathrm{Re} = 400$, 129^2, D2Q9 central moments, the run scored against Ghia *et al.* in §13 ($\max|\Delta u| = 0.0069$). The primary vortex sits above and right of centre and both downstream corner eddies are resolved. The vorticity is clipped hard, at the 88th percentile against a peak twenty times larger, because it is concentrated in the lid boundary layer; without that the interior is blank.



(a) velocity u_x (b) induced field b_x

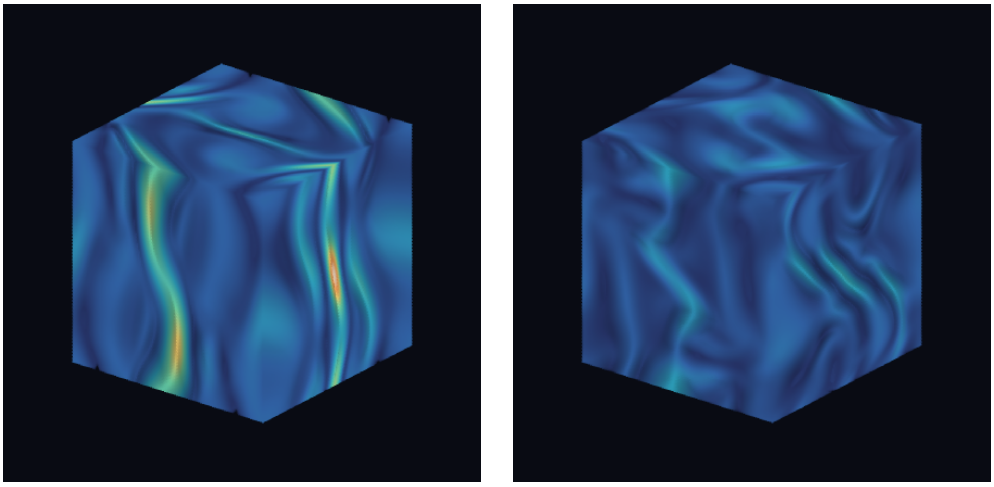
Figure 6: Inlet-driven Hartmann flow, $\mathrm{Ha} = 10$, $L_y = 65$, D2Q9 fluid with D2Q5 induction, central moments (§12.11). The velocity shows the flat Hartmann core with boundary layers of thickness L/Ha; the induced field is antisymmetric about the centreline, vanishing there and at both walls. Both are visibly independent of the streamwise coordinate, which is the property the exact solution has and which the convergence table depends on — any x structure here would be boundary error, and it is how the magnetic outflow defect (§8.4) was found.



(a) $\mathrm{Ra} = 1500$ (b) $\mathrm{Ra} = 10^4$ (c) $\mathrm{Ra} = 10^5$

$Nu = 1.0000$ $Nu = 2.6490$ $Nu = 4.9869$

Figure 7: Rayleigh–Bénard convection, temperature, $H = 48$, one critical wavelength wide, D2Q9 fluid with a D2Q5 scalar (§12.8). The three panels are the same run at three Rayleigh numbers and show what the Nusselt column of that section measures. (a) is *below* the critical $\mathrm{Ra}_c = 1707.762$: the perturbation decays and the field relaxes to pure conduction — a linear vertical gradient, no horizontal structure, and $Nu = 1$ to 6×10^{-13}. (b) is supercritical, and the same perturbation grows into a steady convection cell with a hot plume rising against cold fluid descending. By (c), at $58.6\,\mathrm{Ra}_c$, the thermal boundary layers at the two plates have thinned and the core is nearly isothermal — which is the mechanism behind Nu rising to 4.99, since Nu is precisely the wall gradient normalised by the conductive one. All three share a colour scale.



(a) $t = 1$, peak current

(b) $t = 2$

Figure 8: Three-dimensional Orszag–Tang vortex, current magnitude $|\nabla \times \boldsymbol{b}|$ through the whole 128^3 box, D3Q27 with D3Q7 induction (§13.7). Volume rendered rather than sliced: a cut plane shows the sheets only where they happen to intersect it and says nothing about how they are oriented, which for this flow is the point. The current organises into curved sheets wrapping the box, and (a) is the instant the peak is reached — the same $t \approx 1$ the J_{max} curve of Fig. 3 locates. *Both panels share one colour scale*, set by the $t = 1$ peak, so the visible dimming from (a) to (b) is the real decay: J_{max} falls from 19.1 to 8.32, less than half, in the same units as Fig. 3. Normalising each panel to its own maximum — the default in most rendering tools — would have made the later, weaker field the brighter of the two. The reference is recomputed at this resolution rather than carried over from the 64^3 volumes: the raw lattice values differ between the two by the factor $\Delta t \propto 1/M$ alone, so reusing the coarse-grid number would have rescaled the pair by about two. Orthographic view, azimuth 38°, elevation 24°; emission–absorption compositing with the weak field left transparent, the local gradient used as a surface normal for a diffuse term so the sheets read as surfaces rather than fog, and 2× supersampling. Rendered by `doc/fig/vol3d.py`, which is self-contained.

What can and cannot be reproduced. That paper is about a *coupled* FVM–LBM framework. Its domain is split into an LBM subregion, an FVM subregion and an overlap, and most of its tables measure the coupling: the width of the overlap, the number of LBM sub-iterations per FVM step, the continuity of mass flux across an interface. M3LB has no FVM side, so none of that is testable and none of it is attempted. What is testable is the physics each case was posed to check, against the same external references the paper itself is measured against.

The comparison is therefore three-cornered, and the three corners are not equivalent. Where a table below carries a column for the paper's own coupled result, it is there for context and not as a target: a monolithic solver pays no coupling error, so being closer to the reference than the coupled framework is the expected outcome and says nothing on its own. The reference column is the one that matters.

Discretisation. The fluid is D3Q27 with the central-moments operator, shifted storage, Esoteric Pull and Guo forcing in the Boussinesq form; the temperature is D3Q7 with the BGK scalar operator. The two-dimensional cases are three-dimensional runs one cell deep with periodic z. For D3Q27 that reduction is the familiar one (§12.10.3). For D3Q7 it is worth spelling out, because it is less obvious: with $n_z = 1$ the $\pm z$ populations stream onto their own node, so they never transport, and since $c_z \cdot u = 0$ they relax to $w_z\, \delta T$ exactly as the rest population does. The seven-speed set collapses to a five-speed one with $w_0 = \frac{1}{4} + 2 \cdot \frac{1}{8} = \frac{1}{2}$ and $w_{x,y} = \frac{1}{8}$, whose second moment is still $2 \cdot \frac{1}{8} = \frac{1}{4} = c_s^2$. A legitimate D2Q5, just not the usual one, and with the same diffusivity calibration — which case 3.4 below confirms rather than assumes.

The five cases are taken out of the paper's order and in two groups: 3.1 and 3.4 first, because both have exact solutions and so measure the scheme rather than a comparison, then 3.2, 3.5 and 3.6, where the answer is somebody else's number.

Wall conventions, which are the trap in every coupled case. There are two consistent families, and mixing them puts the viscous and the thermal boundary half a spacing apart and corrupts Ra or Gr without any symptom — the argument of §12.8. *Midway*: halfway bounce-back for the fluid and anti-bounce-back for the scalar, both planes at 0.5 and $N - 1.5$, cavity $N-2$ wide. *On-node*: the regularised velocity condition for the fluid and Dellar's moment condition for the scalar, both planes on nodes 0 and $N - 1$, cavity $N - 1$ wide. Cases 3.2 and 3.6 use the midway family throughout. Case 3.5 has a moving lid, which requires the on-node family for the fluid, so it uses the on-node family for the isothermal walls too; its two adiabatic flanks are the one unavoidable exception, since the scalar module has no on-node zero-flux condition.

14.1 A convergence criterion that was wrong by two orders of magnitude

Every case here declares steady state on the whole-field relative change over a probe interval,

$$\mathrm{res} = \frac{\|q(t+\Delta) - q(t)\|_2}{\|q(t+\Delta)\|_2}, \tag{116}$$

taken over u and T together. A per-step residual measures the step size as much as the drift and can be made small by shrinking the step.

It is not enough, and the shortfall was found rather than anticipated. The residual is not the distance to the fixed point. Near steady state the interval change decays geometrically, $\mathrm{res}_{k+1} \approx q\, \mathrm{res}_k$, so the change still to come is

$$\mathrm{drift} = \mathrm{res}\, q + \mathrm{res}\, q^2 + \cdots = \frac{\mathrm{res}\, q}{1-q}, \tag{117}$$

and for a diffusive relaxation with a probe interval far shorter than the diffusive time, q sits within a percent of unity and drift is a hundred times res. Case 3.1 makes the consequence concrete because its exact answer is a straight line: stopping at res $< 10^{-6}$ left the $\kappa = 1$ profile wrong by $1.9 \times 10^{-4}\%$ and called it converged. Every case below thresholds drift; the same case then returns $1.1 \times 10^{-9}\%$, which is round-off. Both numbers are printed, and $q \geq 1$ falls back to drift = res so that a run still growing can never be declared converged.

14.2 3.1 Conjugate heat conduction: exact

A layer of length $L = 1$, solid over $0 \leq x \leq 0.2$ and fluid beyond it, held at $\theta = 1$ and $\theta = 0$ at the two ends and periodic elsewhere. With ρc_p uniform — which is what the paper's Eq. (36) assumes, writing the balance in terms of α — the conductivity ratio $\kappa = \lambda_s/\lambda_f$ is also the diffusivity ratio and the steady solution is piecewise linear,

$$\theta = \begin{cases} a_s x + 1, & 0 \leq x \leq L_s, \\ \kappa a_s x + b_f, & L_s < x \leq 1, \end{cases} \qquad a_s = \frac{-1}{L_s + (1 - L_s)\kappa}. \tag{118}$$

This is the one case that needed a new capability. `ScalarBGK` carried a single relaxation rate; it now carries an optional field of them (`omega_of`), and `ScalarSolver` asks for the rate per node through `Collision::omega_at(n)`. Nothing imposes the interface condition: continuity of θ and of $\alpha\,\partial_n\theta$ emerges from the streaming, with the effective interface midway between the last node of one material and the first of the other. Whether that is exact or merely second-order was the question.

It is exact. Table 7 reports the relative error of Eq. (38), the two branch slopes fitted by least squares, and the abscissa at which the two fitted lines cross. The slopes reproduce a_s and κa_s to six figures and the crossing lands on 0.2000000 at every ratio from $\kappa = 0.1$ to $\kappa = 100$; driving the tolerance from 10^{-11} to 10^{-13} takes the error from $10^{-9}\%$ to $10^{-10}\%$, so what is being measured is the residual tolerance and not a discretisation floor.

κ	τ_s	τ_f	ϵ [%]	slope$_s$	slope$_f$	x_{int}	Ref. [16] [%]
0.1	0.5632	1.1325	1.5×10^{-9}	-3.571429	-0.357143	0.2000000	1.23
1	0.7000	0.7000	1.1×10^{-9}	-1.000000	-1.000000	—	0.48
10	1.1325	0.5632	2.4×10^{-9}	-0.121951	-1.219512	0.2000000	0.37
100	2.5000	0.5200	1.5×10^{-8}	-0.012469	-1.246883	0.2000000	0.35

Table 7: Conjugate conduction at $L = 200$ cells, drift tolerance 10^{-11}. Exact slopes are $a_s = -1/(0.2 + 0.8\kappa)$ and κa_s; the crossing point is undefined at $\kappa = 1$, where the branches are parallel. The last column is the coupled framework's own error at a finer resolution (1/500).

One choice in the table deserves naming because it is a choice and not a property of the problem. The ratio is imposed as $D_s = D_0\sqrt{\kappa}$, $D_f = D_0/\sqrt{\kappa}$, a geometric split about $D_0 = 0.05$. Putting the whole ratio on one side would either drive one τ to the 1/2 floor — and with it the run length to $L^2/D_{\min}$, a hundred times longer at $\kappa = 100$ — or to tens, where the closures whose error grows as $(1/\omega - 1/2)^2$ stop being accurate.

14.3 3.4 Normal plate velocity: second order, and a hypothesis that died

A uniform stream u_y crosses an isothermal layer, injected at the cold bottom and withdrawn at the hot top, with periodic flanks. The steady solution is the paper's Eq. (44),

$$\theta(Y) = \frac{e^{\mathrm{Pe}\,Y} - 1}{e^{\mathrm{Pe}} - 1}, \qquad \mathrm{Pe} = \mathrm{Re}\,\mathrm{Pr} = \frac{u_y H}{\alpha}, \tag{119}$$

so the three cases the paper labels $\mathrm{Pr} = 0.1, 1, 10$ at $\mathrm{Re} = 10$ are $\mathrm{Pe} = 1, 10, 100$ here: nothing in the scalar module can see Re and Pr apart from their product. Both wall families are run, because anti-bounce-back is derived assuming no normal velocity through the wall and this case has exactly that, whereas the moment condition makes no such assumption.

Pr	Pe	u_y	ϵ anti-b.b. [%]	ϵ moment [%]	Ref. [16] [%]
0.1	1	3.75×10^{-4}	0.00060	0.000036	0.41
1	10	3.75×10^{-3}	0.0233	0.0151	2.00
10	100	3.75×10^{-2}	2.331	1.971	3.65

Table 8: Normal plate velocity at $H = 200$, the paper's resolution, $\tau = 0.8$ fixed so that u_y follows Pe.

The $\mathrm{Pe} = 100$ entry is two per cent, and the obvious suspect was wrong. This equilibrium is only first order in u (§5.5), so a defect growing as u^2 — and therefore as Pe — looked like the explanation. Re-running $\mathrm{Pe} = 100$ at $\tau = 0.8$, 0.65 and 0.575, a factor of four in u_y and in D at fixed u_y/D, moves the error in no printed digit: 2.3313% three times over. At fixed Pe the steady discrete solution is a function of u_y/D alone, so the whole two per cent is grid resolution of the boundary layer — which at $H = 200$ and $\mathrm{Pe} = 100$ is two cells thick. The refinement ladder confirms it:

H	100	200	400	800	1600	order
ϵ anti-b.b. [%]	9.353	2.331	0.5824	0.1456	0.0364	2.00
ϵ moment [%]	11.267	1.971	0.4218	0.0983	0.0238	2.12

Table 9: Grid convergence at $\mathrm{Pe} = 100$. The order is fitted over $H \geq 200$; at $H = 100$ the boundary layer is one cell and neither family is in its asymptotic range.

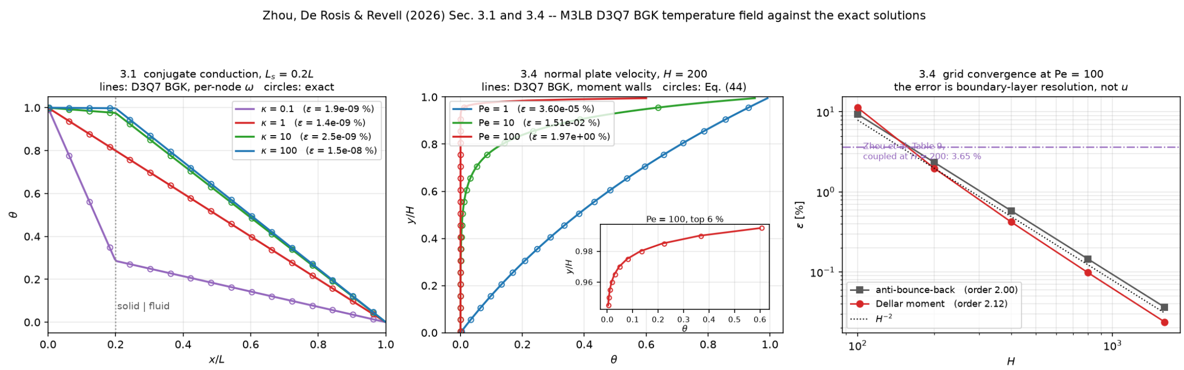


Figure 9: The two cases with exact solutions. Left: conjugate conduction, four conductivity ratios, against the piecewise linear steady state; the kink is the material interface at $x/L = 0.2$. Centre: the normal-plate profile at $\mathrm{Pe} = 1, 10, 100$ against Eq. (44), with the top six per cent inset because at $\mathrm{Pe} = 100$ that is where the entire profile lives. Right: the refinement ladder, second order in both wall families, with the coupled framework's 3.65% at $H = 200$ marked for scale. Rendered by `doc/fig/zhou_analytic.py`.

14.4 3.2 Natural convection in a square cavity

The differentially heated square cavity again, but this time on D3Q27 and D3Q7 rather than the D2Q9 and D2Q5 of §12.7, and at the two Rayleigh numbers the paper runs. The reference is de Vahl Davis [7].

Two remarks on the diagnostics before the numbers, because the first version of this case got one of them wrong.

The hot/cold Nusselt difference is not a convergence check. The configuration is invariant under $(x, y) \mapsto (L - x, L - y)$ with $T \mapsto -T$ and $u \mapsto -u$; the boundary conditions, both lattices, bounce-back and anti-bounce-back all respect that map, and the initial state $T = 0$, $u = 0$ is a fixed point of it. The discrete solution is therefore centro-symmetric at every step, the two wall Nusselt numbers agree identically — the measured imbalance is $10^{-13}\%$, on a run and on a run that has barely begun alike — and their difference says nothing whatever about convergence. It is still worth printing, as a check that the symmetry is intact, but it was originally presented as a conservation test and it is not one.

The independent estimator is the volume one. At steady state the horizontal heat flux through every vertical plane is equal, and averaging that statement over x removes the wall gradient entirely:

$$\mathrm{Nu}_{\mathrm{vol}} = 1 + \frac{\langle u_x \theta \rangle H}{\alpha}, \tag{120}$$

the average taken over the fluid volume, with $\langle u_x \rangle = 0$ in a closed cavity so that the temperature gauge drops out. It shares no arithmetic with the wall stencil. The runs report $\langle u_x \rangle H/\alpha$ alongside it, at 10^{-15} or below, because the identity is worthless if the cavity is not closed to round-off.

Peak velocities are the plain maximum, not the largest magnitude. The two extrema on a centreline are nearly equal and opposite, so an absolute-value search flips sign between resolutions — it did, between $N = 66$ and $N = 98$ — and the reported location then stops matching de Vahl Davis, who tabulates the positive one.

N	H	τ_f	$\mathrm{Nu}_{\mathrm{wall}}$	$\mathrm{Nu}_{\mathrm{vol}}$	spread	dev. from 4.519
50	48	0.5384	4.5147	4.4999	0.330%	−0.10%
66	64	0.5512	4.5177	4.5107	0.156%	−0.03%
98	96	0.5767	4.5196	4.5173	0.051%	+0.01%
130	128	0.6020	4.5203	4.5192	0.023%	+0.03%

Table 10: Natural convection at $\mathrm{Ra} = 10^5$, $\mathrm{Pr} = 0.71$, $u_c = 0.1$. The spread between the two estimators falls by roughly N^{-2} and is the honest discretisation error; the deviation column is measured against de Vahl Davis's Richardson-extrapolated 4.519.

The collision operator, and the paper's Figs. 13–14. That paper's central algorithmic claim is that a central-moments collision keeps driving the residual down where BGK stalls — at 10^{-9} for $\mathrm{Ra} = 10^5$ and 10^{-8} for 10^6 in its Fig. 13. Re-measuring it here needs one thing changed and nothing else, and since the temperature field is D3Q7 BGK in both runs, what is being compared is the fluid operator alone.

The accuracy half of the claim holds and is not marginal: central moments are four to five times closer to the benchmark at both Rayleigh numbers. The convergence half does not reproduce. Both operators drive the residual smoothly to round-off with no plateau anywhere, in step counts that agree to four per cent, and the two histories are indistinguishable on a log scale. The floor in the paper's Fig. 13 is therefore unlikely to be a property of BGK; the natural candidate is the FVM–LBM exchange, which injects a fixed perturbation every FVM step. One caveat on the comparison: that paper's residual is per step and this one is per 2000-step interval, so the two are not the same number — but a per-step residual of 10^{-9} is four to seven orders above round-off however it is normalised, and nothing here sits above round-off at all.

quantity	this work	dVD83 [7]	dev.	Luan [17]	Ref. [16]
$\mathrm{Ra} = 10^5$, $N = 130$					
$\overline{\mathrm{Nu}}$	4.5203	4.519	+0.03%	4.507	4.505
$\mathrm{Nu}_{\max}$	7.7019	7.717	−0.20%	7.738	7.742
$(y/L)_{\max}$	0.0820	0.081	—	0.083	0.082
$\mathrm{Nu}_{\min}$	0.7319	0.729	+0.4%	0.746	0.700
$u^*_{\max}$	34.733	34.73	+0.01%	—	—
$v^*_{\max}$	68.654	68.59	+0.09%	—	—
$\mathrm{Ra} = 10^6$, $N = 98$					
$\overline{\mathrm{Nu}}$	8.8151	8.800	+0.17%	8.807	8.803
$u^*_{\max}$	64.846	64.63	+0.33%	—	—
$v^*_{\max}$	219.65	219.36	+0.13%	—	—

Table 11: Natural convection against de Vahl Davis. The extremum locations are reported as $(y-0.5)/H$, which cannot reach 0 or 1: the nearest sample to a corner is half a cell away, which is why both this work and Luan et al. place $(y/L)_{\min}$ near 0.997 rather than at 1.000.

Ra	op	steps	final residual	$\overline{\mathrm{Nu}}$	dev. from dVD83
10^5	BGK	108 000	1.2×10^{-13}	4.5129	−0.14%
	CM	104 000	1.8×10^{-13}	4.5177	−0.03%
10^6	BGK	228 000	2.1×10^{-13}	8.7894	−0.12%
	CM	228 000	2.8×10^{-13}	8.8022	+0.03%

Table 12: $N = 66$, $u_c = 0.1$, drift tolerance 10^{-13}.

14.5 3.5 Thermal lid-driven cavity, and two conclusions drawn too early

A square cavity with the lid sliding right, the other three walls no-slip, the *bottom* hot and the top — the moving lid — cold, and adiabatic flanks. Three groups are independent: $\mathrm{Pr} = 0.71$, $\mathrm{Gr} = 10^6$ and $\mathrm{Ri} = \mathrm{Gr}/\mathrm{Re}^2$, swept over 10, 1 and 0.1, so $\mathrm{Re} = 316.2$, 1000 and 3162.3. Setting $u_c = \sqrt{g\beta\Delta T L}$ fixes $\nu = u_c L/\sqrt{\mathrm{Gr}}$, and then the lid speed is *not* free: $U = u_c/\sqrt{\mathrm{Ri}}$. That is the whole numerical difficulty of the case. At $\mathrm{Ri} = 0.1$ the lid must run 3.16× the buoyancy scale, so u_c has to come down to keep the lid subsonic, and $\tau = \frac{1}{2} + 3u_c L/1000$ comes down with it: $u_c = 0.03$ caps the Mach number at 0.16 and then $L \geq 128$ is needed to clear $\tau = 0.51$. The two constraints pull opposite ways, and the largest u_c each case tolerates is also the cheapest, because $\langle\theta\rangle$ relaxes on $L^2/(\alpha\,\mathrm{Nu})$ and α grows with u_c — so u_c is chosen per Ri (0.15, 0.08, 0.03) rather than held fixed, with a separate control on that choice reported below.

Three things had to be fixed before this case produced a number worth quoting, and they are worth recording because two of them were invisible in the residual.

A body force along both side walls. The moving lid forces the fluid side into the on-node family, so the isothermal walls use the moment condition. The obvious partner for the adiabatic flanks was `ScalarAdiabatic`, and it is wrong here: bounce-back puts the insulated plane at 0.5, which makes the boundary node a ghost outside the fluid, and `ScalarSolver` says so by writing $T = 0$ there. In cases 3.2 and 3.6 that costs nothing, because those nodes are `Solid` for the fluid and never collide. Here they are `RegWall` nodes — fluid nodes that do collide and do get forced — so `BoussinesqGuo` read $T = 0$ against $T_0 = 0.5$ and applied $-\frac{1}{2}g\beta$ down two whole columns. The gauge is now symmetric about zero, which makes a node reporting $T = 0$ neutrally buoyant instead of maximally cold, and `-flank` selects between bounce-back and `ScalarOutflow`, which is on-node and zero-gradient and reports the real temperature. This is the concrete cost of

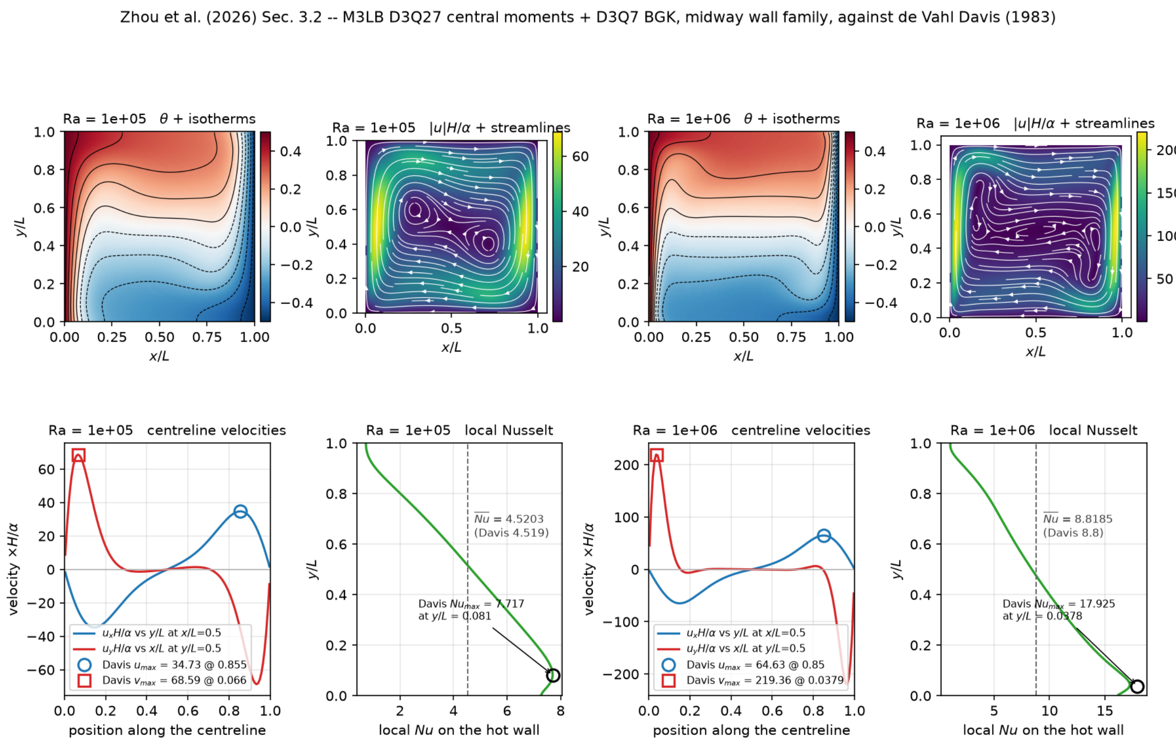


Figure 10: Natural convection in a square cavity, Ra $= 10^5$ and 10^6. Top: the temperature field with isotherms, and the speed with streamlines. Bottom: the two centreline velocity profiles and the local Nusselt number on the hot wall, with de Vahl Davis's tabulated peaks marked — they land on the curves. Rendered by `doc/fig/zhou_natural_convection.py`.

mixing the two wall families, and it is now in `CLAUDE.md`.

The hot/cold Nusselt imbalance is real here, and it was the mean temperature. Unlike case 3.2, the moving lid breaks the centro-symmetry, so the two wall Nusselt numbers are independent and their difference is an honest check. It read -1.5% to -4% and would not go away. It was not a leak at either flank: the domain-mean temperature was still falling, and

$$\frac{\mathrm{d}\langle\theta\rangle}{\mathrm{d}t} = (\mathrm{Nu}_{\mathrm{hot}} - \mathrm{Nu}_{\mathrm{cold}})\,\frac{\alpha}{L^2} \tag{121}$$

predicted -0.128 against a measured imbalance of -0.111 — fifteen per cent, so the imbalance *is* the drift and nothing is being destroyed at a wall.

Which means the convergence criterion had to change. The mean temperature relaxes on $L^2/(\alpha\,\mathrm{Nu}) \sim t_{\mathrm{diff}}/\mathrm{Nu}$, about 10^5 steps at $N = 65$ and growing with N, because α itself grows with N here. Several hundred lid turnovers is nowhere near it: after 190 000 steps $\langle\theta\rangle$ was 0.386 and still moving. And the whole-field residual of §14.1 cannot see the problem: it is dominated by the velocity field, which has already settled, so it floors while $\langle\theta\rangle$ is still walking. So this case declares steady state on the heat budget itself: $|\mathrm{Nu}_{\mathrm{hot}} - \mathrm{Nu}_{\mathrm{cold}}|/\mathrm{Nu}$ below a tolerance for three consecutive probes, and not before twenty probes have elapsed — a state that is still *symmetric* balances the budget trivially, which is how a seed with no asymmetry satisfied the test within a few thousand steps before the flow existed. By Eq. (121) the criterion is exactly the statement that $\langle\theta\rangle$ has stopped moving.

A seed that cost 130 000 steps, and two wrong conclusions. The relaxation above is the expensive part, so `-tseed` replaces the conduction profile with a well-mixed core at a chosen

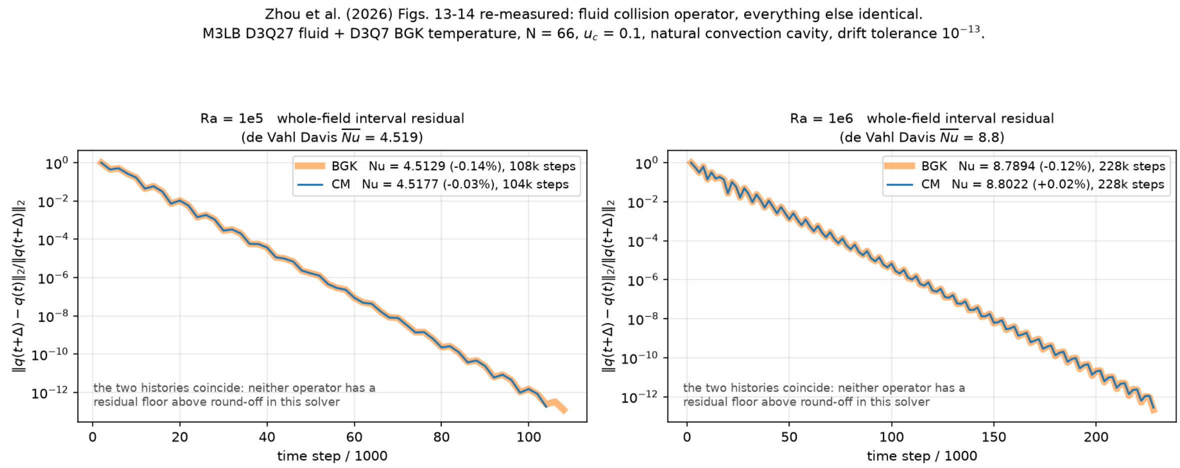


Figure 11: The paper's Figs. 13–14, re-measured. BGK is drawn wide and pale underneath, central moments thin and dark on top, because the two histories coincide — a same-width dashed line simply vanishes under the solid one. Neither operator has a residual floor above round-off in this solver. Rendered by `doc/fig/zhou_operator_residuals.py`.

mean. It cannot change the answer — the fixed point is unique — but the first pass chose that mean badly: 0.38, taken from a coarse $N = 65$ run that was itself still moving. With the core seeded some 0.08 too cold the hot-wall gradient is inflated, and Ri = 10 read Nu $\approx$ 7 for 130 000 steps, looking quasi-steady, before falling back to 4.7 as $\langle\theta\rangle$ climbed to its real value of 0.463.

Two conclusions were drafted from that state and both were wrong: that the case has no steady state, and that the buoyancy-dominated cases disagree with Cheng & Liu by some 44%. Nothing in the field residual flagged either. The imbalance did, at the 2–4% level, which is the whole reason it is the criterion. The runs below are seeded at the measured mean, use a four-fold tighter tolerance, and average over sixty intervals.

Ri	Re	u_c	U	τ_f	Ma	$\langle\theta\rangle$	$\overline{\mathrm{Nu}}$	[18]	[19]	[16]	dev.
10	316.2	0.15	0.0474	0.5576	0.098	0.463	4.689 ± 0.021	4.860	4.848	4.588	−3.5%
1	1000.0	0.08	0.0800	0.5307	0.139	0.373	5.749 ± 0.008	5.750	5.739	5.402	−0.0%
0.1	3162.3	0.03	0.0949	0.5115	0.164	0.248	12.323 ± 0.723	12.161	12.138	12.610	+1.3%

Table 13: Thermal lid-driven cavity, $N = 129$, on-node isothermal walls, bounce-back flanks. $\overline{\mathrm{Nu}}$ is the mean of the two wall values over the last sixty probe intervals and the quoted spread is half their gap; the fluctuation within the window is smaller still, 0.05% to 0.5% of the mean, so these are steady states rather than time averages of a wandering one. Deviations are measured against Cheng & Liu.

The agreement is close at all three Richardson numbers, and the flow structures are the paper's. Its Fig. 23 describes, at Ri = 10, two counter-rotating vortices stacked vertically with the lower one larger and a secondary vortex in the lower-left corner; at Ri = 1 two of comparable strength with the buoyancy-driven one distorted; at Ri = 0.1 a single primary vortex with three small corner vortices. Fig. 12 has all of it, feature by feature.

The Ri = 0.1 error bar is the one exception worth naming. Its two wall Nusselt numbers differ by 12% (13.05 against 11.60) and that gap is *not* a heat-budget violation: Eq. (121) turns it into a mean-temperature drift a hundred times larger than the 5×10^{-4} actually measured over the window. It is a difference between the two wall-gradient stencils in the thinnest thermal layer of the set — Nu = 12 over $L = 128$ puts it at five cells — one of which sits under a sheared lid. The mean is quoted with the gap as the uncertainty.

The u_c control did not do its job, and what it found instead. Re-running $\mathrm{Ri} = 10$ at $u_c = 0.03$ rather than 0.15 was meant to show that the answer does not depend on the compressibility knob. It came back at $\overline{\mathrm{Nu}} = 7.439$ with $\langle\theta\rangle = 0.394$, against 4.689 at 0.463 — a different state, with its own balanced heat budget, reached because that run was seeded near 0.394 and stayed there. So the control is inconclusive on u_c; what it demonstrates instead is that this configuration admits more than one steady state and that the seed selects among them. A second reading of the same thing: with the lid stopped altogether, a seed carrying one half-wavelength across the box settles at $\overline{\mathrm{Nu}} = 5.71$ while one carrying two settles at 3.94, both with a balanced budget and $\langle\theta\rangle$ back at 0.500 as the symmetry of the lid-off problem requires. The $\mathrm{Ri} = 10$ row above is the branch the physical evolution actually reached, having drifted there on its own from a colder seed. Anyone using this case should expect branch sensitivity and say which state they are reporting.

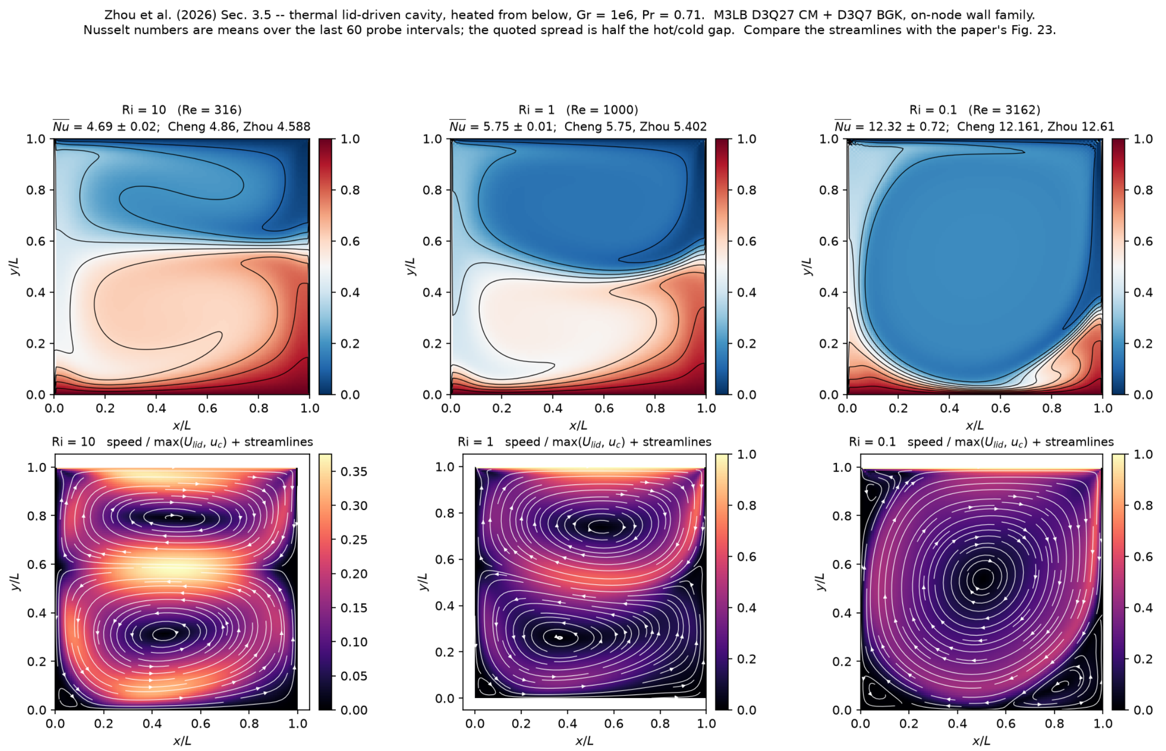


Figure 12: Thermal lid-driven cavity, $\mathrm{Gr} = 10^6$, $\mathrm{Pr} = 0.71$. Top: the temperature field with isotherms at $\Delta\theta = 0.1$. Bottom: the speed with streamlines. Compare with the paper's Fig. 23 — two counter-rotating vortices stacked vertically with the lower one larger and a secondary vortex in the lower-left corner at $\mathrm{Ri} = 10$; two of comparable strength at $\mathrm{Ri} = 1$; a single primary vortex with three corner vortices at $\mathrm{Ri} = 0.1$. Rendered by `doc/fig/zhou_lid_cavity.py`.

14.6 3.6 Natural convection in a cubic cavity

The same problem in three dimensions: hot at $x = 0$, cold at $x = L$, the other four walls adiabatic and no-slip. Cases 3.2 and 3.6 share one routine here, because they are the same problem one cell deep and N cells deep, and running them through the same code is what makes the 2-D and 3-D numbers comparable.

The paper gives no table for this case — it compares centreline profiles on the $z = L/2$ plane against Fusegi et al. [20], plotted as symbols in its Figs. 27 and 29. So the reference numbers below are the peaks *read off those plots*, which makes them good to about five per cent and no better. They are used anyway, for two reasons: they come from the source rather than from a remembered table, and the paper's own normalisation $u_{\mathrm{ref}} = \sqrt{g\beta h \Delta T}$ is exactly the u_c these runs are pinned with, so the comparison needs no conversion.

Ra	N (H)	$\mathrm{Nu}_{\mathrm{wall}}$	$\mathrm{Nu}_{\mathrm{vol}}$	u_x/u_{ref} at y/L	u_y/u_{ref} at x/L
10^5	50 (48)	4.3296	4.3154	0.1409 at 0.844	0.2429 at 0.073
	Fusegi via Figs. 27, 29:			$\sim$0.14 at $\sim$0.85	$\sim$0.22 at $\sim$0.075
10^6	50 (48)	8.5716	8.4499	0.0809 at 0.865	0.2450 at 0.031
	Fusegi via Figs. 27, 29:			$\sim$0.11 at $\sim$0.88	$\sim$0.26 at $\sim$0.055

Table 14: Cubic cavity, $\mathrm{Pr} = 0.71$, $u_c = 0.2$. The hot/cold Nusselt difference is 10^{-14}% here as well — the centro-symmetry of §14.4 survives the third dimension — so $\mathrm{Nu}_{\mathrm{vol}}$ is again the independent number.

At $\mathrm{Ra} = 10^5$ the agreement is as good as reading a plot allows, and the two estimators sit 0.33% apart — the same figure the 2-D case reaches at the same H. Note also that the 3-D Nusselt number, 4.330, is four per cent *below* the 2-D value 4.520 at the same Rayleigh number, which is the expected effect of the two extra no-slip walls.

At $\mathrm{Ra} = 10^6$ the same resolution is not enough and the diagnostics say so without needing the reference: the two estimators separate to 1.44%, and the u_x peak — which sits at $y/L = 0.865$, inside a boundary layer roughly $H/2\mathrm{Nu} \approx 2.8$ cells thick — comes in 26% low. This is a resolution shortfall, not a model failure, and it is stated rather than presented as a result: $H = 48$ at $\mathrm{Ra} = 10^6$ in three dimensions is under-resolved, and the resolution that would fix it is beyond what this machine can reach in a session.

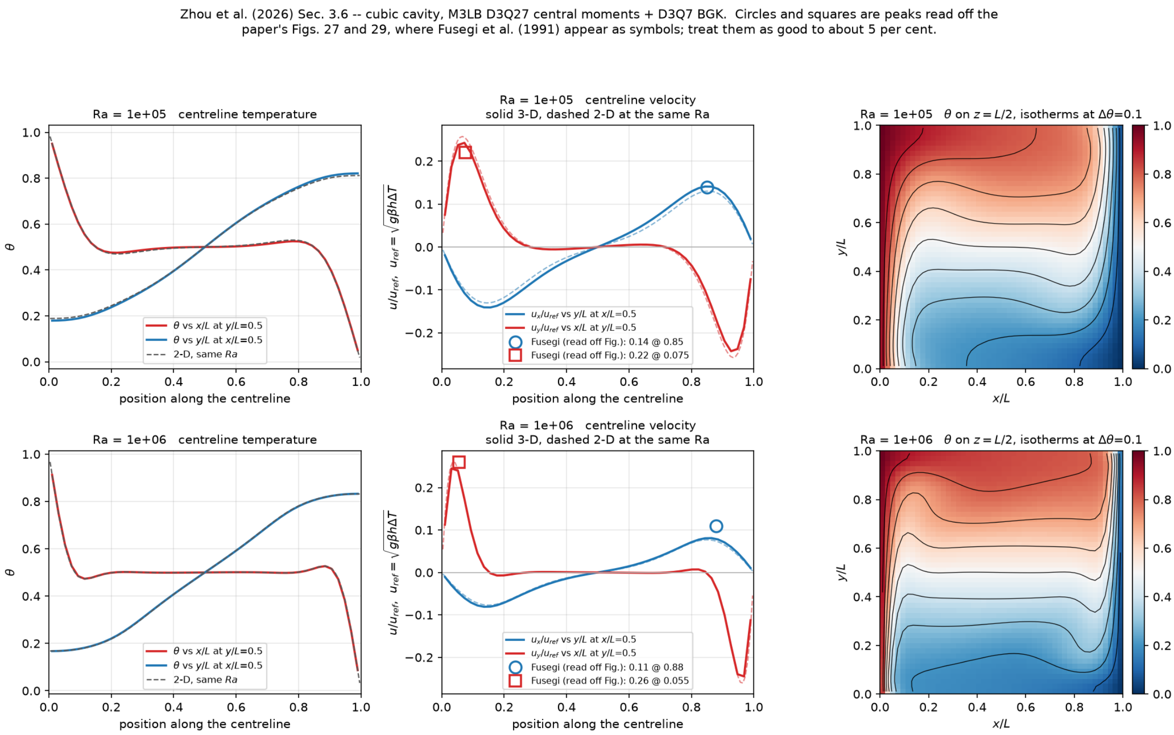


Figure 13: Cubic cavity, the same quantities the paper's Figs. 26–29 plot. Velocities are on the paper's own axis, u/u_{ref} with $u_{\mathrm{ref}} = \sqrt{g\beta h\Delta T}$, which is exactly the u_c these runs are pinned with; the circles and squares are the Fusegi peaks read off those figures and are good to about five per cent. The dashed curves are the 2-D result at the same Rayleigh number. At $\mathrm{Ra} = 10^6$ the u_x peak sitting well below its marker is the resolution shortfall discussed in the text, visible without needing the reference. Rendered by `doc/fig/zhou_natural3d.py`.

15 Demonstrator: flow in a patient-specific aorta

Everything else in this document runs on a geometry described by a formula. This one loads a voxel array off disk. It is included as a *demonstrator*, not as a validation case, and the distinction is worth stating precisely because the figures are seductive: there is no analytic solution here, no reference data is compared against, and **no number in this section is a claim of accuracy**. What it demonstrates is that the solver runs unmodified on a geometry with no symmetry and no closed form — `set_geometry` already takes an arbitrary predicate, so real geometry needed a reader, not a new solver.

15.1 Geometry and its verification

`src/io/VoxelGeometry.hpp` reads the flat binary produced by voxelising a SimVascular surface mesh (case `0074_H_AO_H`): $109 \times 184 \times 361 = 7\,240\,216$ voxels at a pitch of 0.0616 cm, tagged 0 solid, 1 fluid, 2 inlet, 3 outlet. Solid is 83.65% of the box; the lumen is 16.31%. The optional second inlet normal is detected by *file size*, not by stream state — guessing from the stream would silently read 24 bytes of the tag array as doubles on a file that does not carry one.

Three checks were run on the geometry before any flow was believed.

1. **Connectivity.** A six-connected flood fill seeded from an inlet voxel reaches 1 184 057 of 1 184 057 non-solid voxels: one lumen, no isolated pockets. This check exists because a single constant-x plane through this vessel cuts it into *two* disconnected regions — the aortic root and the arch — which looks exactly like a hole in the mesh and is not one. The projected silhouette is a single connected region, as the flood fill requires it to be.
2. **Caps.** Under six-connectivity the inlet fragments into 303 pieces and the outlets into 270. This is not damage: an oblique plane on a voxel grid is diagonally connected, not face-connected. By centroid they are one inlet and four outlets — three arch branches and the descending aorta.
3. **Domain edges.** 1283 fluid voxels sit on the box boundary carrying no cap tag. Were the domain periodic these would wrap, connecting the descending aorta to the arch and corrupting the field silently. It is not: the case constructs `Domain(nx,ny,nz,false,false,false)`, and independently there are *zero* fluid-to-fluid wrap pairs on any axis. The mesh is clipped flush against the box there and those voxels act as wall.

15.2 Boundary conditions

The vessel wall is halfway bounce-back on solid voxels, which under Esoteric Pull costs nothing: bounce-back is the identity on the storage, so solid cells are skipped outright — no load, no store, no arithmetic (§3.2). The population a fluid node emits toward the wall sits untouched for one step and is read back reversed on the next, placing the no-slip surface midway between the last fluid node and the first solid node. Halo cells are flagged `Excluded` and skipped by the same mechanism, so the box faces behave identically.

The inlet is a regularised velocity wall along the cap normal stored in the file. The cap is oblique to the voxel axes, so it has no single face normal; it is given `NrmCorner`, whose unknown set is built geometrically per node rather than from an axis. Using a face normal instead would drive flow into the wall.

The outlets are left at fixed pressure rather than a prescribed flow split, so the division of flow between the branch vessels is an *output* rather than an input. Imposing $\boldsymbol{u} = 0$ there would close the domain — measured as mass rising monotonically, $+8.5 \times 10^{-3}$ in 2000 steps.

Conserving outflow. The outlet overwrites its nodes every step, so whatever density is imposed is injected regardless of what arrived: pinning $\rho = 1$ makes the boundary a mass source and sink, and inlet and outlet fluxes then differ by $\sim$60% with no way to tell which is wrong. Closing a loop on *total mass* instead makes it conserving — raise the outlet density when mass is lost, lower it when mass accumulates. The fixed point is the density at which what leaves equals what enters, and it is found rather than assumed. Two details are load-bearing. The error must be normalised by total mass, not by outlet-node count: normalising by the latter gives an error of order 3, which saturates the clamp, reverses the flow and diverges the run in 1500 steps. And the controller is held off until the vessel has filled, since during filling mass *should* be changing and the loop otherwise fights the physics.

Note that the finite-difference corner-stress path is disabled for this case. It walks a two-node stencil off each wall node; in a box those neighbours are fluid or wall, but in a vessel they are frequently *solid*, whose populations Esoteric Pull never updates, so the stencil reads stale memory and feeds garbage stress into the inlet. The local closure is used instead until that path is made geometry-aware.

15.3 Steady inflow

$\mathrm{Re} = 50$ on the inlet cap's equivalent diameter $D = 2\sqrt{A/\pi} = 32.9$ lattice units, $U = 0.02$, $\nu = 1.3167 \times 10^{-2}$, $\tau = 0.5395$, D3Q27 central moments. After 13 500 steps the run is steady: $|\boldsymbol{u}|_{\max}$ flat at 7.2×10^{-2}, outlet density settled at 0.9934, and the mass error oscillating about zero at $\sim 1 \times 10^{-5}$ — which is the conserving outflow working.

The inlet flux is 12.98 in lattice units, and this is worth one sentence because measuring it is less obvious than it looks. Summing over *every* exposed face of the cap gives $|\sum \boldsymbol{n}_{\text{face}}| = 653.2$ pointing within 0.65° of the normal stored in the file, so $UA\cos\theta = 0.02 \times 653.2 \times 0.9935 = 12.98$. Filtering the faces by direction to exclude the vessel wall was tried and is *wrong* — it double-weights by the projection and drove the figure down to 10.38. The lateral wall faces already cancel in the vector sum; that is what makes the identity $\sum \boldsymbol{n}_{\text{face}} = A\boldsymbol{n}$ hold.

The outlet flux is **not** a valid measure for this boundary and is not used as one: the condition overwrites populations, so a flux summed through it reports the imposed state rather than what the flow delivered. Mass conservation is the trustworthy check.

15.4 Pulsatile inflow

The inlet is driven by the waveform used in the source project, reproduced unchanged so the two are comparable:

$$\frac{U(t)}{U_0} = \min\left(1, \frac{t}{t_{\text{ramp}}}\right)\Big[0.15 + 0.85\, w(\phi)^{3/2}\Big], \qquad w(\phi) = \tfrac{1}{2} + \tfrac{1}{2}\sin\Big(2\pi\phi - \tfrac{\pi}{2}\Big), \tag{122}$$

with $\phi = (t \bmod T)/T$: a ramp from rest, then a cycle between a diastolic floor at 15% of peak and a systolic peak, the upstroke sharpened by the 3/2 power. $\phi = 0$ is the diastolic minimum and the systolic peak falls at $\phi = 0.5$. Here $T = 2000$, $t_{\text{ramp}} = 800$, and U_0 is the systolic peak.

Since the direction is fixed by the cap normal and only the magnitude varies, the whole drive is a scale on the deduplicated wall-velocity table (`set_wall_velocity_scale`), not a per-node update, and costs nothing per step. The scale multiplies a saved copy of the original table rather than the live one: scaling in place would compound, and a waveform bounded in $[0, 1]$ would become a product of its own history and decay to zero within a few hundred steps — which would look like a plausible physical relaxation and is not one.

The parameter that characterises a pulsatile flow is not Re but the Womersley number, which fixes how far the oscillating shear layer penetrates in a beat:

$$\alpha = \frac{D}{2}\sqrt{\frac{2\pi}{T\nu}} = 8.04, \tag{123}$$

in the physiological range for an aorta. The drive is exact beat after beat: Q_{in} reaches -12.98043 at every systolic peak and -1.947065 at every trough, identical to five digits across six cycles.

Convergence. The viscous diffusion time across the vessel is $R^2/\nu = 20\,552$ steps, which is 10.3 beats. This run is 60 000 steps, 2.92 diffusion times, and it is converged: sampled at matched phase over the final three beats, *every* statistic of the speed field repeats to eight significant figures.

ϕ	step	mean	median	p_{99}	$\lvert\boldsymbol{u}\rvert_{\max}$
0.0	54000	0.00364663	0.00238384	0.0153191	0.0565515
0.0	56000	0.00364663	0.00238384	0.0153191	0.0565515
0.0	58000	0.00364663	0.00238384	0.0153191	0.0565515
0.5	55000	0.00483408	0.00495437	0.0104059	0.0320153
0.5	57000	0.00483408	0.00495437	0.0104059	0.0320153
0.5	59000	0.00483408	0.00495437	0.0104059	0.0320153

The residual differences are in the eighth digit, which is round-off. The outlet density has stopped moving at 0.99699, so the mass controller has found its fixed point rather than still walking toward one. An earlier 12 800-step run — 0.62 diffusion times — was reported in a previous draft of this section as *not* converged, with mean and median still rising monotonically through six beats. That was accurate, and it is what a run of that length looks like; it is superseded rather than corrected.

Phase relationships, and a storage effect that is not Womersley. Over a converged beat the *mean* speed peaks at $\phi = 0.60$ against a drive peaking at $\phi = 0.50$: a lag of 0.10 of a cycle, 36°, the expected sense and order for $\alpha = 8$.

The fast tail does the opposite. Both p_{99} and $\lvert\boldsymbol{u}\rvert_{\max}$ peak at $\phi = 0$ and reach their *minimum* at systole — nearly antiphase with the drive. This is not a transient: it repeats identically across the final three beats, and it is visible in Fig. 15, where at the diastolic trough the root has gone pale while the descending aorta carries the fastest flow in the vessel.

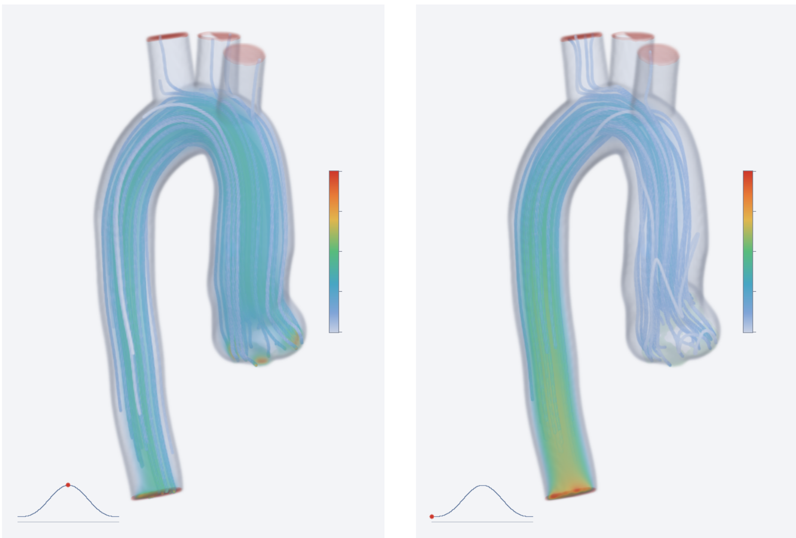
The mechanism is storage, not phase lag. Total mass is not constant over a beat: it swings by $\pm 2.7 \times 10^{-3}$, with its minimum at $\phi = 0.25$ and its maximum at $\phi = 0.70$, accumulating through systole and discharging through diastole. The fast tail peaks where that discharge is steepest. A transit-time explanation was considered and does not survive arithmetic — at these speeds a fluid particle needs several beats to cross the vessel.

What this data does *not* settle is where the storage lives. Two candidates produce the same signature and are not distinguished by the dumps taken here, which carry velocity but not density: the lattice fluid's own $O(\mathrm{Ma}^2)$ compressibility acting as a compliance chamber, or the fixed-pressure outlet acting as a reservoir. Either way the effect is numerical rather than physiological — the wall is rigid (§15, and the Known Limitations) — so the antiphase should not be quoted as a Womersley result, and the fast tail should not be read as a haemodynamic observation. $\lvert\boldsymbol{u}\rvert_{\max}$ in particular is a single staircase-corner voxel and should not be used at all.

16 Demonstrator: urban pollutant dispersion

The second demonstrator, and the second geometry source. A passive scalar is released into a city and transported by an atmospheric wind over building geometry read from a height field (§16.1). Like the aorta, this is a demonstrator and not a validation case: there is no analytic solution for a plume in a real city and no measurement to compare against, so nothing here can fail except by crashing. What *can* be checked is reported as a number, and one of those numbers turned out to matter.

Figure 14: Steady inflow at Re $= 50$, speed, after 13 500 steps. Volume rendering: a pale translucent shell wherever the lumen mask has a gradient, diffuse-shaded off that gradient, with speed-coloured interior showing through. Inlet cap green at the aortic root, outlet caps red. A single plane through this vessel cuts it into disconnected islands, which is why the whole volume is cast rather than sliced.



(a) systolic peak, $\phi = 0.5$ (b) diastolic trough, $\phi = 0$

Figure 15: Pulsatile inflow, converged, $\alpha = 8.04$, shared colour scale, with streamlines traced from the inlet cap through the current field. The inset traces (122) with a marker at the current phase. At systole the root drives a visible jet; at the diastolic trough it has gone pale while the descending aorta carries the fastest flow in the vessel — the antiphase fast tail of §15.4, which is a storage-and-release effect and not a Womersley lag. Both panels are from the same beat, at 2.9 viscous diffusion times, where every speed statistic repeats to eight significant figures from beat to beat.

16.1 Geometry from a height field

Urban geometry does not arrive as a voxelised surface mesh. It arrives as building footprints with heights, so the reader for it is a different one from the aorta's (§15.1): a two-dimensional map of building height in metres, plus a small JSON of grid metadata, with a cell solid when its centre lies below the local height. That keeps $O(n_x n_y)$ on disk instead of $O(n_x n_y n_z)$ — for Manchester, 640 kB rather than the 9.6 million cells it expands to.

Two decisions in that reader are about failure modes rather than features, and both are there because *none* of the ways it can go wrong produce an error. A transposed index order rotates the city; a mis-parsed cell size rescales it; an off-by-one in the solid rule shaves a layer off every roof. All three run to completion and produce a plausible plume. So a missing key in the metadata is an error rather than a default — falling back to a built-in grid size turns a typo into a silently wrong domain — and a vertical spacing that disagrees with the horizontal one is rejected rather than silently stretching the city. The regression test compares against independently known values throughout, using a deliberately asymmetric height pattern so that a transpose changes what is read back.

On the real data the reader is checked against an independent implementation: Manchester city centre, $400 \times 400 \times 60$ at 5 m, 171 375 solid cells of 9.6 million (1.79%), tallest column 200.5 m — matching cell for cell.

16.2 The model, and its principal limitation

The wind is prescribed: a neutral-stability logarithmic surface layer $u(z) = (u_*/\kappa)\ln((z+z_0)/z_0)$, horizontal, along a fixed compass bearing, simply zeroed inside buildings. It has no street-canyon recirculation, no building wake and no channelling along avenues — which are precisely the phenomena an urban dispersion study is about. The scalar side is the validated one: D3Q7 transport, adiabatic buildings and ground, and the open boundaries of §8.5.

The time step follows from capping the fastest cell at a chosen lattice velocity, and the relaxation rate then follows from the physical diffusivity. Deriving Δt from the wind and ω from Δt, rather than choosing τ and letting Δt fall out, is what keeps the advective Courant number bounded — the constraint that actually binds in an advection-dominated problem.

How not to measure the limitation. A log profile zeroed inside buildings is not divergence-free, and it is tempting to look for the damage in a mass budget. That budget cannot see it. Bounce-back is conservative and $\sum_i w_i \boldsymbol{c}_i = 0$, so the advective term contributes nothing to $\sum_i h_i$: the budget closes to round-off however wrong the wind is. The error appears in the concentration instead, because $\nabla\cdot(\boldsymbol{u}C) = \boldsymbol{u}\cdot\nabla C + C\,\nabla\cdot\boldsymbol{u}$ and the second term is spurious — it concentrates the scalar where $\nabla\cdot\boldsymbol{u} < 0$ and dilutes it where the divergence is positive. So $\nabla\cdot\boldsymbol{u}$ is the diagnostic, and it is reported separately over the open cells and over the cells adjacent to a wall, where the zeroing bites hardest. On Manchester city centre, $400 \times 400 \times 60$ at 5 m:

$$\text{RMS}(\nabla\boldsymbol{u}) = 9.4\times10^{-4} \text{ over } 9.06\times10^{6} \text{ open cells,} \qquad 8.4\times10^{-3} \text{ over } 1.14\times10^{5} \text{ wall-adjacent cells,}$$

a factor of nine worse next to a building. Those two numbers are the quantitative case for replacing this wind with a solved one, and they are the reason the original project wrote a Navier–Stokes solver as its second stage.

16.3 Agreement with the original implementation, and one disagreement

The same case run through the original hand-written solver and through this one agrees exactly where it should. On a synthetic $60 \times 30 \times 20$ domain both give $\Delta t = 0.0129$ s, $\tau = 0.5646$, and identical divergence statistics — RMS 1.063×10^{-3} over open cells, 9.213×10^{-3} over wall-adjacent cells, maximum 1.867×10^{-2}. Along the plume centreline the two concentration fields agree to within 0.8% through the interior. The wind, the scaling, the geometry and the transport are the same problem.

They part company at the outflow, and it is worth recording which one is wrong. Steady-state mass here is 11.4% higher than the original's, and the excess lies entirely near the open faces: 12.5% of this field is within three cells of one, against 5.9% of the original's. The advective flux through successive planes settles it:

plane x	original	here
30	1.000	1.000
45	0.908	1.052
55	0.840	1.071
58	0.823	1.076
59	0.478	1.076

A steady state cannot destroy half its flux at a boundary, and the original loses from 0.823 to 0.478 across a *single* cell. Its exit is an artificial sink: outgoing populations are discarded and no incoming ones are supplied, which starves the last cell and pulls material out diffusively for some distance upstream. The gradual decline from $x = 30$ onward is that reach.

The 7.6% rise on this side is not error either, or not all of it. The wind is sheared, so scalar mixing upward into faster-moving air genuinely increases $u\,C$ through successive planes while the total flux, advective plus diffusive, is what is conserved. The condition itself was measured in §12.6 against a case with no shear and an analytic answer, where it holds the flux to 0.038% to the last interior cell.

16.4 A stability limit that does not announce itself

The first attempt at the full city ran to completion, reported no error, and wrote four hundred megabytes of nonsense. Mass in the domain grew from 5.6×10^1 at $t = 60$ s to 1.2×10^8 at $t = 90$ s and thereafter by roughly nine orders of magnitude per frame, reaching 10^{70} — without ever producing a NaN, which is why the finiteness check in the output loop passed every time.

The mechanism is the spurious term of the previous section read in the other direction. Since $\partial_t C \supset -C\,\nabla\cdot\boldsymbol{u}$, the scalar *grows* exponentially wherever the divergence is negative, at rate $|\nabla\cdot\boldsymbol{u}|$. What opposes it is diffusion, which damps a cell-scale blob at approximately $D_{\mathrm{lat}}\pi^2$. Writing the lattice diffusivity out,

$$D_{\mathrm{lat}} = \frac{D\,\Delta t}{\Delta x^2} = \frac{D\,u_{\mathrm{lat}}^{\max}}{u_{\mathrm{peak}}\,\Delta x}, \tag{124}$$

which contains no Δt: **shortening the time step buys no margin at all**, because Δt cancels between the diffusivity and the velocity scaling. Only a larger diffusivity, a slower wind or a finer grid helps.

case	$D_{\mathrm{lat}}\pi^2$	max $\lvert\nabla\cdot\boldsymbol{u}\rvert$	margin	rms $\lvert\nabla\cdot\boldsymbol{u}\rvert$	margin	outcome
synthetic	0.159	1.87×10^{-2}	8.5×	1.06×10^{-3}	150×	stable
Manchester, E, $D = 5$	0.041	2.32×10^{-2}	1.8×	9.42×10^{-4}	44×	diverged
Manchester, E, $D = 20$	0.166	2.32×10^{-2}	7.2×	9.42×10^{-4}	176×	stable
Manchester, NE, $D = 20$	0.166	3.28×10^{-2}	5.1×	9.21×10^{-4}	180×	stable
Manhattan, $D = 20$	0.152	3.35×10^{-2}	4.6×	2.43×10^{-3}	63×	**exponentiated**
Manhattan, $D = 35$	0.267	3.35×10^{-2}	8.0×	2.43×10^{-3}	110×	stable

The maximum is the wrong statistic, and it took a second city to see it. Read the max-based column on its own: the failures sit at 1.8× and 4.6×, the successes at 5.1×, 7.2×, 8.0× and 8.5×. There is no threshold to draw — 4.6 diverges and 5.1 does not, eleven per cent apart, and those two runs have the same peak divergence to within two per cent. Read the rms-based column instead and the failures are at 44× and 63×, the successes at 110×, 150×, 176× and 180×, with a clear gap between. That is where the threshold now sits, at 100×.

The reason is not hard to see once the numbers force the question. The spurious source is damped at the cell scale by $D_{\mathrm{lat}}\pi^2$, so an isolated bad cell is held down by diffusion from its neighbours; what outruns diffusion is an *extended* region of divergence. How much of the domain is in that state is exactly what the rms measures and the max does not. The decomposition is exact,

$$\mathrm{rms}_{\mathrm{open}}|\nabla\cdot\boldsymbol{u}| = \sqrt{N_{\mathrm{wall}}/N_{\mathrm{open}}}\ \mathrm{rms}_{\mathrm{wall}}|\nabla\cdot\boldsymbol{u}|, \tag{125}$$

$0.112 \times 8.22 \times 10^{-3}$ for Manchester and $0.183 \times 1.33 \times 10^{-2}$ for Manhattan, both reproducing the table to three figures — so the quantity really is *how much city*, not *how bad is the worst cell*. Manhattan has 4.3 times as many wall-adjacent cells as Manchester, 485 982 against 113 863, from 50.4% of columns built against 29.9%, 6.87% of the domain solid against 1.79%, and buildings to 472 m against 200 m. **The margin is a property of the geometry**, so a diffusivity calibrated on one city does not transfer to another, and the code now says so at setup rather than after fourteen minutes.

Two caveats on the threshold. It is calibrated on *one* failure that was diagnosed properly — Manhattan at 63× — so it is a warning, not a proof, and both margins are printed because the max is still worth seeing. And the rms is taken over open cells, so a domain carrying a great deal of empty air above the rooflines dilutes it; the decomposition above is how to check whether that has happened.

Neither is $u_{\mathrm{lat}}^{\mathrm{max}}$ a lever, which is worth stating because it looks like one. It scales D_{lat} through Eq. (124) and it scales $\nabla\cdot\boldsymbol{u}$ in lattice units by exactly the same factor, so it cancels out of the ratio entirely.

The undershoot is the early warning; the mass test is not

Mass in the domain cannot exceed what has been injected by any meaningful factor, and that — not finiteness — is what catches a run that has already failed. It is a poor *early* warning. The Manhattan run at 4.6× never came close to tripping it: mass tracked injection to within a per cent for every one of the nine frames it was allowed to write.

What did move, from the first frame, was the undershoot. A first-order equilibrium with no limiter produces small negative concentrations at sharp gradients; those are bounded and do not grow, and a few parts in ten thousand of the peak are expected. An exponentiating $C\,\nabla\cdot\boldsymbol{u}$ is not bounded, and it appears here first by a wide margin:

t (s)	peak C	$\min C$	$\lvert\min C\rvert$/peak
18	6.40×10^{-2}	-2.23×10^{-5}	0.03%
54	8.63×10^{-2}	-2.81×10^{-5}	0.03%
90	1.07×10^{-1}	-4.74×10^{-4}	0.44%
126	1.15×10^{-1}	-3.57×10^{-3}	3.09%
162	1.24×10^{-1}	-1.31×10^{-2}	10.53%

A factor of two every eighteen seconds, with the count of cells below -10^{-4} growing from 53 to 1137 over the same span and centred at $z \approx 150$–190 m, where the tower tops make the divergence largest. The run is therefore stopped on the undershoot exceeding half the peak, and warns at one twentieth of it. At $D = 35$ the same diagnostic reads -9.5×10^{-9} at $t = 108$ s against -1.5×10^{-3} at $D = 20$: five orders of magnitude, and flat rather than growing.

This is worth stating plainly as a limitation of the prescribed wind rather than a tuning inconvenience. A larger D is defensible — it is a calibrated stand-in for unresolved mixing, not a measured constant, and $20\,\mathrm{m^2 s^{-1}}$ at 5 m resolution in a city is not extravagant — but the reason it is *needed* is that the wind is not divergence-free. A solved wind removes the spurious source altogether, and with it this constraint.

16.5 Thirteen per cent that was never missing

The Manhattan run reported 87% of the release still in the domain and held there, with the plume nowhere near a boundary. Summing the written field directly confirms it: the mass within three cells of any face is zero to twenty decimal places. A steady 13% sink, in the interior, in a scheme whose collision conserves $\sum_i h_i$ exactly at every node — $\sum_i w_i = 1$ and $\sum_i w_i \boldsymbol{c}_i = \mathbf{0}$, so $\sum_i h_i^{\mathrm{eq}} = C$ whatever $\boldsymbol{u}$ is — and whose streaming is a permutation.

Three controls localised it correctly and could not explain it. The synthetic case with no wind retained 99.0%; a flat height field with the wind on, where $\nabla \cdot \boldsymbol{u}$ is 0.000 to machine precision because $\boldsymbol{u} = (u(z), 0, 0)$, retained 98.6%; Manhattan with the same wind retained 87.2%. The difference is the buildings, not the divergence. But every one of those numbers was produced by the same estimator, and none of them could say whether material was being destroyed or merely hidden.

A closed box can. `validation/scalar_mass.cpp` puts a source inside a domain whose every face is insulating, so what went in is what must be there, and adds one feature at a time:

case	injected	in fluid	error	every slot
closed box	400.000000	397.155123	−0.711%	400.000000
+ interior block	400.000000	390.406451	−2.398%	400.000000
+ advection	400.000000	372.499608	−6.875%	400.000000
+ wind zeroed in solids	400.000000	372.499608	−6.875%	400.000000

Nothing is lost. The last column is the injection to round-off in every case. What is short is the sum of the *macroscopic field* over fluid nodes, which is what the demonstrator was reporting. Under Esoteric Pull a population emitted toward a wall spends one step in a slot the *wall* node owns, and `compute_field` sets the field to zero at an adiabatic node by definition, so that material belongs to no node's field while it is in flight. Measuring the same sum one step later shows the shortfall is steady rather than alternating with the parity, so it is a standing boundary layer: it scales with wall area (0.7% for a bare box, 2.4% once there is a block in it) and with the advective flux into walls (6.9%). Over Manhattan's 6.87% solid fraction it is 13%. Manchester's 1.79% is why this never surfaced before.

Two details are worth keeping. First, the third and fourth rows agree to the last digit. Zeroing the wind inside solids — the thing that makes the prescribed field divergent in the first place — changes the budget by exactly nothing, because solid cells are never read and the velocity stored there is never used. That confirms the claim of §16.3 that no mass budget can see the divergence; what that claim did not say is that the fluid-only sum carries a wall-area bias of its own. Second, `scalar_walls.cpp` covers Dirichlet walls, where mass is deliberately not conserved — that is what a fixed-value boundary is for. Nothing covered the insulating one, which is precisely the boundary that does nothing at all, and "costs nothing" turned out to be a strong claim to leave untested.

The budget now uses `ScalarSolver::total_population`, which sums every slot in the lattice, and reports the fluid fraction beside it rather than in place of it. With that change the demonstrator reports 100.0% retained and $\mathrm{d}M/\mathrm{d}t$ of exactly 1.000 until material actually reaches an outlet, which is what the column always claimed to mean.

16.6 The city

Manchester city centre, $400 \times 400 \times 60$ cells at 5 m — 2 km square and 300 m deep, 6058 buildings — with a $5\,\mathrm{m\,s^{-1}}$ westerly at 10 m, $z_0 = 1$ m, $D = 20\,\mathrm{m^2s^{-1}}$ and a continuous release of one unit per second. The source cell falls inside a building and is lifted clear of it, so the release is from a roof at 32.5 m rather than from the street. Five minutes of physical time is 14 280 steps and runs in 801 s on four threads of an M1, or 171 MLUPS.

The plume approaches steady state monotonically: $\mathrm{d}M/\mathrm{d}t$ falls from 0.97 to 2.0×10^{-3} over the run while the mass in the domain plateaus at 1.409×10^2. It has not fully arrived — the derivative is three orders down but not zero — which is honest for a five-minute release into a two-kilometre domain at $5\,\mathrm{m\,s^{-1}}$: the leading edge has crossed roughly 1.5 km.

A second configuration, and what the first one hid. Releasing at the centre of the domain into a due-east wind gives only 1 km of fetch, and the plume reaches steady state at $t = 5$ min — three fifths of that run is static. Moving the source to an open pocket in the south-west at $(140, 135)$ m and turning the wind onto the diagonal, toward the north-east, gives 2.6 km of fetch; steady state then arrives at 10.8 min and the plume develops throughout. The source also sits at street level, 7.5 m, rather than being lifted onto a roof as the central cell was.

That change exposed a defect the first configuration could not. The count of cells used to normalise the release and the set of cells actually injected into were separate predicates, and they disagreed at $z = 0$: the count required $z > 0$ while the injection covers every bulk cell, which since §16 includes the ground layer. A box centred at $z = 1$ therefore had its per-cell rate computed for three layers and applied to four — 33% over-injection, visible within eighteen seconds as more scalar in the domain than had been released, and invisible for any source clearing $z = 2$. The two are now one named predicate, asked twice.

The diagonal wind appears to cost stability margin: zeroing both horizontal components inside buildings raises $\max|\nabla\cdot\boldsymbol{u}|$ from 2.3×10^{-2} to 3.3×10^{-2}, and the max-based margin of §16.4 falls from 7.2 to 5.1. That was described here as "comfortably clear of the 1.8 that diverged", on the strength of a mass budget that stayed sensible for the whole run — which, as §16.4 now records, is not a judgement a mass budget can make.

The configuration was therefore repeated and its undershoot examined. It is stable, and not marginally so: $\min C$ is bit-identical at -1.11×10^{-7} for six consecutive frames from $t = 45$ to $t = 120$ s, while a Manhattan run at nominally the same margin had grown its undershoot twentyfold over the same span. What the max-based number was hiding is that the diagonal wind barely changes the rms at all — 9.42×10^{-4} to 9.21×10^{-4}, a 2% *fall* — because it redistributes divergence among the same wall-adjacent cells rather than adding any. On the statistic that predicts the outcome this is the safest configuration in the table, at 180×.

Negative concentrations. The scheme undershoots. A first-order equilibrium with no limiter produces small negative concentrations at sharp gradients: 3.2 million cells of 9.6 million hold values between 0 and -7.9×10^{-5}, against a section peak of 5.4×10^{-2} — a few parts in ten thousand. They are bounded and do not grow, and `min C` is now reported every frame so that they are visible rather than assumed away. The practical consequence is for anything downstream that takes a logarithm or treats a negative value as a flag: the plan-view renderer for Figure 16 initially tested `v < 0` for the building mask, which is how it came to paint three million undershooting cells as buildings and hide most of the city.

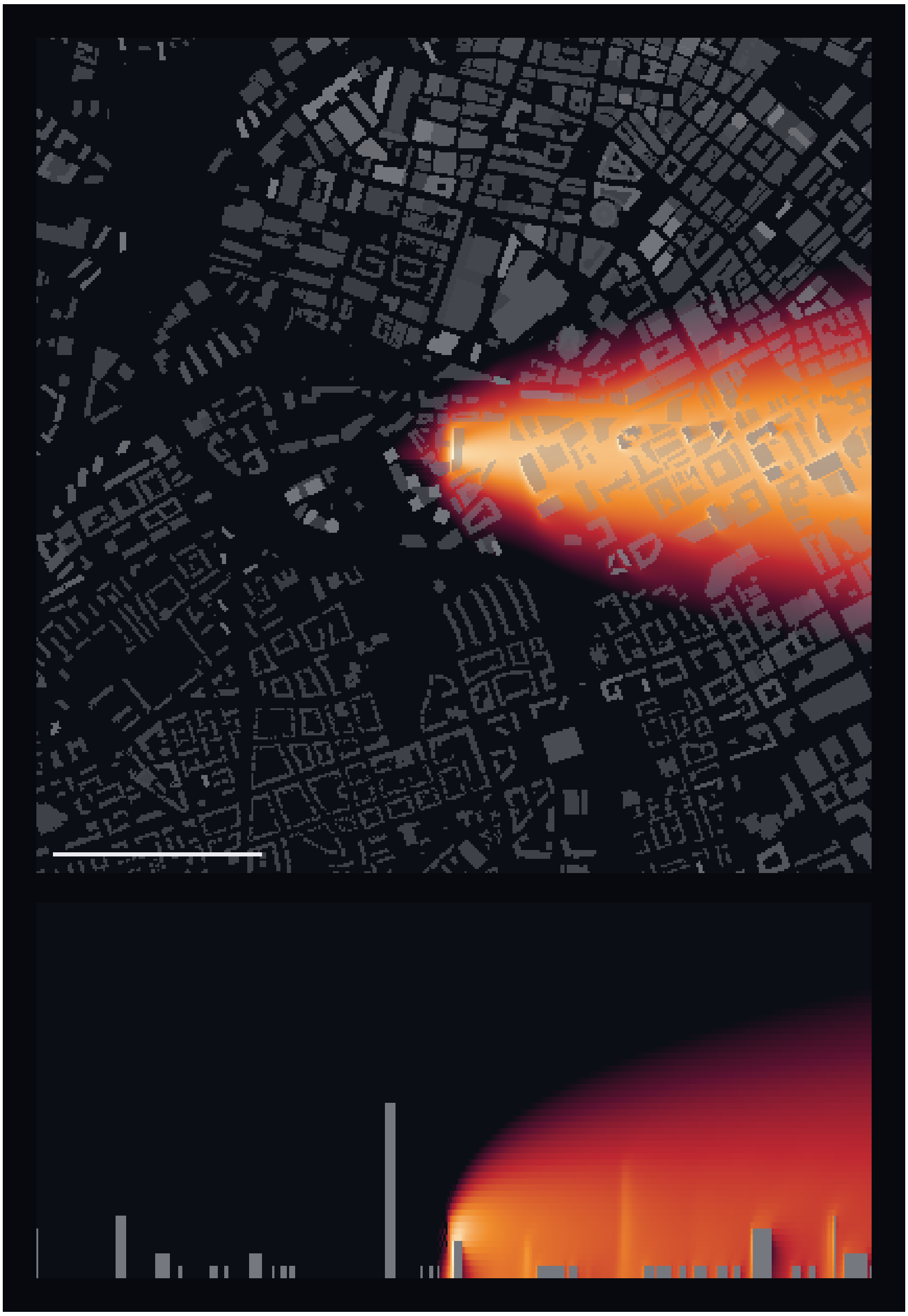


Figure 16: Urban plume over Manchester city centre after five minutes of continuous release from a rooftop at 32.5 m. *Top*: plan view, building footprints in grey, column-integrated concentration on a logarithmic scale; the bar is 500 m. *Bottom*: vertical section along the plume centreline, vertical scale exaggerated three times, showing the release lifting as it enters faster-moving air aloft. The wind is a $5\,\mathrm{m\,s^{-1}}$ westerly, so the plume runs east. Note what the figure does *not* show: the wind is prescribed, so there is no recirculation behind any of these buildings and no channelling along the streets the plan view makes so visible.

16.7 A second city

Midtown Manhattan, $400 \times 400 \times 100$ cells at 5 m — the same 2 km square, 500 m deep, 10 210 buildings — with the same $5\,\mathrm{m\,s^{-1}}$ wind at 10 m and $z_0 = 1$ m, and the same continuous release of one unit per second from street level at 7.5 m. Everything that could be held fixed against §16 was. What could not is the diffusivity: $D = 35\,\mathrm{m^2 s^{-1}}$ against Manchester's 20, for the reason §16.4 sets out. Fifteen minutes of physical time is 46 665 steps and 51 frames.

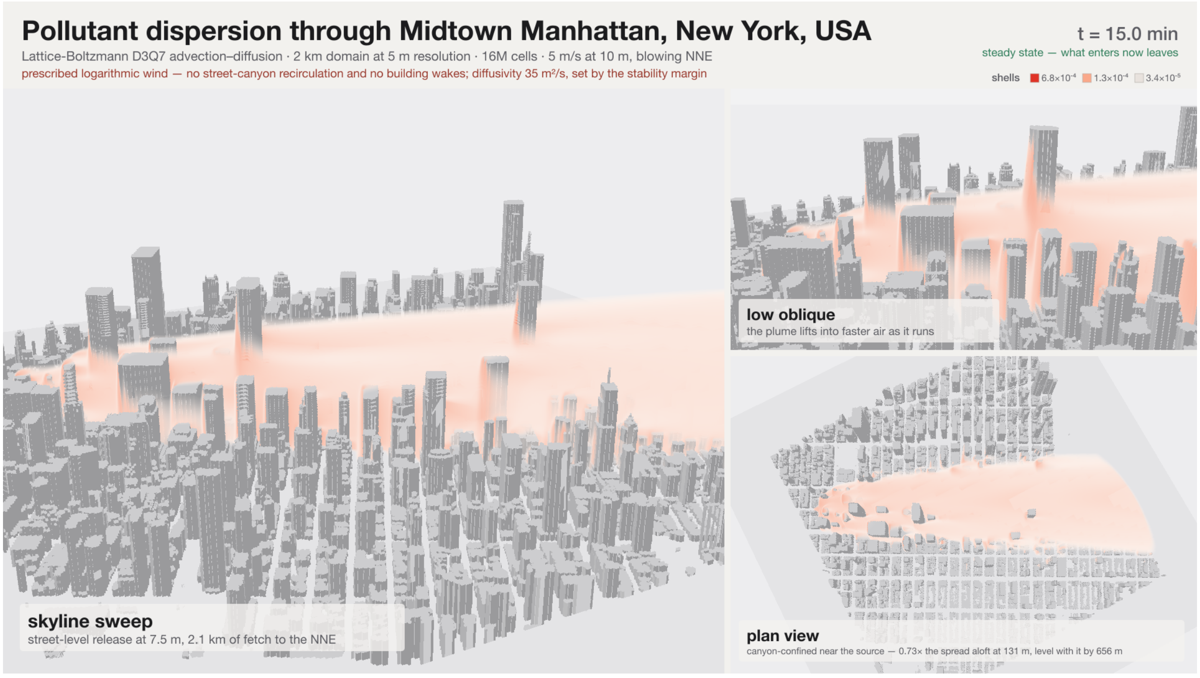


Figure 17: Midtown Manhattan at steady state, fifteen minutes into a continuous street-level release. Three shells of concentration, at the levels enclosing 95%, 80% and 50% of the plume's mass — chosen from this frame rather than carried over from another run, since the same release spread over 2 km carries an order of magnitude less than it does over 200 m. *Left*: the skyline broadside to the plume, which runs left to right along the avenues. *Top right*: the same view from street level, showing the release threading between the towers. *Bottom right*: plan view, with the Commissioners' grid visible; the wind bearing was measured out of the height field, not looked up. Every figure in the caption strip is read from the run's own log, and the plan-view line is measured from this frame. The animation is `doc/fig/manhattan_dispersion.mp4`.

The wind direction was measured, not assumed. Manhattan's street grid is not aligned with north, and rather than look the angle up it was recovered from the height field: rays marched from every open column, in every direction, recording how far they travel before meeting a building. The mean free run peaks at two orthogonal bearings, 28° and 118° — 385 m along the second, 285 m along the first — which is the Commissioners' Plan of 1811, extracted from a 400×400 array of heights with no knowledge of what city it is. The wind was then aimed along 28°, up the avenues, giving 2.10 km of fetch from a source in an open pocket in the south of the domain.

Steady state, and a budget that now says what it means. The plume reaches steady state at about 11.4 min, where $\mathrm{d}M/\mathrm{d}t$ falls below 2% of the injection rate; by the last frame it is 1.8×10^{-3} and the mass in the domain has plateaued at 3.670×10^2 against 8.817×10^2 injected. That 41.6% retained is a real statement about where the release has gone — the remaining 58% has left through the outlets — which it was not before §16.5: the same run reported under the old fluid-only sum would have shown 87% and held there while the plume was still in the middle of the domain. The companion column reads 92.0% for the whole run, unchanging: that is the fraction the plotted field sees, and the 8% gap is Manhattan's wall area.

The undershoot is -8.08×10^{-9} and does not move for any of the 51 frames — the total negative mass is 3×10^{-7}, or 0.00% of the plume. At $D = 20$ the same configuration reached -1.3×10^{-2} by $t = 162$ s and was doubling every eighteen seconds.

The canyons hold the plume, but only near the source. The plan view of a city plume invites a claim about street channelling, and the caption on the Manchester animation made one: *isotropic lateral spread — the streets do not steer it.* On a regular grid of canyons with the wind aimed along one of its axes that is worth measuring rather than asserting, so `doc/fig/plume_stats.py` bins the last frame by downwind distance along the plume's own axis and compares the concentration-weighted crosswind spread below and above the rooflines:

s (m)	σ_n street	σ_n aloft	ratio	$\langle z \rangle$ street	street fraction
131	39.5	53.8	0.73	26.2	75.5%
394	57.7	72.1	0.80	31.5	47.8%
656	78.1	80.6	0.97	31.9	33.8%
919	99.3	94.9	1.05	32.1	32.4%
1181	99.8	99.9	1.00	33.2	25.7%
1706	120.0	123.0	0.98	31.3	22.7%
1969	130.3	120.4	1.08	33.0	19.7%

Within the first 130 m the plume is 27% narrower below the rooflines than above them, and by 650 m the two are the same. The last column says why: the fraction of the plume still below roof level falls from 76% to 20%, so there is progressively less left for the canyons to hold. The street-level material that remains does not rise — its mean height sits at 32 m the whole way — while the part aloft climbs from 91 m to 164 m as it mixes into faster air.

That is a confinement effect and *not* a channelling one, and the distinction matters here more than it would in a solved wind. The prescribed field blows in a straight line by construction: nothing in this model can steer a plume along a street. What the buildings do is block, and being opaque to the scalar is on its own enough to produce the near-field narrowing. A solved wind would add the channelling this one cannot represent, and the honest reading of the table is as a lower bound on how much the geometry matters.

The caption on the animation is generated from exactly this measurement rather than written by hand, for the reason the Manchester caption illustrates: pointed at a second city, a caption made of literals states the first city's name, cell count and fetch with complete confidence.

16.8 Solving the wind

The wind can instead be solved: D3Q27 TRT Navier–Stokes on the same geometry, sharing the Domain, Δx and Δt with the scalar, so the fluid's lattice velocity *is* the scalar's and the two couple with no conversion. TRT rather than BGK because the effective viscosity puts τ^+ within a few thousandths of 1/2, where BGK is unusable and TRT's free antisymmetric rate is exactly the lever required; $\Lambda = 3/16$ additionally places the bounce-back wall halfway, so a voxelised building is the size it looks. Starting from the undisturbed profile rather than from rest leaves only the wakes to develop, and the solve converges to a relative change of 2×10^{-7} in 7000 steps.

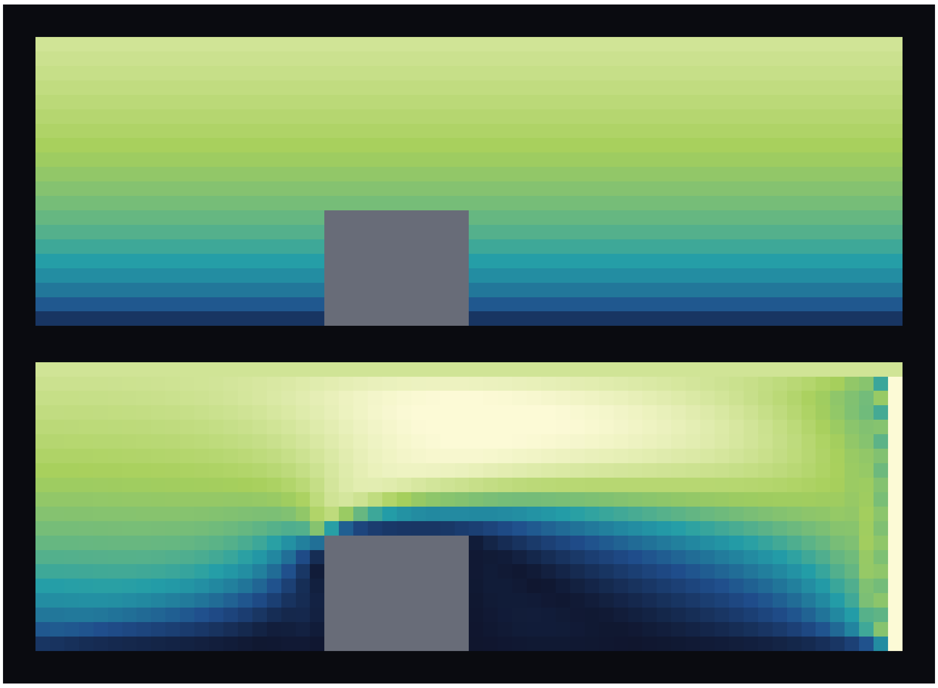


Figure 18: Vertical section through a building, wind speed, both panels on the same scale. *Top*: the prescribed logarithmic profile — flat horizontal layers, unaware that the building is there. *Bottom*: the solved field — upstream stagnation, acceleration over the roof, and a slow wake in the lee recovering over roughly twenty cells. Along the centreline at mid-building height the prescribed field is $4.34\,\mathrm{m\,s^{-1}}$ everywhere, while the solved one falls to 1.37 approaching the building and 0.30 behind it. The bright stripe at the right edge is the outlet-corner artefact discussed below, left visible rather than cropped. The colour scale is the 98th percentile, not the maximum: that artefact reaches $29\,\mathrm{m\,s^{-1}}$ and scaling to it renders every physical feature in the bottom quarter of the ramp — the same trap as the aorta volume render, where p_{100} was 2.7 times $p_{99.9}$.

The interior is right, and is the point. *All the figures in this and the next paragraph are the synthetic* $60 \times 30 \times 20$ *case*, which for a long time was the only geometry this mode had ever been run on; the city results are in §16.9. Three cells clear of every face, the divergence falls from RMS $= 1.27 \times 10^{-3}$ prescribed to 2.12×10^{-4} solved — a factor of six — and the peak speed is $8.80\,\mathrm{m\,s^{-1}}$ against a $7.74\,\mathrm{m\,s^{-1}}$ free stream, a 14% speed-up over the building. That is a result the prescribed profile cannot produce at all, and the divergence drop is the spurious source of §16.4 shrinking, which is what drove both the concentration error and the stability floor.

The domain edges are not right. Where the pressure outlet meets the velocity-prescribed lateral and top faces the corner is over-determined, and a jet forms along it: $31.6\,\mathrm{m\,s^{-1}}$ and $\nabla\cdot\boldsymbol{u} = 9.5 \times 10^{-2}$, against 8.8 and 7.0×10^{-3} in the interior. This is why the global maxima look worse than the prescribed field's while every interior statistic is better, and it is the reason the diagnostics now report interior-only figures and say whether the worst divergence sits on a domain face — without which the two were not separable.

Two remedies were tried. Giving the edges the free-stream profile removed the sixty-eight inert outflow nodes, whose donor lay on an adjoining wall, but did *not* remove the jet, which locates the cause in the mixed condition itself rather than in the donor lookup. The identified fix is periodic lateral boundaries, the standard atmospheric arrangement, which deletes two of the four offending faces outright.

16.9 The same mode on a city, and the two things that stopped it

Everything above is the synthetic case. Pointed at Manchester the mode produced a NaN within 500 steps — in FP64 as well as FP32, so not a precision effect — and it did so at the τ^+ the synthetic case had converged at. Bisecting by geometry:

case	buildings	on faces	outcome	peak / free stream
synthetic $60 \times 30 \times 20$	1	no	converges, 5.7×10^{-7}	4.1×
flat $200 \times 200 \times 150$	none	—	no NaN, residual 6.5×10^{-2}	4.4×
Manchester $400 \times 400 \times 60$	6058	**yes**	**NaN by step 500**	—
Manchester, 8-cell clear margin	2083	no	no NaN, residual 2.7×10^{-1}	2.6×

A face cell shadowed by a building is ill-posed. The regularised condition reconstructs the density at a wall node from the populations that streamed into it, split by the wall normal into known and unknown, and that split assumes the known ones arrived from the interior. When the cell one step inward is a building, what actually arrives is bounce-back off that building, the reconstructed density is not a density, and the solve diverges. Manchester has 3975 built columns touching the domain edge; clearing an eight-cell band removes the NaN outright, which is what identified the cause. Only 141 cells are *actually* shadowed — a face cell with solid immediately inward — and those 141 were enough to destroy a nine-million-cell solve. They are now closed as `Solid`, which extends the building by the one cell it was already effectively occupying and turns an ill-posed node into a well-posed wall. That also subsumes the "outflow node has no fluid neighbour and is inert" case: the same geometry seen from the outlet side, previously left frozen rather than closed.

Periodic laterals, and what they cost. `-lateral periodic` makes the y faces periodic, deleting two of the four junctions where a velocity-prescribed wall meets a pressure outlet. Periodicity is handled inside `Domain::fill_neighbours`, so neither solver needed changing; the divergence diagnostics did, because $y = 0$ and $y = n_y - 1$ become ordinary interior cells whose neighbours wrap, and omitting them would drop 0.5% of the domain from an rms the stability threshold is built on.

The two fixes are independent, and it is worth being explicit that periodic laterals alone do **not** cure the NaN — they leave the inlet and outlet faces, where the same shadowing occurs. Both are required.

What periodicity costs is a modelling choice rather than accuracy, and the run now states it rather than leaving it implicit: the domain becomes an infinite strip, the city repeats every 2 km across the wind, and a plume drifting sideways out of one edge re-enters at the other. The seam is not smooth either, since real building heights at $y = 0$ and $y = n_y - 1$ were never chosen to match: 220 of Manchester's 400 edge columns disagree, worst by 48 m. That count is printed, and a separate warning fires if the wind has a cross-wind component, because the plume is then advected through the seam rather than merely diffusing across it.

A solved wind inverts which statistic misleads. With both fixes, Manchester at 8000 flow steps — one flow-through time at $u_{\text{lat}}^{\max} = 0.05$, so a developing field and not a converged one:

	prescribed	solved	change
rms, whole domain	9.422×10^{-4}	2.638×10^{-3}	0.4×
rms, 3 clear of a face	9.614×10^{-4}	7.401×10^{-5}	**13.0×**
rms, wall-adjacent	8.407×10^{-3}	2.045×10^{-3}	4.1×
max, whole domain	2.318×10^{-2}	8.351×10^{-2}	0.3×
max, 3 clear of a face	2.318×10^{-2}	1.016×10^{-2}	2.3×

The interior is 13× better, more than double the factor of six the synthetic case showed. The global figures are 2.8× *worse*, and the worst cell sits at $(398, 342, 58)$: the outlet/top edge, the one junction periodic laterals cannot delete.

So the two fields fail differently, and the statistic that reads correctly for one misreads the other. A prescribed field spreads its error through the city, so its whole-domain and interior rms agree to 2% (9.42×10^{-4} against 9.61×10^{-4}, margins 44× and 43×) and the test of §16.4 is the whole story. A solved field is divergence-free everywhere it is solved and carries its entire error at one boundary, so the whole-domain figure is set by that edge and understates the interior by more than an order of magnitude: 16× against 561×. Both are now printed, the run says so explicitly when they differ by more than a factor of three, and the warning stays on the whole-domain figure because only that one is a guarantee.

The diffusivity floor is a property of the prescribed profile. This is what solving the wind buys, stated at the diffusivity that matters. $D = 5\,\mathrm{m}^2\mathrm{s}^{-1}$ is the value that diverged under the prescribed wind, at an interior margin of 43×. Under the solved wind the same $D = 5$ gives 561×. Every urban run in this document has been constrained to $D \geq 20$ by a term that the wind model introduced and the physics does not contain.

Two caveats stand. The field above is one flow-through time, not a converged urban wind — three to five would be the usual requirement, and at the measured rate on this machine that is four to seven hours. And the outlet/top edge is still over-determined, so a plume must be kept clear of it. The prescribed wind therefore remains the default.

16.10 Coupling the wind, unsteadily

Everything in §16.9 freezes the wind: solve once, hand the converged field to the scalar, transport. A converged field has a steady recirculation behind every building, and a steady recirculation stirs nothing — so nothing in a frozen-wind run can show the plume being torn apart by a building's wake, only advected around its time-averaged shape. `-mode unsteady` spins the flow up as before and then advances it *one step per scalar step*, refreshing `ux/uy/uz` in place: they are the very Views the scalar was handed, so the refresh *is* the coupling and nothing is copied. The added cost is one `compute_macroscopic` pass per step, which is the price of an instantaneous rather than a mean field.

A crop, and a viscosity found the hard way. A coupled run costs a flow step every scalar step, so the full 2 km city is hours rather than minutes; a 1 × 0.75 km, 200 × 150 × 60 crop at the same 5 m resolution keeps what resolves the wakes and spends the saving on footprint. The crop was chosen for density, not the emptiest corner: a 169 m tower, the tallest in the whole city, sits inside it, together with 49% of columns built. The obvious first choice happened to put that tower on the periodic seam, wrapping it into itself; moved 30 m, the seam mismatch on the tallest column falls from 169 m to 42 m.

Effective viscosity had to be found rather than assumed. $\nu = 2$ and $\nu = 4\,\mathrm{m^2s^{-1}}$ both NaN on the crop; $\nu = 8$ runs. During spin-up its residual oscillates — 8.2×10^{-3} at 4000 steps against 7.0×10^{-2} three hundred steps earlier — and it was tempting to read that as an unsteady field. It is not. **The flow settles.** Measured between consecutive output frames, four seconds apart, once the coupling is running:

frames	$\mathrm{rms}\lvert\boldsymbol{u}_{i+1} - \boldsymbol{u}_i\rvert$	as % of $\mathrm{rms}\lvert\boldsymbol{u}\rvert$	max
30→31	1.66×10^{-3}	0.015%	0.022
31→32	1.37×10^{-3}	0.012%	0.014
45→46	5.5×10^{-4}	0.005%	0.006
59→60	3.2×10^{-4}	0.003%	0.005

Monotonically decaying to three parts in a hundred thousand. The oscillating spin-up residual was a decaying transient, not shedding, and the field the plume rides is steady to within round-off by the end. At $\nu = 8$ the 169 m tower sits at $Re \approx 250$ on its own width, which is enough for a free-standing bluff body to shed — but this one is wall-mounted in a rough boundary layer among neighbours, and at 6.7×10^{-3} lattice viscosity the shear layers are damped before they can roll up. Reaching a shedding regime needs a lower viscosity, and $\nu = 4$ is where this configuration stops converging at all. **That is the gap: between the viscosity at which the solve survives and the viscosity at which the flow becomes unsteady, there is currently no overlap on this geometry.**

So this section demonstrates that the coupling MECHANISM works — the flow advances with the scalar, the plume rides an instantaneous field, and $D = 5$ is stable throughout — and does not demonstrate turbulent dispersion. What the plume experiences here is a steady solved wind with recirculation and channelling, which is still something no prescribed profile can produce, and is not turbulence.

$D = 5$ **is comfortable, not marginal.** Under the prescribed wind this diffusivity diverged outright. Coupled and unsteady, $\min C$ — the early warning of §16.4 — holds at -2.5×10^{-4} for the entire second half of a four-minute release, drifting by less than 3% step to step despite riding on a field that is itself fluctuating by tens of per cent. The interior divergence is 319× the damping rate against the prescribed field's 43× at the same D, and the peak interior speed is $15.4\,\mathrm{m\,s^{-1}}$ against an $11.9\,\mathrm{m\,s^{-1}}$ free stream — a 29% speed-up over the buildings, against 14% in the converged synthetic case of §16.8. An unsteady field is not merely divergence-free on average; the instant-by-instant acceleration over a roof is larger than a time-averaged solve ever shows.

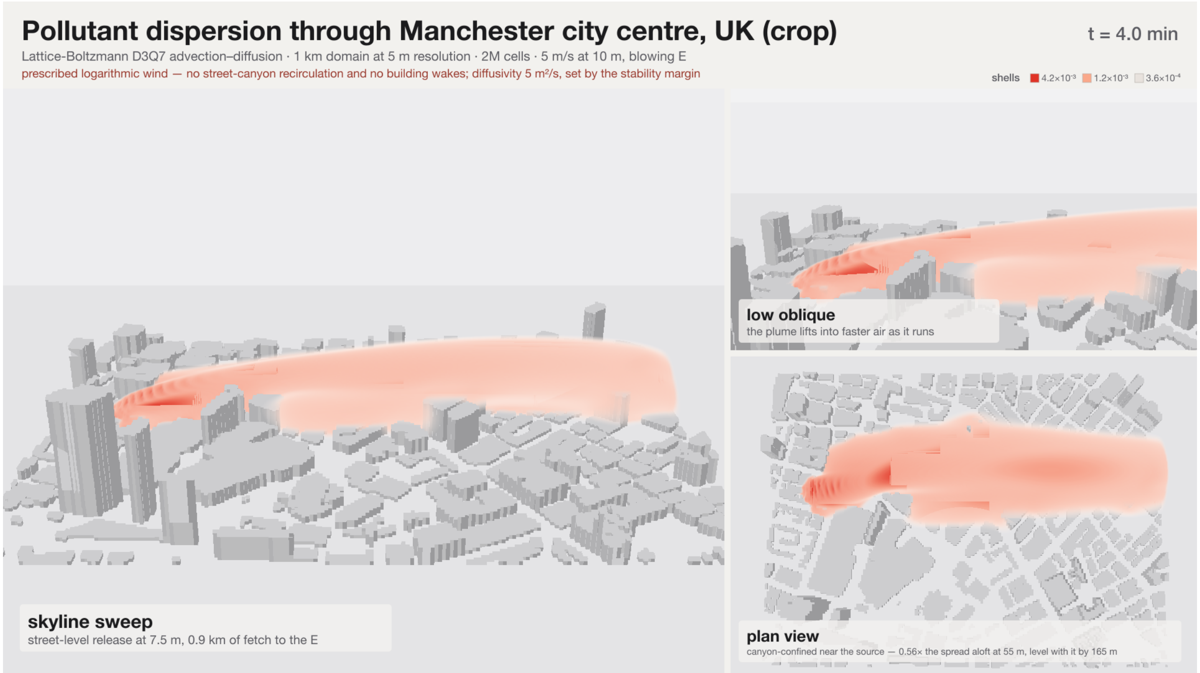


Figure 19: Coupled dispersion, Manchester crop, 3.7 min into a street-level release at 7.5 m with 0.9 km of fetch, at $D = 5\,\mathrm{m}^2\mathrm{s}^{-1}$ — the value that diverged outright under the prescribed wind (§16.4). The flow is advanced one step per scalar step, but it converges rather than shedding (above), so the plume is riding a steady solved wind, not a turbulent one. The animation is `doc/fig/mcr_coupled_unsteady.mp4`.

The confinement result reverses. §16.7 found Manhattan's plume narrower below the rooflines than above near the source, widening to equal by 650 m — confinement, from buildings simply blocking a wind that cannot itself be steered. Measured the same way here:

s (m)	σ_n street	σ_n aloft	ratio
55	23.6	42.2	0.56
165	30.0	24.9	1.20
275	41.9	32.5	1.29
385	52.3	37.5	**1.39**
495	51.7	42.8	1.21
605	42.1	43.4	0.97

The ratio rises *above* one in the near-to-mid field before settling back near it: the street-level plume spreads sideways *faster* than the air above the roofs, rather than merely failing to be confined.

The cause is steady recirculation, not turbulence — the field is steady, as measured above. A prescribed profile is zeroed inside buildings and otherwise flows straight, so it can only fail to confine; a solved field has standing lateral recirculation in the lee of every block, and that alone moves material across the flow faster than diffusion does. It is a weaker claim than turbulent dispersion and it is the one the data supports. Row one, at $s = 55$ m, is still the source's immediate vicinity and reads like the confined case of §16.7 for the same reason — the plume has not yet reached a recirculation zone.

This is one release over four minutes on one crop, and the numbers above should be read as a demonstration that the mechanism is there and measurable, not as a converged statistic. `-mode unsteady` is new in this section and has had one city and one geometry through it — and, as recorded above, has not yet been run at a viscosity where the flow is actually unsteady.

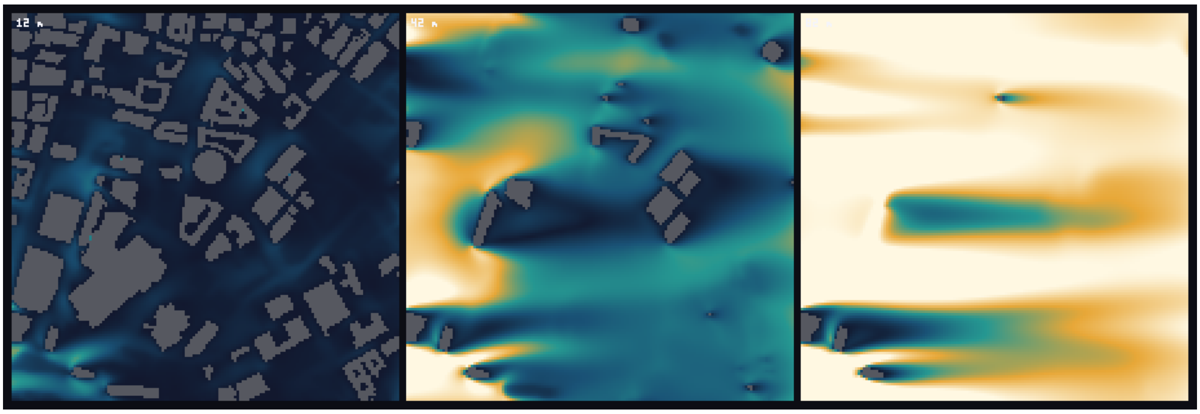


Figure 20: The solved wind at three heights: 12 m inside the canyons, 42 m around roof level, 82 m in the free stream. One colour scale across all three, so they are comparable — dark is slow. The wake streaks trailing downwind of the tall blocks in the 82 m panel are the structure a volume render could not show: a wake field is a boundary layer that fills the lower domain, 45% of fluid cells carrying a positive velocity deficit, so front-to-back compositing accumulates along every ray and returns an opaque slab at any threshold. Stacked slices work where shells do not. The animation is `doc/fig/mcr_wind_3heights.mp4`.

16.11 The top boundary is a modelling choice

A horizontal wind has $u_z = 0$ at the top face, so a zero-gradient condition there passes no advective flux — there is none — and no diffusive flux, because that is what zero-gradient means. The top is therefore a *lid*: material that mixes to it stays in the domain. That is right when the top is far above the plume and wrong when it is not. Holding the top at $C = 0$ instead says the atmosphere above is clean and lets the scalar diffuse into it, which is the more physical statement for a domain whose top is an arbitrary cut through the boundary layer, at the cost of forcing $C = 0$ there. Both are offered; the lid is the default because it is the validated condition. On the synthetic case the choice is worth 0.7% of retained mass, which is small only because the plume does not reach the top — it is not a general bound.

17 Demonstrator: two-phase flow with a moving body

The two cases in this section are demonstrators, and the distinction the repository draws is worth repeating before either of them is shown. A validation case has something to be right against — an analytic solution, a published benchmark, a convergence rate — and fails if it misses it. A demonstrator has none of that: it shows the solver running on a problem the rest of the suite cannot express, and the only thing it can fail is to crash. The two do not share a directory, and no demonstrator is registered with `add_test`. The same rule governed the aorta (§15) and the city (§16), and it governs these: **no number in this section is a claim of accuracy**. The accuracy claims for the multiphase module are the static droplet, the layered channel and the floating box, and they are made where those cases are. What is shown here is an interface that rolls up, reconnects and keeps going, and a rigid body that falls into one under its own weight.

17.1 Rayleigh–Taylor instability

Heavy fluid over light in a box W wide and $4W$ tall, periodic in x, no-slip on the top and bottom, with the interface given a single-mode perturbation at the box wavelength,

$$y_i(x) = 2W + 0.1\,W \cos \frac{2\pi x}{W}.$$

The controlling groups are the Atwood number and a Reynolds number built on the free-fall velocity,

$$\mathrm{At} = \frac{\rho_H - \rho_L}{\rho_H + \rho_L}, \qquad \mathrm{Re} = \frac{W\sqrt{gW}}{\nu}, \qquad t^* = t\Big/\sqrt{\frac{W}{g\,\mathrm{At}}}, \tag{126}$$

and the driver takes $g = U^2/W$ so that U *is* the free-fall velocity and $\nu = WU/\mathrm{Re}$ follows. Reporting time as t^* makes runs at the same At comparable regardless of the lattice numbers underneath. The defaults are $W = 128$, $\mathrm{At} = 0.5$, $\mathrm{Re} = 256$, $U = 0.04$, interface width $W_{\mathrm{int}} = 5$, mobility 0.02 and $\sigma = 1 \times 10^{-4}$, on D2Q9 for both the fluid and the phase field. Only the pressure-based operator is used: it is the only one here that reaches a density ratio at all.

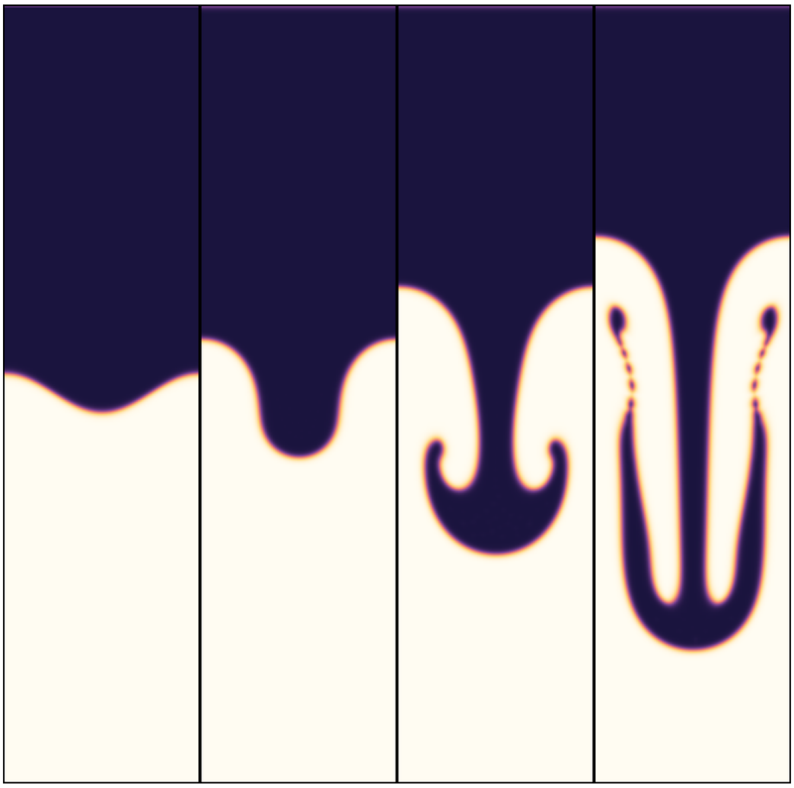


Figure 21: Rayleigh–Taylor instability at $\mathrm{At} = 0.5$, $\mathrm{Re} = 256$, $W = 128$, central-moment operator. Phase field at $t^* = 0$, 1, 2 and 3. The single-mode perturbation steepens into a falling spike, the spike rolls up into a mushroom, and by $t^* = 3$ the stem has gone secondarily unstable — the beads along it are Kelvin–Helmholtz billows on the shear between the descending spike and the rising fluid beside it. Nothing here is a measurement; what it shows is an interface that rolls up, reconnects and keeps going without the phase field being reinitialised.

The initial condition is hydrostatic through the diffuse interface, and it has to be. The pressure is seeded by integrating the *actual* density profile away from the interface rather than a step, which the tanh profile makes closed-form. With $\delta = y - y_i(x)$ and $\phi = \frac{1}{2}(1 + \tanh(2\delta/W_{\mathrm{int}}))$,

$$p = -g\,I(\delta), \qquad I(\delta) = \rho_L\,\delta + \tfrac{1}{2}(\rho_H - \rho_L)\Big[\delta + \tfrac{W_{\mathrm{int}}}{2}\ln\cosh\frac{2\delta}{W_{\mathrm{int}}}\Big], \tag{127}$$

and the populations are seeded with $\tilde{p} = p/(\rho(\phi)c_s^2)$, not with a uniform $\tilde{p}$ — a uniform one starts the box with the whole hydrostatic imbalance as an acoustic transient, larger than the instability it is meant to be watching. Assuming a sharp interface instead leaves an imbalance of order $(\rho_H - \rho_L)\,g\,W_{\mathrm{int}}$ smeared over the interface. That is 1×10^{-4} at $\mathrm{At} = 0.5$ and invisible;

at $\mathrm{At} = 0.998$ it is 0.25, $\tilde{p}$ on the light side starts at 0.75, $|\boldsymbol{u}|$ reaches the free-fall velocity within thirty steps — before the instability could have grown at all — and the run is gone by step 120. A diffuse-interface model deserves a diffuse-interface initial condition. The ln cosh is evaluated as $|z| + \log_1 p(e^{-2|z|}) - \ln 2$ rather than through cosh, which overflows in the far field at any useful box height.

The pressure zero goes at the interface. Only ∇p is physical, so the level is free in the continuum and is not free on the lattice: $\tilde{p} = p/(\rho c_s^2)$, so a constant added to p is not a constant added to $\tilde{p}$ when ρ varies, and both of the large terms that cancel at the interface — $\rho c_s^2 \nabla \tilde{p}$ and $F_p = -\tilde{p}\, c_s^2 \nabla \rho$ — scale with it. Since p is continuous across the interface while ρ is not, $\tilde{p}$ jumps by the density ratio wherever p is not near zero. Two measurements of the cost. Referencing p to $\rho_L c_s^2$, a uniform physical pressure, made this case diverge inside 424 steps at $\mathrm{At} = 0.5$, with $|\boldsymbol{u}|$ at twice the free-fall velocity at step zero, before anything had moved. Referencing to the top of the box is harmless at $\mathrm{At} = 0.5$, where $g\rho_H\, 2W$ is 1×10^{-2}, and is not at $\mathrm{At} = 0.998$, where the same quantity is 3.2 and $\tilde{p}$ in the light fluid starts at 9.6.

The collision operator is not a cosmetic choice at high Reynolds number. BGK relaxes every mode at ω, so as $\omega \to 2$ the non-hydrodynamic modes ring instead of decaying; at $\mathrm{Re} = 30000$ on a 128-wide box $\omega = 1.99795$. Run with `-op bgk` the case diverges at $t^* = 1.5$, where the central-moment operator completes $t^* = 3$. That is the whole of the argument for a moment operator here, and it is why `-op cm` is the default.

A known artefact, kept as evidence. At $\mathrm{At} = 0.998$ — a density ratio of 999 — BGK completes the run and produces a phase field that is smooth and physically sensible, while its velocity field is dominated by a one-cell alternating mode measured 70× stronger than the same case at a density ratio of 3. Grid-scale oscillation that BGK does not damp is exactly what the equations predict at that ω, and this measurement is what motivated the central-moment operator rather than a consequence discovered after it. The clip of that run is retained for that reason and is not a result.

The renderer is a separate program, deliberately. The case writes raw fields, three per frame, in `FieldDump.hpp`'s format, and `render_rt` turns them into pictures. Welding a rasteriser to the simulation makes every change of colour map cost a full re-run — eight and a half minutes at $W = 192$ to alter a hue — against 2.6 seconds to re-render 181 frames. One detail of that renderer is a measurement rather than a preference: the vorticity scale calibrates on the *last* frame, because the first frame of this case is a fluid at rest whose 99.9th-percentile vorticity is round-off, 1.4×10^{-7} against the 6.2×10^{-3} the developed flow reaches, and calibrating there puts the whole film at full saturation. A percentile rather than a maximum, so that one hot cell cannot flatten everything else.

Three clips live in `demonstrator/anim/`. They are not reproduced as figures: a still cannot show a reconnection, which is the thing being demonstrated.

file	what it shows
`rayleigh_taylor.mp4`	$\mathrm{At} = 0.5$, $\mathrm{Re} = 256$ — the reference case
`rayleigh_taylor_re30000.mp4`	$\mathrm{At} = 0.1$, $\mathrm{Re} = 30000$, central moments
`rayleigh_taylor_at998.mp4`	$\mathrm{At} = 0.998$ under BGK — the artefact above

Two things in this case are not to be trusted, both stated in the module headers and both live here. The interface never touches a wall within $t^* = 3$, which is what makes it legitimate to run with no wetting condition; take it further and the spike reaches the floor, and what happens there is not modelled. And the pressure form's F_p and the LBE's own pressure gradient are different discrete operators whose mismatch scales with the density jump and with $1/W_{\text{int}}^2$, so a well-resolved interface matters more here than the phase-field literature's usual $W_{\text{int}} = 4$ suggests.

17.2 Water entry of a square

A square dropped into a free water surface: approach, impact, cavity, splash-up jets, closure. The domain is $6L \times 7L$ for a square of side L, periodic in x, with a floor and a lid, and the free surface flat at $y = 4L$. Gravity is set from the release height, $g = U^2/(2\,h\,L)$ with h the drop in body widths, so the square arrives at the reference speed U; time is reported as tU/L. The defaults are $L = 48$, a density ratio of 50, $\text{Re} = 2000$, $U = 0.05$ and a body twice the density of the water. Both phases are given the same kinematic viscosity rather than the true 15 : 1 air-to-water ratio: the air's only job in this problem is to be light and get out of the way, and its physical viscosity buys nothing while costing stability at these relaxation rates.

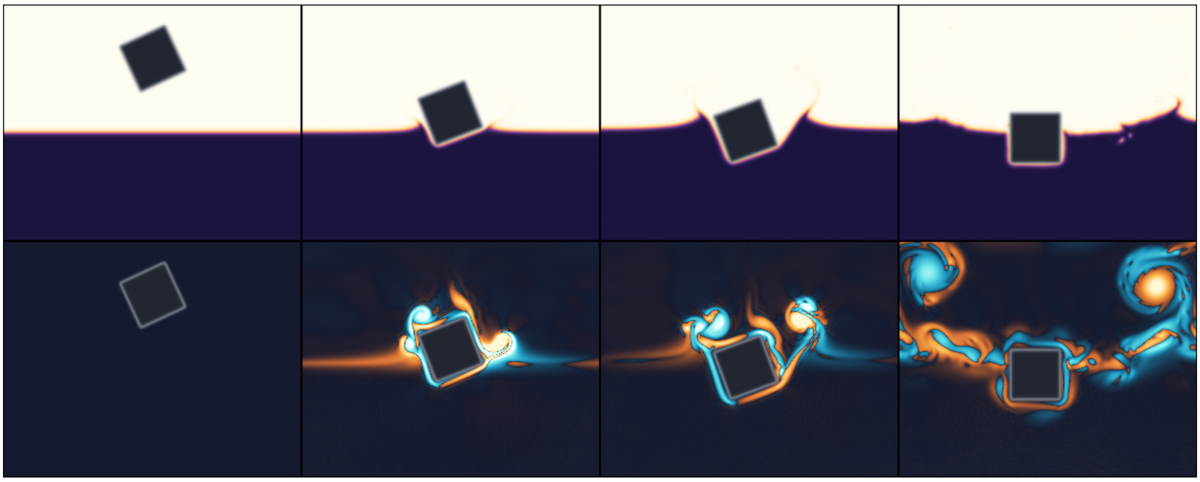


Figure 22: Water entry of a square at 0.6 times the water's density, released at 25° from one body width above the surface; $L = 48$, density ratio 50, $\text{Re} = 2000$. Phase field above, out-of-plane vorticity below, at $tU/L = 0$, 2.4, 3.3 and 16.0. The square strikes on one corner, sheds vortices of *opposite* sign from the two corners — the imbalance between them is what turns it — sinks past its draft, and returns upright and floating. A body lighter than the fluid it displaces is the case the classical fictitious-mass arrangement cannot express at all (§10.7).

It falls rather than being pushed. De Rosis and Enan run this problem with a Peskin immersed boundary and a *prescribed* constant entry velocity. Here the body is coupled by volume penalisation, the reaction on it is known at every node for free, and it closes Newton's equations — so the deceleration on impact is a result rather than an input. That is the harder direction, and it is the reason the coupling had to be arranged so that the body and the fluid are solved at the same instant: the classical fictitious-mass step divides by $m_b - m_f$, which vanishes for a neutrally buoyant body and changes sign for a light one, so the floating case — the only interesting one, since floating is what a density ratio is *for* — is precisely what that arrangement cannot express.

The floating case. Measured at `-rhob 0.6`, the square enters, stops, reverses at $tU/L = 5.0$ and rises through $0.56\,L$ before falling back. Nothing in that sequence is available to a formulation whose denominator has already changed sign, and it is the clearest single demonstration that the coupling is well posed across the whole mass range rather than on one side of it.

The tilt case. `-theta` releases the square rotated, and the roll follows from the same reaction: an off-axis square strikes one corner first, the reaction on that corner is not through the centre, and it slaps flat — the reason a belly-flop hurts. The rigid-body solve is a coupled 3×3 in sway, heave and roll, coupled because the first moments of the fictitious fluid mass about the body centre are non-zero the moment the body straddles an interface, with half of it standing in water and half in air. It is also the harshest test of the coupling here, since the impact torque arrives over a couple of steps. The hydrostatics the case rests on — draft, righting couple, metacentric sign — is checked against Archimedes and metacentric theory in `validation/floating_body`, which is where those numbers are; this case asserts nothing.

One failure that was a driver bug, not a model one. The body's effective mass is $m_b - \int \chi\rho$, and the penalisation reads ρ off the phase field. The conservative Allen–Cahn form keeps ϕ close to $[0, 1]$ but does not guarantee it, and an unclamped interpolant lets that bracket collapse or change sign — which is a runaway in Newton's equation rather than in the flow. The signature was unambiguous once looked for: the fluid stayed finite at $\max|\boldsymbol{u}| = 2.1 \times 10^{-2}$ while the body left the domain at $5 \times 10^{18}\,U$. The driver now clamps the interpolant with the same expression the collision operator's equation of state uses, and for the same reason.

There is no exact answer here, which is why this is a demonstrator. Wagner's slamming theory covers a *wedge*, whose wetted length grows smoothly; a flat-bottomed square has a theoretically singular impact pressure at first contact and no closed form at all. What can honestly be shown is the qualitative sequence and a force history whose shape is physical, not a number to quote. There is also no contact-line model, so the angle at which the free surface meets the square's sides is not physics: read the splash and the cavity, which are inertia-dominated, not the meniscus, which is not.

file	what it shows
`water_entry.mp4`	upright, body twice the water's density, ratio 50
`water_entry_ratio800.mp4`	the same entry at a density ratio of 800
`water_entry_floating.mp4`	`-rhob 0.6 -theta 25`: corner first, slaps flat, sinks past its draft, returns upright

The third is the one the previous formulation could not produce at all, and it is the clearest picture of what the roll does: the square strikes one corner, sheds a vortex pair of opposite signs from the two corners, and the imbalance between them is what turns it.

18 A second implementation, in CUDA

Everything measured so far in this document is the Threads backend. This section records what happened when the code was taken to a GPU, why that made a second implementation necessary, and what the second implementation then measured.

18.1 The result that forced the question

On a Tesla T4 (sm_75, CUDA 12.8, FP32), seven validation executables build and run after six separate fixes for `nvcc` restrictions that neither the Serial nor the Threads backend can see. All six are committed with explanatory comments, and they are worth listing because any future port will meet them:

1. A kernel-launching member function may not be private: an extended `__host__ __device__` lambda's enclosing function must not have private or protected access.

2. A variable may not be first-captured inside an `if constexpr`. In `run_step` this was repaired with a plain `if`, which still folds because the tested value is a compile-time constant.
3. The same restriction in the wall closure, where a plain `if` does *not* work: the captured member does not exist in the unforced instantiation, so the object has to be named before the branch.
4. Lattice constants declared `static constexpr int cx[Q]` are unreachable from device code when indexed at runtime. They are now `KOKKOS_INLINE_FUNCTION` accessors wrapping function-local arrays. **Invariant to preserve: `src/` contains no static constexpr array members.**
5. The same, in `MonomialBasis`.
6. Extended lambdas may not appear inside a *generic* lambda, and function-local types may not be template arguments of one.

BGK and TRT then reach 420–447 MLUPS at 64^3, with mass drift **exactly** 0.000×10^0 on every row — about 8.4× the M1 Threads figure of §19.

But the central-moment operators do not run. The two `MomentCollision` configurations did not finish in **seventeen minutes** at 64^3 over 50 steps, against ~ 0.03 s for BGK. That is roughly four orders of magnitude and it is not explained. The obvious hypothesis was register spilling, and `-Xptxas -v` rules it out: **zero spill bytes**, largest stack frame 96 bytes, 116 registers per thread. Occupancy at 116 registers is about 25% on a T4 — a real effect, and far too small a one.

This matters more than a performance note. The aorta demonstrator and the whole validation campaign of §13 run central moments, so the 447 MLUPS is a **BGK-only** figure and must not be quoted as this code's GPU throughput.

Which leaves one question. Is the collapse a property of the *algorithm* — 27 moments, two transforms and a relaxation schedule per node — or of *this code's route to the device*? Nothing in the Kokkos implementation can answer that, because every hypothesis is tested against the same code. The only decisive experiment is a second implementation.

18.2 The answer

`GPU/` is that experiment: a lattice Boltzmann solver written directly in CUDA, sharing no headers with `src/`, not built by the parent `CMakeLists`, and not a port. Only the physics and the output format are deliberately the same, which is what makes the comparison mean anything.

device	operator	grid	MLUPS	GB/s
Tesla T4	BGK	128^3	950.2	205.2
Tesla T4	central moments	128^3	950.4	205.3
NVIDIA H200	BGK	512^3	17233.4	3722.4
NVIDIA H200	central moments	512^3	17171.4	3709.0

The H200 rows were measured by Yang; see §18.3. Mass drift is exactly 0.000×10^0 on every row. Central moments runs at 99.6% of BGK on two card generations, and 3722 GB/s is 77.3% of the H200's 4814 GB/s peak — i.e. the kernel is memory-bound, which is what a correctly written lattice Boltzmann kernel should be. Taylor–Green at 512^3 (134M nodes, 256 000 steps) completes in 2046 s.

So the collapse of §18.1 is not the algorithm. The same 27-moment separable transform, the same relaxation schedule and the same Esoteric Pull run at BGK speed when written directly. Whatever is wrong is in how the Kokkos code reaches the device, and the next step there is measurement (`ncu -set full` on one kernel), not another hypothesis.

18.3 The H200 measurement, and what it says

The 512^3 figures above were measured by Yang on an NVIDIA H200 (sm_90, CUDA 12.8, FP32). They are the strongest performance statement this project has, and they say three things that the T4 alone could not.

Seventy-seven per cent of peak bandwidth is the result, not the MLUPS. **A lattice Boltzmann kernel does almost no arithmetic per byte moved, so a correct one is bound by the memory system and its only meaningful efficiency measure is the fraction of peak bandwidth it sustains. At 3722 GB/s against the card's 4814 GB/s the kernel reaches 77.3%, which is close to what a plain streaming benchmark achieves on the same hardware. The remaining quarter is the irreducible cost of the neighbour access pattern: 27 directions, each landing in a different cache line. A kernel far below this figure has arithmetic to remove; one far above it is not moving the data the scheme requires.**

The efficiency rose with the card, which is not automatic. **On the T4 the same code sustains 66% of 320 GB/s; on the H200, 77% of 4814 GB/s. The throughput ratio between the two is 17.6× against a bandwidth ratio of 15.0× — the code gains more than the memory system alone gives it. That is the signature of a kernel whose limit is bandwidth rather than latency or occupancy: a latency-bound kernel would have fallen behind the bandwidth ratio on the larger machine, not moved ahead of it. It also means the figure should be read as a floor for future hardware rather than a ceiling.**

Central moments at 99.6% of BGK now holds on two architectures. **One card could be a coincidence of that card's particular ratio of arithmetic throughput to bandwidth. Turing and Hopper are three years and two memory technologies apart, and the operator is free on both. Taken with §18.1, this is what settles the question: the central-moment collision is not intrinsically expensive on a GPU, and the collapse in the Kokkos build is a property of that build.**

Two things the measurement does not say. **It is FP32, and it is the *fluid* kernel alone — no thermal, magnetic or geometry coupling was run on the H200, so the multiphysics figures of §18 remain T4-only. And the problem had to be sized to the card to obtain it: 512^3 is 134 million nodes and 14.5 GB in FP32, which fits only because Esoteric Pull keeps one population set instead of two. The same case in a two-lattice scheme would need 29 GB and would not have run at this size. Taylor–Green at 512^3 over 256 000 steps completes in 2046 s at 16 792 MLUPS, with mass drift exactly 0.000×10^0 throughout.**

18.4 Written so it can be checked without a device

Every function in the numerical core is marked `LBM_HD`, which expands to `__host__ __device__` under `nvcc` and to *nothing* under a plain C++ compiler. That is not portability for its own sake. The per-node update is an ordinary function; the CUDA kernel is three lines of index arithmetic around it; and a host driver calls *the same function* in a serial loop.

Why a serial loop is a valid substitute for a grid of threads. Under Esoteric Pull each storage slot has exactly one writer per step, and that writer is also its only reader (§3.2). The nodes of one step therefore commute: any order gives the same answer, so a `for` loop is not an approximation to the launch, it is the same computation. This is also why the scheme needs no temporary buffer on the device.

What a host build cannot catch is a launch configuration error, a register pressure problem, or a host–device transfer bug. What it catches is wrong physics, which is the failure mode that is expensive to diagnose remotely. It earned its place immediately: `eq_factors` added a c_s^2 that `fwd1d` already subtracts, so equilibrium was not a fixed point of the collision. The flow would have decayed *plausibly but wrongly*.

The property was later extended to whole applications. Each driver is written against aliases that resolve to the CUDA classes under `nvcc` and to the host drivers otherwise, so one source builds and runs both ways with no `#ifdef` of its own, and `cmake -DLBM_HOST_ONLY=ON` configures with no CUDA toolkit present at all.

18.5 Thermal transport, magnetohydrodynamics and geometry

The three capabilities the parent had and this code did not were added on the same principle: written and verified on a machine with no GPU, then built with `nvcc` and run.

- **Scalar transport** on D3Q7 ($c_s^2 = 1/4$), first-order equilibrium, with adiabatic and Dirichlet walls, and Guo forcing in a uniform and a Boussinesq form.
- **Magnetohydrodynamics** by Dellar's vector-valued induction distribution, again on D3Q7, with the Maxwell stress entering the *fluid* equilibrium's second moment rather than as a body force. Both the BGK and the central-moment fluid operators carry it; for the latter the equilibrium central moments of the Maxwell term are obtained by transforming the term itself, which is exact at every order.
- **Geometry** as one byte per node. A solid cell is skipped, which under Esoteric Pull *is* halfway bounce-back: the slot a fluid node emits into toward a wall is a slot the wall owns, the wall never touches it, and the fluid node reads its own emission back one step later, reversed.

The coupling must be simultaneous. This is not an implementation detail, and §9.3 explains why: stepping the fluid against a temperature or a magnetic field from the previous step is a first-order splitting error that does *not* vanish under refinement. The drivers therefore refresh T and $\boldsymbol{b}$, then step the fluid against them, then step T and $\boldsymbol{b}$ against the velocity the fluid just wrote. That costs one extra light pass, and it is the cheaper of the two mirror images: recovering $\boldsymbol{u}$ for the coupled fields instead would need a separate pass over 27 populations, where refreshing T costs 7 and $\boldsymbol{b}$ costs 21.

18.6 What the host verification established

`test/host_physics.cpp` runs whole simulations against closed-form answers. Figures are worst relative error; where the two precisions agree, the number is discretisation error and not round-off.

case	FP32	FP64
Poiseuille between bounce-back walls	1.4×10^{-3}	1.3×10^{-3}
closed box, mass over every slot	1.6×10^{-6}	1.4×10^{-14}
insulating box, scalar conserved	9.7×10^{-9}	1.0×10^{-13}
conduction between Dirichlet walls	1.4×10^{-6}	2.5×10^{-15}
decaying sinusoid (the D3Q7 diffusivity)	8.1×10^{-4}	8.1×10^{-4}
advected sinusoid (the advective flux)	4.4×10^{-4}	4.4×10^{-4}
uniform buoyancy (the Boussinesq path)	6.7×10^{-5}	2.9×10^{-13}
resistive decay (induction alone)	4.1×10^{-3}	4.1×10^{-3}
shear Alfvén wave, BGK	8.1×10^{-4}	3.3×10^{-4}
shear Alfvén wave, central moments	8.4×10^{-5}	3.8×10^{-4}

The wall is exactly where it should be. For Poiseuille the interesting quantity is not the error but how it scales: a wall in the wrong place gives an error going like $1/H$, a correctly placed one leaves the $1/H^2$ truncation error. In FP64 the amplitude error times H^2 is 0.333 at $H = 16$, 32 and 64 — constant to three digits over two doublings. In FP32 the same run reads 0.336 at $H = 16$ and 0.498 at $H = 32$: the raw-population storage runs out of mantissa before the discretisation error does, which is the cost of not having the shifted storage of §4.

Conduction confirms where the thermal plane sits. With Dirichlet layers at $y = 0$ and $y = H + 1$ the measured gradient is $0.062500 = 1/16$ exactly, the distance between the two *planes*; measuring between the *nodes* would give $1/15 = 0.066667$. Both walls — no-slip and isothermal — land on the same two planes, which they must, or H would mean two different things in the Rayleigh number.

The D3Q7 sound speed is $c_s^2 = 1/4$, and the sinusoid proves it. The decay-rate error is -0.327% at $L = 16$ and -0.081% at $L = 32$, a ratio of 4.05. A wrong c_s^2 would be an offset that does not converge.

18.7 What the device then confirmed

The CUDA build was clean on the first attempt: eight targets, 1 m 03 s, zero errors and zero warnings. None of the six restrictions of §18.1 arose, because this code was written already knowing them — structs rather than lambdas, accessor functions rather than static constexpr arrays, no function-local types as template arguments.

Two host predictions could only be tested at resolutions the laptop could not reach, and both held.

Rayleigh–Bénard needs no reference table. Linear stability puts the onset of convection between rigid plates at $\mathrm{Ra}_c = 1707.762$, independent of Pr and of the fluid. Central moments, 15 thermal diffusion times, on a T4:

H	nodes	Ra = 800 (below)	Ra = 5000 (above)	MLUPS
24	4 992	Nu = 0.998848, $\lvert \boldsymbol{u} \rvert_{\max}$ 5.60×10^{-6}	Nu = 2.100998, 1.51×10^{-2}	510
48	19 200	Nu = 0.999703, 1.00×10^{-6}	Nu = 2.109297, 7.60×10^{-3}	1099
96	75 264	Nu = 0.999912, 8.42×10^{-7}	Nu = 2.122282, 3.85×10^{-3}	612

Below onset the exact answer is Nu = 1, and the host runs had argued the deviation was a second-order artefact of a spurious wall-driven flow — a residual that is *independent of the seeded perturbation* (4.263×10^{-5} with no seed at all against 4.271×10^{-5} with a seed a hundred times larger) and linear in Ra. Measured on the device: $\mathrm{Nu}-1=-1.152\times10^{-3}$, -2.97×10^{-4}, -8.8×10^{-5}, convergence exponents 1.95 and 1.76.

Above onset those three numbers are not grid convergence, and say so themselves. They rise by *increasing* amounts, +0.0083 then +0.0130, which no converging sequence does. The $H = 96$ run has not reached steady state: its last five probes read 2.120058, 2.120604, 2.121189, 2.121720, 2.122282, still climbing at $\sim 7\times10^{-4}$ per diffusion time. Nu = 2.1223 **is a lower bound, not a converged value.** A converged one needs a much longer run — fourteen minutes per 15 diffusion times at $H = 96$ on this card. What the three rows do establish is the two-sided bracket of Ra_c, which is what the case is for.

$|\boldsymbol{u}|_{\max}$ halves when H doubles, which is required: with Ra and D fixed in lattice units the velocity scale is D/H, so diffusive scaling demands $u \sim 1/H$. A quantity that did not do that would mean the non-dimensionalisation was wrong.

The throughput is not monotonic in H, and the L2 cache explains it. At $H = 24$ the grid is 39 blocks against 40 streaming multiprocessors — barely one each, so the kernel is occupancy-bound at 510 MLUPS. At $H = 48$ the whole lattice is $19\,200\times27\times4 = 2.1$ MB and fits inside the T4's 4 MB L2: 1099 MLUPS, *faster than the bandwidth-bound* 128^3 *benchmark reaches.* At $H = 96$ it is 8.1 MB, L2 no longer holds it, and it falls back to 612.

18.8 The shear Alfvén wave, and the coupling-order signature

The wave of §12.9 is an exact solution of the full *nonlinear* incompressible MHD equations. Its two measurements fail differently, and that is its diagnostic value: an error in the **Lorentz coupling** shows as the wrong wave *speed*, immediately and at any resolution, while an error in the **coupling order** shows only in the *damping*, and only under refinement.

Diffusive scaling ($v_A \sim 1/L$ at fixed ν and η, so the Reynolds and Lundquist numbers are the same on every grid), FP64, central moments:

L	speed error	ratio	damping error	ratio
32	$+2.222\times10^{-4}$		$+2.080\times10^{-3}$	
64	$+5.570\times10^{-5}$	3.99	$+4.972\times10^{-4}$	4.18
128	$+1.397\times10^{-5}$	3.99	$+1.249\times10^{-4}$	3.98

Both converge at second order: the coupling is simultaneous, not lagged. These values are identical to those the host driver produced, to every digit printed — a serial loop and 128-thread CUDA blocks agreeing to the printed precision.

A warning about running this sweep in FP32. The same sweep in single precision reads $+2.9\times10^{-4}\to-4.9\times10^{-4}\to-1.1\times10^{-3}$ on the speed and $+2.7\times10^{-3}\to+3.8\times10^{-3}\to+5.1\times10^{-2}$ on the damping — errors that *grow* with resolution, which is exactly the signature of the bug the sweep exists to find. They are not. Diffusive scaling puts $v_A \sim 1/L$, so at $L = 128$ the perturbation is 5×10^{-4} on populations of order 0.3, leaving about three decimal digits above the FP32 noise floor. **Run this sweep in FP64 or not at all**, and treat it as a caution about how easily a precision failure reads as a physics failure.

18.9 Orszag–Tang in three dimensions

In both wave cases $\nabla\cdot\boldsymbol{b}$ is structurally zero and reports round-off whatever the scheme does. Orszag–Tang's nonlinear dynamics makes every component depend on every coordinate, so a scheme that generates monopoles shows it. The initial condition is that of §13.7; Re = 100, Ma = 0.034, $\mathrm{Pr}_m = 1$, central moments, to $t = 4$:

M	nodes	steps	$\max\lvert\nabla\cdot\boldsymbol{b}\rvert/k\lvert\boldsymbol{b}\rvert$	energy rise	MLUPS
64	262 144	5 870	7.177×10^{-2}	0.000×10^{0}	359.2
96	884 736	8 805	3.461×10^{-2}	0.000×10^{0}	378.4
128	2 097 152	11 741	2.002×10^{-2}	0.000×10^{0}	391.6

$\nabla\cdot\boldsymbol{b}$ converges at **second order** — exponents 1.80 and 1.90 against 1.5× and 1.33× refinements. (Coarse host runs at $M = 16$–32 had suggested only first order; those grids were deep in the under-resolved regime and the estimate was too pessimistic.) The antisymmetry of the induction equilibrium's first moment is what keeps this bounded, and the unit checks assert that antisymmetry directly.

Ideal incompressible MHD has $\mathrm{d}E/\mathrm{d}t = -\nu|\nabla\boldsymbol{u}|^2 - \eta|\nabla\boldsymbol{b}|^2$, so $E_u + E_b$ must never rise, and it does not at any resolution here. A spurious rise the host had found at $M = 12$ (1.10×10^{-2}, falling to zero by $M = 32$) was under-resolution, as argued, and it is gone.

The $M = 128$ history is the physics. Up to $t \approx 0.8$ the magnetic energy barely moves ($0.4898 \to 0.4673$ of E_0) while the kinetic energy falls by a third ($0.5102 \to 0.3316$): the flow is winding up the field, and the transfer is nearly lossless. After that both decay together — resistive dissipation in the current sheets. And $\nabla\cdot\boldsymbol{b}$ is largest exactly where the sheets are sharpest, peaking near $t \approx 1.6$ and then falling back fivefold. $J_{\max}$ peaks between $t = 1.0$ and $t = 1.4$ on a $\Delta t = 0.2$ sampling, which is consistent with the paper's "maximum at about $t = 1$" and is all the sampling can support; no stronger claim is made.

18.10 A cost estimated wrong by a factor of ten

The cell-flag array is one byte per node against 216 of population traffic in FP32, so its cost was written down from that byte count as half a per cent — "a rounding error in bandwidth". Measured against the commit immediately before geometry existed, on one card, at 128^3:

	BGK	central moments	
before geometry existed	978.15	977.35	
flags loaded on every node	923.99	919.81	−5.5%, −5.9%
geometry behind a template	950.16	950.37	−2.9%, −2.8%

Ten times the predicted cost. What a bandwidth-bound kernel pays for is not the byte but the extra memory *stream*: more transactions, more cache pressure, another address in flight. Making the presence of geometry a compile-time template parameter, so a periodic run never emits the load, recovers about half of it.

The remaining 2.9% is not explained, and it is not the obvious candidates. `-Xptxas -v` on the two builds: the old kernel used 96 registers across its 4 instantiations, the new one uses 63–80 across its 40, and both spill zero bytes. Registers went *down*. What else differs is a much larger kernel parameter struct and a runtime test for the coupled-velocity write. This is recorded as open, and `ncu -set full` is the next step rather than another guess — the same conclusion, and for the same reason, as §18.1.

18.11 Both multiphase engines, ported

The colour gradient of §10.8 and the phase field of §10.1 are both in `GPU/`, as `colour.cuh` and `phasefield.cuh`, with `src/droplet.cu` and `src/bubble.cu` driving a static droplet against Laplace's law. The free surface of §11 is not, and §11.3 is the reason: it `static_asserts` against Esoteric Pull, so porting it means adding a two-lattice storage path plus five kernel launches with a fence between each.

The equilibrium that would not fit in registers. The colour gradient arrived at **20.2** MLUPS at 64^3 in FP32 against the single-phase core's 950. A factor of 47 is far more than the extra arithmetic explains, and the `-DLBM_PTXAS_VERBOSE=ON` build — which exists for exactly this suspicion — said why: zero spills, but a **216-byte stack frame**. Two 27-element arrays did not fit in registers and lived in local memory, read and written at every node.

The cause was written in the operator's own banner, as a deferred decision: Eq. (18) is not the product-form equilibrium, so its moments have no factorised closed form, and the implementation built the equilibrium *populations* and transformed them — "a closed form could be derived; until it is measured to matter, it would be a second thing to keep correct." That was the measurement.

The closed form is separable in a way the raw equilibrium is not. Writing the order-≥ 3 central moment of index (p, q, r) as

$$k_{pqr} = c_N(P_3, \rho)\, u_x^p u_y^q u_z^r - 3S\, D_p(u_x) D_q(u_y) D_r(u_z) + H_{pqr}(\boldsymbol{u}), \tag{128}$$

with $D_0 = 1$, $D_1(u) = -u$, $D_2(u) = u^2 - c_s^2$ and H collecting the thirteen source terms of orders 2, 4 and 6, every moment becomes a product of three one-dimensional factors and a small fixed table. Nothing is an array. One transform in, one out.

	before	after
stack frame	216 bytes	**0 bytes**
spill stores / loads	0 / 0	0 / 0
registers	119	124
throughput, 64^3 **FP32**	**20.2 MLUPS**	**252.5 MLUPS**

That is 12.5×, and 252.5 against 950 means the colour gradient costs 3.8× a single-phase step — for two distributions, three kernels and a heavier collision, which is about what the work implies. FP64 is unchanged to every digit printed, on the device and on the host, so this is the same operator and not a cheaper approximation of it: `test/host_colour.cpp` keeps the old path as `reference_collide()` and asserts the two agree over 60 states to 5.9×10^{-16} in FP64 and 2.1×10^{-07} in FP32, about one ulp.

The phase field was written afterwards and therefore written already knowing this. It runs at **354.6** MLUPS with **0 bytes** of stack frame on every one of its six kernels, by construction rather than after a measurement.

A conservation claim FP32 cannot keep. The conservative Allen–Cahn form conserves $\sum_i \phi$ exactly, and in FP64 the port shows it: 10^{-13} at 150, 600, 2400 and 9600 steps alike. FP32 cannot represent that, and the first version of the test asserted it anyway and failed. Rather than loosen the tolerance to whatever passed, the drift was measured across those four run lengths — -2.9×10^{-6}, -3.4×10^{-5}, -1.0×10^{-4}, -1.4×10^{-4} — fast while the seeded profile relaxes onto the discrete equilibrium, then nearly stopping. Not a leak. The test now bounds drift *per step* against one ulp, which round-off cannot beat and a real bug would exceed by orders of magnitude.

An instability that was the viscosity, not the model. At a density ratio of 100 the phase-field droplet diverged, and it was nearly written up as a capability limit. Matching the *dynamic* viscosity across the ratio leaves the heavy phase with a hundredfold smaller kinematic viscosity, so $\omega = [\mu/(\rho c_s^2) + 1/2]^{-1} = 1.994$ against a limit of 2; matching the *kinematic* viscosity runs the same case to -3.77% with a spurious current twenty times smaller. The driver now prints ν and ω per phase before the run and flags $\omega > 1.9$, because that number was reconstructed in a post-mortem when it could have been on screen beforehand.

18.12 The two models on one case

Both engines on one measurement, at settings matched rather than defaulted: Tesla T4, FP32, 64^3, $R = 16$, $W = 4$, σ asked 10^{-3}, $\nu = 0.1$ in *both phases of both models*, the same $3W$ averaging gap, 60000 steps.

	colour gradient		phase field	
γ	error	spurious $\lvert\boldsymbol{u}\rvert$	error	spurious $\lvert\boldsymbol{u}\rvert$
1	+0.94%	1.98×10^{-5}	−6.17%	6.65×10^{-6}
10	+0.67%	1.22×10^{-5}	−4.32%	1.11×10^{-6}
100	−3.36%	3.51×10^{-5}	−3.77%	2.99×10^{-7}

The colour gradient is the more accurate on Laplace's law at every ratio tested; the phase field carries the smaller spurious current at every ratio, by a factor of 3 at $\gamma = 1$ and of 118 at $\gamma = 100$, where its current *falls* with the ratio while the colour gradient's does not. The accuracy gap closes as the ratio rises, and at $\gamma = 100$ the two are within half a percentage point.

What building the table found. Both engines had already been measured, both agreed with their host references, and neither had anything visibly wrong. Putting them side by side produced a number that could not be right: two effective radii, 16.78 and 16.20, for one seeded $R = 16$. The cause was in the driver rather than in either model — `droplet.cu` seeded $\tanh[(r-R)/W]$ where the paper's Eqs. (41)–(42), the parent's `validation/static_droplet.cpp` and `bubble.cu` all seed $\tanh[2(r-R)/W]$, so every colour-gradient case on the GPU had run at twice the intended interface width.

It hid because a wrong interface width is still a *consistent* simulation. The device matched the host to every digit, all 51 algebraic identities in `host_colour.cpp` held, and σ converged and stayed. Nothing was inconsistent; only wrong. The first version of this table, run before the fix, read −2.50%, −3.54%, −13.61% — a model apparently collapsing under density ratio and losing to the phase field at $\gamma = 100$. Ten percentage points at $\gamma = 100$ were the seed.

Two consequences beyond the table. The port now agrees with the parent to about a tenth of a point at both ratios the parent measures (0.87% and 0.77%), having been threefold worse, with that gap recorded as an open question about the model. And a claim in §18.11's README — that the sign of the error is geometry, +2.5% at $R = 10, W = 2$ against −2.6% at $R = 16, W = 4$, the two constituent errors "evidently crossing" between those resolutions — was itself the artefact. At the corrected seed and fixed $\gamma = 10$ the error is +4.71% at $R = 10, W = 2$ and +0.67% at $R = 16, W = 4$: same sign, monotonically smaller as the interface resolves.

What is still not controlled: the colour gradient has no mobility and the phase field has no recolouring, so the mechanism maintaining the interface differs by construction. That is a difference between the models, not a parameter left unmatched.

18.13 What this implementation does not do

Absent, not merely untested:

- **No shifted storage.** Raw populations only, and the Poiseuille convergence above shows what it costs in FP32.
- **No open boundary for the scalar.** Outflow needs a donor map and a second kernel after a fence; reading a donor inside the main kernel is a genuine race under Esoteric Pull, because the two slots a node reads are the two it writes (§8.5).

- **No physical magnetic wall condition.** A non-fluid cell is skipped, which on this storage is bounce-back on the induction distribution — neither the perfectly conducting nor the insulating condition. Every MHD case here is periodic, and no wall-bounded MHD result should be read off this code.
- **No free surface**, and the reason is streaming rather than effort: §11.3. Porting it means a two-lattice storage path plus five kernel launches with a fence between each.
- **No central moments for the phase field's fluid operator.** The parent has `MultiphaseCentralMoments.hp` what is on the GPU is the BGK potential form. The colour gradient *is* in central moments.
- **No wetting, no contact angle, no external force field, and no open boundary for ϕ.**
- **D3Q27 only**, no D3Q19, no TRT or raw MRT, no regularised walls, no height-field input, and therefore no aorta and no city.

19 Performance

Measurements on an Apple M1, Threads backend, four threads, 96^3. Run-to-run spread is under 2%.

These are CPU figures and are not this code's ceiling. GPU measurements — on a Tesla T4 and an NVIDIA H200, for both the Kokkos code and the independent CUDA implementation — are in §18, together with the reason the two had to be measured separately.

These tables were re-measured on D3Q27. The streaming and storage comparison was originally taken on another lattice, one this document no longer covers, so keeping the old numbers would have meant either mislabelling them or deleting the result; it was re-run instead. Both tables come from the same pair of runs (one per precision) and are therefore mutually consistent, but they are **2–3× faster than the figures this section carried previously** — the earlier numbers predate the optimisations listed at the end of this section. Rates below are the current build.

streaming and storage (BGK, D3Q27)		bytes/node	MLUPS FP64	MLUPS FP32
two-lattice	raw	432	37.8	46.9
Esoteric Pull	raw	216	**39.5**	**52.9**
Esoteric Pull	shifted	216	39.0	49.5

collision operator (Esoteric Pull + shifted)	BGK	TRT	MRT	central moments
D3Q27, FP64	39.2	35.4	20.7	20.7
D3Q27, FP32	52.0	45.2	21.8	21.6

Caveat. These ratios must be read with care. At ~ 2.3–16.3 GB/s against the ~ 68 GB/s the machine can sustain, this platform is not memory-bound at these rates, so the collision cost shows at full price and the streaming advantage is understated. Lattice Boltzmann is bandwidth-bound on real hardware, and Esoteric Pull moves exactly half the bytes, so its asymptotic gain is $\sim 2\times$. **Re-run `lbm_app` on the target hardware before choosing an operator on cost grounds.**

Optimisations found through the benchmark harness. All were measured rather than assumed:

- Restructuring the streaming hot loop to operate on opposite *pairs* rather than per index removed a runtime `i & 1` test: D3Q27 Esoteric Pull went from 13.5 to 14.0 MLUPS, moving ahead of two-lattice.
- `ProductBasis::pi()` was calling a `constexpr` linear search over the velocity set with *loop-variable* arguments, so it ran live: $2Q^2$ comparisons per node. A precomputed table took the moment operators from 1.8 to 6.6 MLUPS — a 3.7× defect, not a design cost.
- The same class of defect reappeared when the operator was made basis-generic: `p_of(n)` as `n/9`, `(n/3)%3` costs three integer divisions per moment with a runtime index. Table lookups recovered 9%.
- Hoisting the equilibrium out of TRT's non-vectorising pair loop gained 37% on D3Q27.

Compacting the neighbour list to the odd half was also tried, made no measurable difference, and was reverted.

20 Building and running

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
    -DKokkos_ENABLE_SERIAL=ON -DKokkos_ENABLE_THREADS=ON
cmake --build build -j8
cd build && ctest --output-on-failure
```

Precision is a configure-time switch, `-DLBM_PRECISION=float` (default `double`). Kokkos is resolved in the order: an installed `find_package` (use this on a cluster, where Kokkos is built once against the correct CUDA architecture), then `extern/kokkos` as a submodule, then `FetchContent`. On a CUDA machine, configure with `-DKokkos_ENABLE_CUDA=ON -DKokkos_ARCH_<ARCH>=ON`.

executable	role
`tests/*`	unit tests (lattice, streaming, equilibrium, collision, moments)
`validation/poiseuille`	walls, magic parameters, resolution behaviour
`validation/decaying_flows`	convergence, isotropy, cross-lattice consistency
`validation/galilean`	frame invariance across operators
`validation/thermal`	scalar transport + one coupled convection case
`validation/mhd`	resistive decay, Alfvén wave, Orszag–Tang divergence
`validation/laplace`	$\Delta p = \sigma/R$, spurious currents, density-ratio sweep
`validation/layered_poiseuille`	two-fluid channel: $\mu(\phi)$, $\boldsymbol{F}_\nu$, the pressure force
`validation/floating_body`	Archimedes' draft and the sign of the metacentric height
`validation/grid_convergence`	full lattice × operator sweep, CSV
`validation/natural_convection`	Rayleigh sweep against the benchmark
`validation/orszag_tang`	published comparison and stability runs
`lbm_app`	throughput benchmark

The four in the lower block are analysis runs, not regression tests: they take minutes and are excluded from `ctest`.

21 Known limitations

Stated plainly, because each has been measured.

1. **The central-moment operators are unusable on a GPU in this code.** Two `MomentCollision` configurations did not finish in seventeen minutes at 64^3 over 50 steps, where BGK took $\sim 0.03\,\mathrm{s}$ — roughly four orders of magnitude, with register spilling ruled out by `-Xptxas -v` (§18.1). Since the campaign of §13 and the aorta both run central moments, the 447 MLUPS BGK figure is *not* this code's GPU throughput and must not be quoted as such. The cause is not the algorithm: the same scheme runs at 99.6% of BGK in the independent CUDA implementation. It is therefore something in this code's route to the device, and it is open.

2. **The CUDA implementation of §18 covers a fraction of this one.** D3Q27 only; no shifted storage; no open boundary for the scalar; no physical magnetic wall, so every MHD case there is periodic; no D3Q19, TRT, raw MRT, regularised walls or height-field input. It cannot run the aorta or a city. Both immiscible models of §10 are there (§18.11); the free surface is not, and cannot be without a two-lattice path. It exists to answer one question and to carry the multiphysics forward on a device, not to replace this code.

3. **No MPI.** The code is single-rank. `Domain` carries an `mpi_halo` hook and the halo machinery exists, but domain decomposition is not implemented. Note that Esoteric Pull will require a two-way, parity-dependent halo exchange (§3.3).

4. **Convergence studies must watch the Mach number.** The inlet Poiseuille of §13.6 reaches a velocity-dependent error term that does not refine away, and at $U_0 = 0.005$ it dominates by $L_y = 641$ and pulls the fitted rate from 2.067 down to 1.964. This is the cubic aliasing defect $c_{i\zeta}^3 = c_{i\zeta}$ discussed in [12], not a boundary-condition problem: it survives moving the outlet, changing its order, and refining. Any new convergence test should sweep U_0 before a shortfall in the fitted rate is blamed on the scheme.

5. **The scalar module has no shifted-storage default.** T_{ref} must be set to the mean temperature by the user for FP32 accuracy; leaving it at 0 reproduces unshifted storage.

6. **The scalar module has no on-node zero-flux condition.** `ScalarAdiabatic` is bounce-back, so an insulated wall always sits half a cell inside the boundary node. That is consistent with halfway bounce-back for the fluid and inconsistent with the regularised condition, which is on-node — so a case whose *velocity* boundary must be on-node, because a wall moves, cannot have an on-node adiabatic wall to go with it. Case 3.5 of §14 is exactly that case: its isothermal walls sit on nodes and its two adiabatic flanks half a cell in, making the cavity $N-1$ tall and $N-2$ wide. The aspect-ratio error is $1/(N-1)$ and refines away, but it is a gap in the boundary set rather than a property of the problem.

7. **Conjugate heat transfer varies the diffusivity, not the heat capacity.** `ScalarBGK::omega_of` gives every node its own relaxation rate, which reproduces a conductivity ratio only where ρc_p is uniform: the scheme transports $\sum_i g_i$, and that sum is the temperature rather than the enthalpy. A jump in volumetric heat capacity needs a different transported variable and is not implemented. Where ρc_p *is* uniform the interface is exact, not merely second order — §14 measures $10^{-9}\%$ at ratios from 0.1 to 100.

8. **$\nabla\cdot\boldsymbol{b}$ is not preserved to machine precision** (order 1.65 convergence). No divergence cleaning is implemented.

9. **Magnetic walls impose a prescribed $\boldsymbol{b}$ only.** The moment condition of §8.4 sets $\boldsymbol{b}$ to a given value at the wall, which is what Maxwell's equations give for the Hartmann geometry. Genuinely conducting and insulating walls — where the wall's conductivity determines the field rather than the field being prescribed — are not implemented and are not faked.

10. **FP32 cannot resolve the finest convergence tests.** At $H = 64$ the discretisation error falls below the FP32 round-off floor; the affected assertions are FP64-only by design.

11. **Stability results are bounded by the run length.** "Stable through $t = 1.20\,\mathrm{s}$" means no divergence was detected before that cap, not that the run is unconditionally stable.

12. **The regularised condition has a curvature-induced wall slip.** Whenever the tangential velocity has wall-normal curvature the effective wall is displaced by

$$\text{offset} \times W = \frac{4}{3}\frac{\omega - 1}{\omega^2} \qquad \Longleftrightarrow \qquad \delta u_{\text{slip}} = -\frac{2}{3}\frac{\omega - 1}{\omega^2}\frac{\partial^2 u_\parallel}{\partial n^2}. \tag{129}$$

It is zero for a linear profile — Couette is exact to 4×10^{-15} at every τ from 0.6 to 2.0 — and zero at $\omega = 1$, but it is the whole of the offset column in §12.11 and it limits Ra_c for the on-node wall pair in §12.8. It is intrinsic to truncating $f^{(1)}$ at second Hermite order, which is what "regularised" means: a property of the method, not a defect of this implementation. No correction is applied, but since (58) is exact a user needing better than $O(h)$ wall placement in a curved profile can run at $\omega = 1$ or subtract it.

Derivation. The source is in $f^{(2)}$ — curvature cannot appear in $f^{(1)}$, which carries only first derivatives:

$$f^{(2)} \supset \frac{1 - \omega/2}{\omega^2}(\boldsymbol{c}_i \cdot \nabla)^2 f^{\text{eq}} \supset \frac{1 - \omega/2}{\omega^2}\frac{w_i}{c_s^2} c_{iy}^2 c_{ix}\, \rho\, u'', \tag{130}$$

odd in $\boldsymbol{c}_i$, hence exactly what the bounce-back scaffold reverses. Its half-space contraction is an exact lattice sum,

$$\sum_{c_{iy} > 0} w_i c_{ix}^2 c_{iy}^3 = \frac{1}{18} \qquad \Longrightarrow \qquad \delta\Pi_{xy} = -\frac{1}{3}\frac{1 - \omega/2}{\omega^2}\rho\, u''. \tag{131}$$

Decomposing f^{neq} at an interior node of a converged channel reproduces this source with ratio 1.000 at $\tau = 0.6, 0.8, 1.0, 1.5$, the probe first validated against $\sum_i f^{\text{neq}} = 0$ (to 10^{-17}) and $\sum_i \boldsymbol{c}_i f^{\text{neq}} = -\boldsymbol{F}/2$.

For the wall itself, group the populations by c_{iy} and take the x-odd combination $g_{c_y} = f_{(1,c_y)} - f_{(-1,c_y)}$. The steady recurrences give $(1 - \frac{\omega}{2})r^2 - (2-\omega)r + (1 - \frac{\omega}{2}) = 0$, i.e. $(r-1)^2 = 0$: a double hydrodynamic root and *no* kinetic boundary layer, so the discrete solution is exactly the parabola and the whole effect sits in the closure. With $u = 3(g_+ + g_-) + C$, $C = F[(2-\omega)/\omega + 3/2]$, the reconstruction giving $G_{c_y} = -w_{c_y} F/c_s^2 + 2 w_{c_y} c_y \Pi_{xy}/c_s^4$ and the corrected scaffold giving $\Pi_{xy} = -2\hat{g}_- - F/6$, the wall closes to $g_+(0) + g_-(0) = -F/6$.

The step a one-way argument cannot see is that node 0 collides against the *imposed* $\boldsymbol{u}_w = 0$ rather than the bulk-extrapolated velocity, so continuity with node 1 requires $g_+(0)|_{\text{BC}} = p(0) + \omega U/[6(1-\omega)]$. Eliminating,

$$U\frac{2-\omega}{6(1-\omega)} = \frac{C}{3} - \frac{F}{6}, \qquad 2C - F = \frac{4F}{\omega} \qquad \Longrightarrow \qquad U = \frac{4F(1-\omega)}{\omega(2-\omega)}, \tag{132}$$

and dividing by $du/dy|_0 = AW$ with $A = 3F\omega/(2-\omega)$ cancels both F and $(2-\omega)$, leaving (58). The coefficient $\frac{4}{3} = 2 \times \frac{2}{3}$ is rational because both factors are exact lattice quantities. Measured: exact at $\tau = 0.6, 0.7, 0.9, 1.0$, within 0.05% at $\tau = 0.8$, 1% at $\tau = 2$ where $\nu = 0.5$ is far outside the hydrodynamic regime assumed.

Note that the force corrections of §8.1 *raise* the reported Poiseuille error at $\tau = 0.8$, from 9.07×10^{-3} to 1.71×10^{-2}, because the force defect and the curvature slip have opposite signs there and were partially cancelling. That cancellation was accidental and τ-specific; at $\tau = 1$ the corrected condition is exact.

13. ω_3 **(bulk) in the MHD central-moment scheme** is not specified in [6]; the value 1 was used here and is exposed as a flag. The stability comparison should be read as close rather than exact reproduction.

14. **No wall compliance.** The arbitrary-geometry path (§15) treats the wall as rigid and static: halfway bounce-back on solid voxels, with no fluid–structure coupling. For the aorta this is the largest single departure from the physics being depicted. A real aorta distends in systole and stores volume elastically, and the pulse wave propagates along the compliant wall at $\sim$5 m/s; that Windkessel behaviour is the dominant feature of aortic haemodynamics. With a rigid wall the vessel responds essentially instantaneously and there is no pulse wave at all — the pulsatile case of §15.4 produces a pulsatile *flow rate*, not a pulse *wave*, and should not be read as the latter. Nothing in the solver approximates compliance, and none is faked.

15. **Voxelised walls are a staircase, and wall shear stress from them is not trustworthy.** The surface is quantised to half-integer lattice planes. Halfway bounce-back is second order for a wall that genuinely sits at the halfway position, but an oblique vessel wall does not, so the geometric error is $O(h)$ and the effective radius is uncertain by about half a voxel. No interpolated curved-boundary treatment (Bouzidi, Filippova–Hänel) is implemented. Two consequences are measured rather than asserted: the staircase corners produce the speed outlier of §15.4 ($p_{99.9} = 0.026$ against $p_{100} = 0.070$), and because the central-moment operator relaxes all moments of order ≥ 3 to equilibrium there is no free rate left to set the magic parameter — the equivalent $\Lambda = (\tau - \frac{1}{2})/2 = 0.0197$ sits nearly ten times below the 3/16 at which the bounce-back wall is τ-independent (§12.1), so the effective no-slip plane is offset and moves with viscosity. Voxel-scale gradients at the wall should therefore not be used to report WSS.

16. **Performance figures are from a compute-bound machine** and do not predict GPU behaviour (§19).

17. **There is no contact-line or wetting model.** `PhaseWall` is zero-flux on the phase populations, which is correct for the transport but sets no contact angle, and the gradient stencil reads whatever ϕ the wall node holds; a penalised body is invisible to the phase field, which is advected by the fluid velocity alone. The consequence is measurable rather than notional: in `validation/floating_body` the draft taken from the fictitious mass and the draft taken from the geometry differ by 1.8%, and that gap *bounds* how far the waterline slides on the hull without separating it from the discretisation of either measurement. For a small enough body there is also no surface-tension contribution to the draft, so a Bond number of order one is out of scope. An interface should not be put against a wall, and the Rayleigh–Taylor demonstrator is legitimate only because its spike does not reach the floor within $t^* = 3$.

18. **The phase field has no open boundary.** The scalar's donor machinery (§8.5) would port, with the same second pass and the same fence, but it is not written. Every multiphase case here is therefore periodic in one direction and closed by no-slip walls in the other; an interface that must leave the domain is not available.

19. **There are two pressure normalisations and neither dominates.** With $p = \rho(\phi)c_s^2\tilde{p}$ the recovered mismatch is $F_p = -\tilde{p}\, c_s^2 \nabla\rho$, which scales with the pressure *level* and is amplified by the density ratio; with a constant $p = \rho_0 c_s^2 \tilde{p}$ it becomes $c_s^2(\rho - \rho_0)\nabla\tilde{p}$ and scales with the gradient. On the layered channel the first does not converge at all — fitted order -0.21, with $\max|F_p|/G$ at 220 and 408 for $H = 32$ and 64 — while the second converges at 0.95; a gauge shift in $\tilde{p}$ that changes no physics moves the error seventeenfold under the first and about twofold under the second. It is not a free win, and what does *not* improve is the size of the term: F_p/G is *larger* under ρ_0, 379 against 220 at $H = 32$. The recovered pressure

term becomes $-(\rho_0/\rho)c_s^2\nabla\tilde{p}$, so one phase always pays: $\rho_0 = \rho_L$ diverges at a Laplace jump at a ratio of 10 and above, on a case the local-ρ form handles at a ratio of 100, and $\rho_0 = \rho_H$ gives the light phase an effective sound speed of 1.83 lattice units per step at a ratio of 10 and diverged inside 500 steps. The default stays local-ρ because that is what every validated result in this tree was obtained with.

20. **The multiphase central-moment operator does not run on D3Q19.** It is built on `ProductBasis`, and the `static_assert` says so: a product lattice is required — D2Q9 or D3Q27 — and D3Q19 would need the monomial basis and its own equilibrium central moments. The pressure-form BGK operator carries no such restriction, asking only for a Navier–Stokes lattice, so a D3Q19 two-phase run is possible at single-relaxation-time quality — which is exactly the quality that failed at Re = 30000 in §17.1.

21. **Shifted storage is unavailable to the multiphase operators.** `ShiftedPopulations` centres the stored variable on $\tilde{p} = 1$, which is precisely the pressure gauge that must be avoided at a density ratio, and it assumes a reference density of exactly 1 — at a ratio of 10 the stored populations are no longer small and the shift stops buying anything. `MultiphaseCentralMoments` therefore declares `RawPopulations`, and the FP32 accuracy argument that shifted storage exists for does not apply to these cases.

22. **The free surface has no surface tension** (§11). ρ_G is uniform, so the gas pressure carries no curvature term, and every case at a Bond number where surface tension competes with gravity is out of scope. That excludes most droplet problems; it excludes none of the dam-break family, which is what the module is for.

23. **The free surface's moving obstacle is not reliable**, and the cause is located rather than suspected. `mass_exchange()` conserves mass by antisymmetry across a face, which assumes both cells agree on the mask for the whole pass; `move_obstacle()` changes the mask between passes, so every cell the body sweeps breaks the assumption. `transfer_covered_mass()` keeps the total right without restoring the per-face antisymmetry. Bounce-back, momentum exchange and refill are all in place and are not the problem. The fix is to make the mass exchange itself aware of the moving boundary, and it has not been attempted.

24. **The free surface cannot use Esoteric Pull**, and therefore cannot be ported to the CUDA implementation without a two-lattice storage path and five kernel launches with a fence between each (§11.3). It `static_asserts` against it rather than failing at run time.

25. **Neither immiscible model dominates the other** (§18.12). On one matched static-droplet case the colour gradient is more accurate on Laplace's law at every density ratio measured and the phase field carries the smaller spurious current at every ratio, by up to two orders of magnitude. The choice is a trade, and the measurement covers one case — a static droplet — not a flow.

26. **The rigid body is two-dimensional.** `PenalisedBody` carries one angle and one angular velocity about z, and says so in the compiler's voice as well as in prose: instantiated on a 3D lattice it would silently model an infinite prism. A three-dimensional body is a 6×6 system with a rotating inertia tensor and a quaternion, and does not exist here. The fluid side would not change.

27. **The body is of uniform density.** Its centre of mass is taken to be the centre of the rectangle, which is what lets its own weight drop out of the roll equation and leaves $-(\boldsymbol{S} \times \boldsymbol{g})$ as the whole righting couple. So the metacentric height follows from the shape and the draft alone — $+13.61$ for the raft and -6.80 for the pillar in `validation/floating_body`, the two boxes differing only in which side is longer — and ballast low in a hull, the standard

way to make a tall body float upright, is exactly what cannot be expressed. It would need a second centre, a metacentric height that no longer follows from the draft, and a weight torque restored to the roll row.

28. **There is no collision model.** Two bodies, or a body meeting a wall, will interpenetrate: nothing computes a contact force. The floor stop in the water-entry demonstrator is a clamp in the driver on the square's diagonal reach, and it is a clamp, not physics.

29. **The only body shape is a rectangle.** `Rect::chi` is the sole indicator provided. Anything with a signed distance function drops straight in, and nothing else does.

30. **Penalisation has no sub-cell surface.** The body is resolved to the smoothing width of its indicator, so a pressure read at the face of the square is a smoothed pressure over a cell or two, and the draft is resolved to the interface width. Integrated force is far more trustworthy than local pressure here, which is the usual situation with this family of methods. The smoothing is not free of bias either: at a width of 1.5 cells the indicator's second moment exceeds the sharp rectangle's by 2.55%, which is why mass and inertia are taken from χ rather than from the nominal dimensions.

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